WASTEWATER TO BIOGAS GROUPED PROJECT

Document Prepared by Green Lagoon Technology Sdn Bhd and Eco-Ideal Consulting Sdn Bhd

Project title	Wastewater to Biogas Grouped Project
Project ID	4887
Crediting period	01-Apr-2024 to 31-Mar-2031
Original date of issue	17-Nov-2023
Most recent date of issue	20-Dec-2023
Version	1.0
VCS Standard Version	4.5
Prepared by	Green Lagoon Technology Sdn Bhd and Eco-Ideal Consulting Sdn Bhd

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Grouped Project Description

Project description and location: Wastewater to Biogas Grouped Project (hereinafter referred to as the project) is a grouped project which consists of biogas recovery projects located in Malaysia. The project is owned by Green Lagoon Technology Sdn Bhd. (hereinafter referred to as the project owner), and implementation of the projects are by the individual plant operators of each project activity instance (PAI). The purpose of this Grouped Project is to recover biogas generated from wastewater treatment process which would have otherwise, been emitted into the atmosphere resulting in GHG emissions. The recovered biogas is expected to be utilized for various purposes:

a. To upload to grid, and/or

b. Use in transportation applications, and/or

c. Bottled and distributed to end users, and/or

d. Hydrogen production, and/or

e. Excess biogas will be flared in open/enclosed flare.

Project Activity Instance 1 (PAI1) Description

The first PAI1 is located at Tian Siang Oil Mill (Triang) Sdn Bhd ("Tian Siang POM") in the State of Pahang, Malaysia, about 6 KM Off Jalan Triang Ke Kemayang, Triang, Pahang and 180km north of Kuala Lumpur. GLT Maju Sdn Bhd is the owner of the biogas plant. Green Lagoon Technology Sdn Bhd has signed an agreement with GLT Maju Sdn Bhd, who has the legal rights to operate PAI1. The mill operator, Tian Siang POM has signed an agreement with GLT Maju Sdn Bhd. for ongoing access to the palm oil mill effluent (POME), in addition to permitting the construction and operation of PAI1. GLT Maju Sdn Bhd is owned by Green Lagoon Technology Sdn Bhd.

Baseline Scenario: The wastewater is distributed into open anaerobic lagoons for anaerobic treatment. After the anaerobic treatment, the treated wastewater is discharged to the river. The anaerobic environment within the anaerobic lagoons, resulting in baseline methane generation from the organic content of the wastewater. The sludge from the lagoons is disposed of via land application within the vicinity of the project area.

Project Scenario: The project involves the improvement of wastewater treatment by introducing anaerobic digester, a membrane is installed over the lagoon to capture the methane emitted. After the digesters, the residue will be further treated through a series of aerobic processes before finally discharged to river. The biogas generated from the digesters will be captured and utilized for power generation to be uploaded to grid. The estimated annual average and total

GHG emission reductions during the first seven (7) year crediting period of the PAI is tabulated below:

No	Project activity instance	Annual Average GHG Emission (tCO_2e)	Total GHG emission reductions for first crediting period (tCO_2e)
PAI1	Tian Siang POM	28,440	199,079

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	01-04-2024 to 31-03-2031	VCS	Control Union Certifications Germany GmbH	7 years

1.3 Sectoral Scope and Project Type

The project activity under consideration is a grouped project activity. The project activity instances as part of the grouped project are defined as follow:

Sectoral scope[1]	13 - Waste handling and disposal
Project activity type	III - Other project activities AMS-III.D (Version 21.0) - Methane recovery in animal manure management systems AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment AMS-III.O (Version 02.0) – Hydrogen production using methane extracted from biogas AMS-III.AQ (Version 02.0) – Introduction of Bio-CNG in transportation applications

Sectoral scope	1 - Energy industries (renewable/non-renewable sources)
Project activity type	I - Renewable energy projects AMS-I.D (Version 18.0) – Grid-connected renewable electricity generation

[1] Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes

1.4 Project Eligibility

1.4.1 General eligibility

As per section 2.1.1 of VCS Standard (version 4.5), the scope of the VCS Program includes:

1. The seven Kyoto Protocol greenhouse gases: The proposed grouped project is designed to treat the wastewater by the installation of new biogas digester and avoid methane emission from open anaerobic lagoons during the baseline scenario. Methane is one (1) of the seven (7) Kyoto Protocol greenhouse gases, therefore the project is applicable to this scope.
2. Ozone-depleting substances (ODS): Not applicable.
3. Project activities supported by a methodology approved under the VCS Program through the methodology development and review process: Not applicable.
4. Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions): The applied methodologies AMS-III.H (Version 19.0) and/or AMS-III.D (Version 21.0) and/or AMS-III.AQ (Version 02.0) and/or AMS-I.D (Version 18.0) of the grouped project are methodologies approved under CDM Program, which is a VCS approved GHG program.
5. Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: Not applicable.

As per section 2.1.2 of VCS Standard (version 4.5), the scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction: The proposed grouped project involves the utilization and destruction of biogas generated from the biogas digester to replace the existing open anaerobic lagoons, which does not generate GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction.

As per section 2.1.3 of VCS Standard (version 4.5), the VCS Program also excludes the following project activities under the circumstances indicated in Table 1: Excluded Project Activities. The proposed grouped project involves the utilization and destruction of biogas generated from the biogas digester to replace the existing open anaerobic lagoons, which does not fall under the exclusion listed in table 1.

The project is expected to start in operation on 01-Apr-2024, it shall complete validation before 31-Mar-2026.

According to VCS, the projects seeking registration in the VCS Program that comply with all VCS Program rules may use methodologies from other approved GHG programs, including Clean Development Mechanism (CDM). The methodologies to be applied for the grouped project are adopted from CDM methodologies which are AMS-III.D, AMS-III.H, AMS-III.O, AMS-III.AQ, and AMS-I.D. For small scale projects, the measures are limited to those that result in aggregate emission reductions of less than or equal to 60 $ktCO_2$ equivalent annually from all Type III components of the project activity. The PAIs under the grouped project activity will result in less than or equal to 60$ktCO_2$ equivalent annually from all Type III, as demonstrated in section 3.2 of PD.

1.4.2 AFOLU project eligibility

Not applicable as the grouped project activity is not AFOLU project.

1.4.3 Transfer project eligibility

Not applicable as the grouped project activity is not transferred from other GHG Programs.

1.5 Project Design

The project has been designed as*:*

☐ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☒ Grouped project

1.5.1 Grouped project design

The project is a grouped project and following are the eligibility criteria for inclusion of new project activity instances into the grouped project activity:

- **Applicability Conditions:** The project activity instances shall meet applicability conditions for applicable methodologies (AMS III.H version 19.0 and/or AMS-III.D version 21.0 and/or AMS-III.O version 02.0 and/or AMS-III.AQ version 02.0 and/or AMS I.D version 18.0) as defined in section 3.2
- **Geographical Area:** The project activity instance locates in the geographic boundary for the grouped project, i.e., Malaysia, and owned by the project owner or the subsidiary of the project owner.
- **Baseline scenario:** All Project Activity Instances shall meet the baseline definition as defined in respective methodology and as explained in section 3.4
- **Technology type:** The project activity instances to be included in the grouped project activity is the implementation of biogas recovery, flaring and/or utilization systems in palm oil mill or dairy farms using either anaerobic digester or covered anaerobic lagoon, and gas engine for power generation and/or transportation applications and/or hydrogen production.

- **Additionality:** The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology. As per tools Positive lists of technologies - Version 04.0, A positive list of technology and project activity types that are defined as automatically additional if the PAIs apply such technologies and meet specified conditions. The positive list of technology including the methane recovery in wastewater treatment where the specific conditions to be met is listed below:
  - The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation;
  - There is no regulation in the host country that requires the management of biogas from domestic, industrial and agricultural sites;
  - There is no capacity increase in the wastewater treatment system;
  - No other alternative economic activity is expected to be undertaken on the land of the existing lagoon;
  - The biogas is used to generate electricity in one or more power plants, and the total nameplate capacity is below 5 MW.
- **Start Date:** The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.

The following table demonstrates how the initial project activity instances being included in the grouped project activity fulfil above mentioned eligibility conditions:

Table 1: PAIs Eligibility Criteria

No	Eligibility Criteria	PAIs Eligibility
1	**Applicability Conditions:** The project activity instances shall meet applicability conditions for applicable methodology (AMS III.H version 19.0 and/or AMS-III.D version 21.0 and/or AMS-III.AQ version 02.0 and/or AMS-III.O version 02.0 and/or AMS I.D version 18.0)	All initial project activity instances meet respective applicability conditions of methodology AMS III.H version 19.0 and/or AMS-III.D version 21.0 and/or AMS-III.AQ version 02.0 and/or AMS-III.O version 02.0 and/or AMS I.D version 18.0. Hence this eligibility criteria are fulfilled. Please refer section 3.2 of this document for applicability
2	**Geographical Area:** The project activity instance locates in the geographic boundary for the grouped project, i.e., Malaysia, and owned by the project owner or the subsidiary of the project owner	All initial project activity instances being included in the grouped project are located within geographic boundaries of Malaysia. Hence this condition is fulfilled. Please refer to KML file for the Geographical area of project activity

No	Eligibility Criteria	PAIs Eligibility
3	**Baseline scenario:** All Project Activity Instances shall meet the baseline definition as defined in respective methodology	All initial project activity instances have the same baseline as per methodology as detailed in subsequent sections. Hence this eligibility criteria are fulfilled. Please refer section 3.4 of this document for baseline scenario
4	**Technology type:** The project activity instances to be included in the grouped project activity is the implementation of biogas recovery, flaring and/or utilization systems in palm oil mill or dairy farms using either anaerobic digester or covered anaerobic lagoon, and gas engine for power generation and/or transportation applications and/or hydrogen production.	All initial project activity instances being included in the grouped project activity are the implementation of biogas recovery, flaring and/or utilization systems in agriculture or farming industries. The recovered biogas is converted for various purposes which include supplying electricity generated to grid, converted to use for transportation or bottled for end users and/or to produce hydrogen. Hence this condition is fulfilled.
5	**Additionality:** The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology	The initial project activity instances being included in grouped project activity are small scale projects and their additionality is demonstrated as discussed in section 3.5 of this document. Hence this condition is fulfilled. Please refer section 3.5 of this document for additionality.
6	**Start Date:** The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence	The start date of grouped project is considered on 01-Apr-2024 which is the date of the biogas plant in operation for PAI1. Hence this condition is fulfilled. Please refer section 1.8 of this document or commissioning certificate for start date of PAIs.

1.6 Project Proponent

Organization name	Green Lagoon Technology Sdn Bhd
Contact person	Chan Sow Keong
Title	Director
Address	H-3A-01, Pusat Perdagangan Bandar Bukit Jalil, Persiaran Jalil 1, 57000 Kuala Lumpur, Malaysia
Telephone	+603 2733 2960
Email	chansk@glt.my

1.7 Other Entities Involved in the Project

Organization name	GLT Maju Sdn Bhd
Role in the project	Project Operator of PAI1
Contact person	Chan Sow Keong
Title	Director
Address	H-3A-01, Pusat Perdagangan Bandar Bukit Jalil, Persiaran Jalil 1, 57000 Kuala Lumpur, Malaysia
Telephone	+603 2733 2960
Email	chansk@glt.my

1.8 Ownership

Grouped Project

Green Lagoon Technology Sdn Bhd is the sole project owner of the project. Every additional PAI to be added to this Grouped Project is required to provide evidence of project ownership.

PAI1

GLT Maju Sdn Bhd is the owner of the biogas plant. Green Lagoon Technology Sdn Bhd has signed an agreement with GLT Maju Sdn Bhd, who has the legal rights to operate PAI1. The mill operator, Tian Siang POM has signed an agreement with GLT Maju Sdn Bhd. for ongoing access to the palm oil mill effluent (POME), in addition to permitting the construction and operation of PAI1. GLT Maju Sdn Bhd is owned by Green Lagoon Technology Sdn Bhd.

1.9 Project Start Date

Project start date	01-Apr-2024
Justification	According to VCS Standard, version 4.5, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. The Grouped Project start date is 01-April-2024, which is the commencement of emissions reductions at PAI1 when the biogas plant is in operation.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ *Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)*
Start and end date of first or fixed crediting period	01-Apr-2024 to 31-Mar-2031

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO_2e/year (project)

☐ ≥ 300,000 tCO_2e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
01-April-2024 – 31-December-2024	21,427
01-January-2025 – 31-December-2025	28,440
01-January-2026 – 31-December-2026	28,440
01-January-2027 – 31-December-2027	28,440
01-January-2028 – 31-December-2028	28,440
01-January-2029 – 31-December-2029	28,440
01-January-2030 – 31-December-2030	28,440
01-January-2031 – 31-March-2031	7,013

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
Total estimated ERRs during the first or fixed crediting period	199,079
Total number of years	7
Average annual ERRs	28,440

1.12 Description of the Project Activity

Grouped Project

The Grouped Project will consist of multiple PAIs for the implementation of biogas recovery, flaring and/or utilization systems from palm oil mill or dairy farm wastewater in Malaysia. The Grouped Project will reduce greenhouse gas (GHG) emissions from sources of palm oil and/or dairy farm wastewater by capturing the biogas that would otherwise have been generated and released to the atmosphere from the baseline wastewater treatment systems.

PAI1

The PAI1 is named the Tian Siang Biogas Plant. It is implemented at Tian Siang Palm Oil Mill ("Tian Siang POM") in the State of Pahang, Malaysia, about 6km from the Temerloh and 160km north of Kuala Lumpur. Tian Siang POM is an established palm oil mill that is processing approximately 198,891 MT/yr of FFB. The processing of FFB creates a significant volume of wastewater that requires treatment, due to a high level of organic matter, as measured by the wastewater's Chemical Oxygen Demand (COD). The existing wastewater treatment system includes a series of ponds (anaerobic and aerobic). The breakdown of COD in an anaerobic environment result in the emission of methane into the atmosphere.

In PAI1, the biogas digester is installed in a new lagoon, and a membrane is installed over the lagoon to capture the methane emitted. By capturing methane emissions and combusting the biogas, the PAI leads to reduced emissions of greenhouse gases. This methane is converted into biogas before processing for use in the generation of electricity.

PAI1 will include the following technologies:

1. An anaerobic digester – The In-Ground Bio Reactor System with Biogas Storage (IGBR) System is approximately 104m x 44m x 8m depth, with an effective volume of around 30,000m^3. The reactor uses an in-ground pond and covered High-Density Polyethylene Geomembrane (HDPE) membrane, which also acts concurrently as a biogas storage facility, capable of storing up to 21,000m^3 of biogas. The hydraulic retention time is 35 days. The reactor is also integrated with hydraulic sludge mixers and a sludge removal system inside the digester.

2. Biogas treatment system - This includes 1 unit of scrubber and a dehumidifier and chiller. The biological scrubbers with a capacity of 650m^3/hr are used to oxidize hydrogen sulphide (H_2S) to sulphate. The result of the scrubbing process is clean biogas with H_2S content reduced to less than 200 ppm. The biogas is passed from the scrubber to a

dehumidifier/chiller system to reduce the water content in the biogas, from 75-90% to 40%, and to cool the biogas to the point it can be used in the biogas engine.

3. Biogas engine – A single Caterpillar biogas engine model CG170-16, which has a gross electrical output of 1,560kW is installed at GLT Maju Biogas Plant. An open flare is also installed, although it is not expected to be needed due to the storage capacity of the biogas membrane.

4. Control and monitoring system – All parameters necessary for the operation and management of the biogas plant are integrated to a Supervisory Control and Data Acquisition (SCADA) system.

1.13 Project Location

Grouped Project

Project Activity Instances within the border of Malaysia are allowed to be added to and included in this Grouped Project. Geodetic coordinates are provided as a KML file.

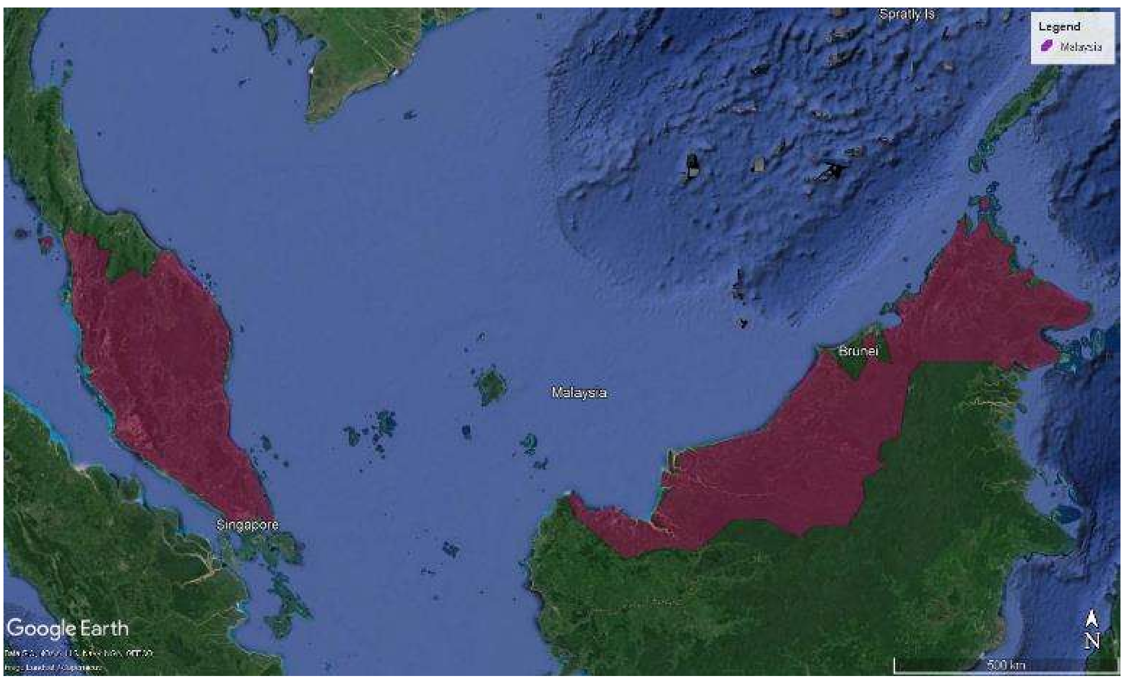

Figure 1: Grouped Project Location

PAI1

PAI1 is situated at the site of Tian Siang Oil Mill (Triang) Sdn Bhd in the State of Pahang, Malaysia, about 6 KM Off Jalan Triang Ke Kemayang, Triang, Pahang and 160km north of Kuala Lumpur. The geographical coordinates for the site are N 3°10'58.6" and E 102°27'27.3". The exact location of the site is shown in Figure below:

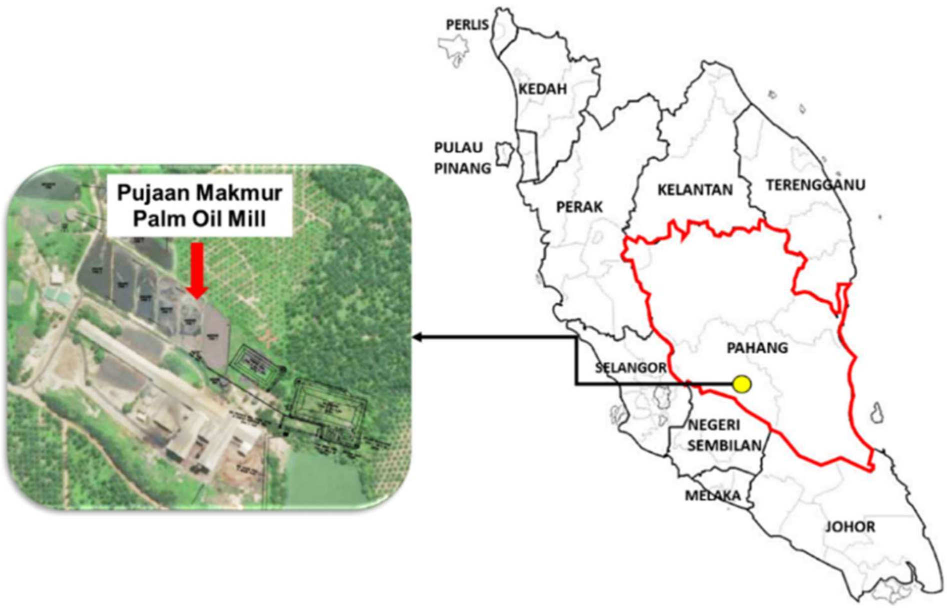

Figure 2: PAI1 - Project Location

1.14 Conditions Prior to Project Initiation

The existing scenario prior to the start of the implementation of each project activity instance is: The wastewater was left at uncovered open lagoon and methane is emitted to the atmosphere directly without any methane recovery and destruction facility. The conditions existing prior to the project initiation are the same as the baseline scenario, please refer to Section 3.4 (Baseline Scenario) for further details.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Grouped Project

Every additional PAI to be added to this Grouped Project is required to demonstrate compliance with all and any relevant local, regional and national laws, statutes and regulatory frameworks.

PAI1

PAI1 is in compliance with the conditions of local laws and regulations of Malaysia, including the following of particular relevance:

- Environmental Quality Regulations (Designated Premises) (Crude Palm Oil) 1977 – P.U.(A) 342/77 Regulation 6 - Making Changes that change the quality of the effluent [2]

[2] Peraturan Kualiti Alam Sekeliling (Premis Yang Ditetapkan) (Minyak Kelapa Sawit Mentah) 1977 – P.U.(A) 342/77 Peraturan 6 - Membuat Perubahan-perubahan yang mengubah kualiti effluen

- Written Declaration on the Design and Construction of Air Pollution Control Systems [Regulation 7(5)] DOE ASPUB[3]
- Construction Site Registration - DOSH JKJ 103[4]
- Environmental Quality (Prescribed Premises) (Crude Palm Oil) Regulations 1977 - DOE[5]
- Labor Act 1995[6]

There are no regulations in Malaysia that requires management of biogas from PAI1. Condition in the criteria and guidelines[7] for an application for licensing by the Malaysian Palm Oil Board (MPOB) require new palm-oil mills constructed after 01-Jan-2014 to install wastewater treatment systems with methane capture. The requirement was also placed on existing mills who applied for capacity extensions after 1-Jan-2014, although an exemption effective 22-May-2021 was given to those existing mills who applied for capacity extensions below 270,000 MT FFB per year, noting this threshold has been revised several times.

Tian Siang POM was established in 2002, and not a mill constructed after 01-Jan-2014 and has not undergone a capacity expansion since 01-Jan-2014. Therefore, the new criteria do not apply and there is no obligation to recover the methane and/or use the biogas.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

[3] Perisytiharan Bertulis Mengenai Reka Bentuk Dan Pembinaan Sistem Kawalan Pencemaran Udara [Peraturan 7(5)] DOE ASPUB
[4] DOSH JKJ 103
[5] Peraturan-Peraturan Kualiti Alam Sekeliling (Premis Ditetapkan) (Minyak Sawit Mentah) 1977
[6] Akta Buruh 1995
[7] Kriteria dan Panduan Permohonan Lesen MPOB, available from: https://e-lesen.mpob.gov.my/document/CRITERIA%20AND%20GUIDELINES%20ON%20MPOB%20LICENSE%20APPLICATION_EnglishVersion_latest.pdf

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The project captures the methane release to produce biogas for electricity generation. The project achieves significant GHG emission reductions by destroying methane gas that would have been vented into the atmosphere directly in the absence of the project activity. The project also provides long term job opportunities to local people and contributes to the economic prosperity.

The Project will contribute to sustainable development in the following ways:

SDG	Project Activity Contribution
7 AFFORDABLE AND CLEAN ENERGY — SDG 7 "Ensure access to affordable, reliable, sustainable and modern energy for all"	The biogas generated from the project activity will be used for electricity generation for the project activity and upload to the grid. The biogas plant will capture the methane to convert into renewable energy and turn in producing valuable climate and clean air energy solution

SDG	Project Activity Contribution
SDG 8 "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all"	During the construction and operation of the project will provide new job opportunities which will increase the income of local residences and accelerate economy development in the areas.
SDG 13 "Take urgent action to combat climate change and its impacts	The project will achieve a GHG emission reduction during the crediting period. The project will recover and destroy methane gas that would otherwise be released directly into the atmosphere, The expected total emission reductions are 199,079 tCO_2e during the first 7-year crediting period, with an average annual emission reduction 28,440 tCO_2e

During this monitoring period, the Project will contribute to sustainable development in the following ways.

SDG	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Activity Contribution	Contributions over project lifetime
7	7.2	Amount of electricity generation	Implemented activities to increase	An average of 8,000 MWh per year of power is uploaded to the grid.	A total of 56,000 MWh of power will be uploaded to the grid during the first crediting period
8	8.5	Number of job opportunities provided	Implemented activities to increase	A total number of 8 job opportunity will be provided to the local	At least total number of 8 job opportunity will be provided to the local
13	13.0	Tons of greenhouse gas emissions avoided	Implemented activities to increase	An annual average emission reduction of 28,440 tCO_2e was achieved from the project activity	A total of 199,079 tCO_2e of emission reduction will be achieved during the first crediting period

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

In accordance with AMS III.D, Ver. 21, as per "Project and leakage emissions from anaerobic digesters" (version 02.0), the leakage emissions include emissions associated with storage of digestate and composting of the digestate. The project does not involve anaerobic composting of digestate. Therefore, leakage emissions of the project associated with the anaerobic digester is not accounted for the Grouped Project or to PAIs.

In accordance with AMS III.H, Ver. 19, no leakage emissions are considered as the technology is not using equipment transferred from other activity.

In accordance with AMS-III.AQ Ver. 02.0, there are two sources of leakage to be considered, competing use of biomass and upstream leakage emissions associated with fossil fuel use. As the biomass for this grouped project are by-products/residues from industrial and/or dairy farm which is being treated and discharged to land application or river, there is no competing use of biomass to be considered for leakage. Therefore, only upstream leakage emissions are considered for leakage, and calculated in accordance with "Upstream leakage emissions associated with fossil fuel use", Ver. 02.0.

In accordance with AMS I.D, Ver. 18, "General guidance on leakage in biomass project activities" shall be followed. As grouped project and PAIs are not demonstrating competing use of biomass, there will be no leakage to be considered.

1.19.2 Commercially Sensitive Information

Grouped Project

Where applicable, PAIs to be added to this Grouped Project shall briefly describe the items to which commercially sensitive information has been excluded from the public version of the Project Description if this is relevant.

PAI1

This is not applicable to PAI1 as there is no commercially sensitive information that is or needs to be excluded from the public of the project description.

1.19.3 Further Information

Not applicable. There is no additional relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information that may have a bearing on the eligibility of the project, the GHG emission reductions or carbon dioxide removals, or the quantification of the project's reductions or removals.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The process to identify stakeholders is by extending a radius of 10 km from the project site (Tian Siang Oil Mill (Triang) Sdn Bhd,), assuming that this is the area likely to be impacted by the project. The area within 10km boundary of the project site was found to include local residents, schools, government agencies, housing estates and neighboring plantation estates, The list of stakeholders identified to be included in the invitation list are: • Pejabat Daerah Dan Tanah Bera • Communities residing within MRSM Bera • Communities residing within SK LKTP Triang II • Communities residing within SJK (C) Mengkuang • Nearby local residents – Felda Triang II • Felda Triang and Felda Mengkuang Satu millworkers and community residing within the mills • Tian Siang Oil Mill (Triang) Sdn Bhd workers and community residing within the mill • Bera town and housing estates residents The stakeholders identified above are considered adequate to fulfil the criteria to represent communities affected by the project activity within 10km radius of the project site.
Legal or customary tenure/access rights	The land for the project activity is owned by Tian Siang Oil Mill (Triang) Sdn Bhd. Surrounding area also has plantation area owned by Felda plantation estate and its settlement. There is a small town, Bera located at the north-eastern part of the 10-km perimeter border of the area. Stakeholders within the area are comprised of different ethnicities having diverse cultural diversity, synonym with Malaysian mix of different racial groups. Though having different diversity of culture with different economic and social groups who make up the population, there is no known conflict amongst the community and

	therefore interactions between the stakeholder groups are harmonious. The nearest indigenous people settlements are located near Tasek Bera which is 40km away from the site[8]. Thus, there is no legal or customary tenure/access rights involved or expected.
Stakeholder diversity and changes over time	Triang is under Bera District Council. The population estimation according to ethnic group in Bera District Council for 2020[9] are as below:

Ethnic Group	Total	Percentage (%)
Bumiputera	63,400	63
Chinese	25,400	25
Indian	3,900	4
Others	500	1
Foreigner	6900	7
Total	**100,100**	**100**

The population estimation according to ethnic groups in 2005[10] are as below:

Ethnic Group	Total	Percentage (%)
Bumiputera	48,184	54.2
Chinese	24,981	28.1
Indian	4978	5.6
Others	10,757	12.1
Total	**88,900**	**100**

There was an increase of 13% of the population from 2005 to 2020. Based on the Population Census 2020, the population in Mukim Triang consist of 5,322 people with the distribution number of male and female, 2,748 and 2,574, respectively.

[8] https://www.mdbera.gov.my/en/mdbera/profile/background
[9] https://statsgeo.mycensus.gov.my/geostats/mapv2.php#
[10] https://www.mdbera.gov.my/en/mdbera/profile/background

	Within the period of 15 years between the two census periods, there seem to be an increase of the ethic group 'others'. The ethnicity 'others' make up for people who are not categorized in the major three ethnicities in Malaysia, i.e. Malay, Chinese or Indian. These could be people from other races, i.e. foreigners who have been brought into Malaysia for employment as laborers and plantation workers. This is due to increase in numbers of workers to meet the demand of plantation jobs[11] and is a trend within Malaysia.
Expected changes in well-being	No major changes to the well-being and stakeholder characteristics for Bera district since the district where the project is located, is an agricultural area located within the central of Peninsular Malaysia. Due to the location of the town being in the rural part of Peninsular Malaysia, it is not expected to show significant development compared to other more centrally located cities within Peninsular Malaysia in the next decade. Over time, though there may be minimal development, the area where the project is located will still remain to be less-developed compared to cities. Therefore, the project will provide a positive impact to the stakeholders by bringing in investment and more work opportunity to the local community.
Location of stakeholders	The majority of the stakeholders consist of local communities and institutions who inhabit or are located nearby the project activity site. The project activity is surrounded by palm oil plantation, and the nearest large communities identified are located approximately at the edge of the 10km perimeter from the project site.
Location of resources	The area identified within the 10km radius of the proposed project are located in the district of Bera. The area is largely surrounded by palm oil plantation except the north-eastern part where the town and housing estates of residents are located. Felda palm oil plantation are owned by a public listed company. The housing estates are privately owned properties. Therefore, there is no customary access land within the area.

[11] https://documents1.worldbank.org/curated/en/892721588859396364/pdf/Who-is-Keeping-Score-Estimating-the-Number-of-Foreign-Workers-in-Malaysia.pdf

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	31-Oct-2023
Stakeholder engagement process	The stakeholder meeting announcement was opened to the public and widely promoted through postings on Green Lagoon company website (https://glt.my/updates-media/) on 23rd October 2023, and also social medias of GLTSB (Green Lagoon Technology Sdn Bhd) in Facebook and LinkedIn on 21st October 2023. In addition to public announcement, invitation by letter to a number of identified organizations and individuals relevant to the project was also issued. The invitations were either delivered by hand or e-mailed. Examples of the invitation means are available as evidence.
Consultation outcome	<u>Duration of the meeting</u> The stakeholder consultation meeting started at 10.00am and adjourned at 1130am. <u>Location of the meeting</u> The stakeholder consultation was held in Pahang Rest House at Bera. <u>Meeting participants</u> The meeting was attended by local residents and representatives from the following stakeholder categories: i. Local residents, ii. Government representatives, iii. Schools representative, iv. Felda plantation staff, and v. Tian Siang POM employees. A total of 23 participants attended the meetings. <u>Language</u> Documentation and meeting were held in both English and local Malay language. <u>Meeting procedure</u> The agenda of the meeting include the followings: I. Welcoming remarks II. Overview of biogas plant (project design, implementation, regulations and benefits etc.) III. Introduction to climate change and GHGs IV. Introduction to VCS project activities, validation & verification process

	V. Questions and Answers session <u>Meeting documents and protocols</u> All public comments received during the meetings were compiled and documented. The overall response to the project from all invited stakeholders, was encouraging and positive to local communities, seen as improving social-economic conditions. By recovering the wastewater in a more efficient way to eliminate the odour of emissions from the lagoons, the project will also create job opportunities for the locals. Overall, there were no adverse reactions, comments or clarifications received during the Stakeholder consultation process. There were also no comments or issues raised on the needs to make any changes to the project design. As a current practice, the project proponent encourages the locals/ participants to actively provide feedback or complaints regarding operation of the project activity.
Ongoing communication	The details for this mechanism were shared with the attendees during the stakeholder consultation. The project proponent shall make sure any current feedback is provided by the locals in fair and transparent manner. Any grievances/ communication between project proponents and all stakeholders can be reachable via the company website: www.glt.my
Stakeholder input	Most of the questions were related to the environmental effect of the project activity and benefits from project to the locals.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The project activity does not affect any IPs, LCs, and customary rights holders. Therefore, there is no ongoing or unresolved conflicts in relation to the project activity The nearest settlements of IPs, LCs are located near Tasek Bera, which is 40km from the project site[9].
Outcome of FPIC	No consent is needed from the IPs, LCs and customary right holders as the project activity does not involve any encroachment of Indigenous Customary land rights. The project activity was constructed within the existing Tian Siang POM existing boundary

2.1.4 Grievance Redress Procedure

<table>
<tr><td>Development process</td><td>The grievance redress procedure applies to comment/feedback/complaints received from stakeholders, and will be channeled by GLT to the POM. The receipt and investigation will be incorporated with the existing customer complaint procedure of the POM.</td></tr>
<tr><td>Grievance redress procedure</td><td>
<table>
<tr><th>Flow/ Process</th><th>Description</th></tr>
<tr><td>1. Receipt of comment/ feedback</td><td>Complaints/ comments/ feedback received verbally or in writing shall be kept in a register.
The comment/feedback received from stakeholders shall be acknowledged with the respondent, prior to channeling to the Mill Head.</td></tr>
<tr><td>2. Investigation and initiate actions</td><td>Mill Head shall identify the response owner upon receiving the comment/feedback.
The response owner shall do an investigation, fill up existing “Complaint form” and along with any feedback documents.
The response owner shall carry out investigation and revert to mill Head upon completion of the investigation. The response shall include actions required, if deemed necessary.</td></tr>
<tr><td>3. Review Feedback</td><td>Mill head shall review the response and agree on the actions, if deemed necessary. Response shall be forwarded to GLTSB.</td></tr>
<tr><td>4. Follow- up & Verify actions & effectiveness</td><td>The response owner shall do a follow-up with the implementation of the actions proposed and the effectiveness.
Verification results shall be recorded in:
Corrective/Preventive Action Report</td></tr>
</table>
</td></tr>
</table>

		▪ Complaint Record ▪ Complaint Summary
	5. Update Procedure	• Updating relevant procedures/ documentation, including the risk(s) and/or opportunities (only if deemed necessary), by the response owner.
	6. Feedback to respondent	• GLTSB shall revert to the respondent via post/fax/email, the completed "Corrective/Preventive Action Report" and/or the "Complaint Record", unless the respondent had specifically required GLTSB to reply via pre-designed forms supplied by the latter.
	7. Compile Data	• Information received from the mill shall be updated in the register, along with any required data for monitoring purpose.

2.1.5 Public Comments

The overall response to the project, from all the involved stakeholders, was encouraging and positive to local environment and social-economic conditions by treating the wastewater in efficient way, avoiding odour emissions from the lagoons and also creating job opportunities to the locals.

Comments received	Actions taken
Will biogas generated be connected to Tenaga National Berhad (TNB) grid or micro-grid?	It will be connected to the grid after the voltage goes through step-down. This is a normal process to reduce voltage before connecting to a grid.
what is the type of gas released during flaring?	Burning of methane by flaring in high temperature >500 degree centigrade will release carbon dioxide.
Where is the location of the project and when will it be expected to start operation? Will the project be open for public/school children to visit?	The biogas plant is located in Triang and is expected to start operation in April 2024. As for the possibility of visiting the plant, they (the stakeholders or any interested parties) are more than welcome to visit. However, prior approval is

	required and therefore they need to contact the plant manager. There is no safety issue that the public need to worry because methane capture projects is common these days, and this is not the first project in Malaysia. No untoward incidence has been recorded so far in Malaysia.
Is the project status under Verra Registry?	Not yet. The PD is still currently under development, pending the information related to stakeholder consultation before being uploaded to VERRA website.

There has been no request to project design update brought up during the meeting. All issues raised are summarised above, and no further actions on the need to update project design are required.

2.2 Risks to Stakeholders and the Environment

	Risks identified	Mitigation or preventative measure taken
Risks to stakeholder participation	No risk identified	Conduct briefing on the project activity to the involved stakeholder to inform on the project. All stakeholders have been informed during the stakeholder consultation meeting on the safety concern of the project activity
Working conditions	No risk identified	Policy related to safety of working conditions was established. Emergency Plan and First Aid Management is available in the biogas plant
Safety of women and girls	No risk identified	Safety at workplace is highly emphasized. In the safety work procedure has stated that pregnant woman is not advisable to handle harmful reagents
Safety of minority and marginalized groups, including children	No risk identified	Safety procedure for visitors including children to the biogas plant is also available and therefore safety issues is not a concern for visitors
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	No risk identified	It is expected that the by-product of the project activity shall not release any harmful pollutants that might degrade the environment

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Discrimination and sexual harassment	There are no discrimination or sexual harassment occurred or expected to occur in the project site.
Management experience	Green Lagoon Technology Sdn Bhd is a Malaysian business entity which was established in May 2010. The Company's goal is to play a role as an environmental solutions provider, contributing to the environmental movement. The Company initially participated in Clean Development Mechanism ("CDM") projects in the palm oil mill industry by providing design, construction, operations, and management services. This laid the foundation for the company to expand into new green products and business models. By converting biogas from palm oil milling into various uses, such as domestic consumption and electricity sales through Malaysia's Feed-in-Tariff (FiT) program, the Company can monetize its slogan, "Powering Waste into Profit." With involvement in more than 60 projects, including investment and ownership in these power assets, the company has become one of Malaysia's top biogas developers.
Gender equity in labor and work	There is no gender discrimination in terms of labor in this project activity. Equal job opportunities were provided to both genders and demonstrated equal pay to both. In GLT Headquarter, the staff ratio of 60% to 40% for male to female. Overall, at the Group level, a gender ratio of 95% to 5% for male to female, with males being dominant in biogas plants operation. Men work in upstream and technical activities, such as putting feedstock into the digester, cleaning clogged feedstock channels and biogas pipe distribution. Besides, most of the operational plants are located in rural areas where safety is one of the utmost concerns. That is why all the trained personnel that work at operational plants are men.
Human trafficking, forced labor, and child labor	Not applicable as the project activity does not involve any human trafficking, forced labor, and child labor in the operation.

2.3.2 Human Rights

Not Applicable as the project activity does not involve any rights of IPs, LCs, and customary rights holders.

2.3.3 Indigenous Peoples and Cultural Heritage

Not Applicable as the project activity does not involve any cultural heritage.

2.3.4 Property Rights

Rights to territories and resources	Not applicable as the PAI1 does not involve any legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights, held by stakeholders. The land for the PAI1 is owned by the mill owner
Respect for property rights	Not applicable

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Not applicable for PAI1 as the project does not impact any property rights
Summary of the benefit sharing plan	Not applicable for PAI1 as the project does not impact any property rights
Approval and dissemination of benefit sharing plan	Not applicable for PAI1 as the project does not impact any property rights

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	No risk identified	No formal impact on biodiversity due to the sitting of the PA1 which is located on the existing mill. No new areas were explored.
Soil degradation and soil erosion	No risk identified	No formal impact on the soil applicable to the project.
Water consumption and stress	No risk identified	No impact on water stress and consumption as there were no water consumed throughout the PAI1.
Usage of fertilizers	No risk identified	No fertilizers applied throughout the PAI1.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

2.4.2 Introduction of species

Not applicable as the Grouped Project and PAI1 is not introducing any species.

2.4.3 Ecosystem conversion

Not applicable as the Grouped Project and PAI1 does not involve ARR, ALM, WRC or ACoGS projects.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Grouped Project

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.D	AMS-III.D.: Methane recovery in animal manure management systems	21.0
Tool	TOOL03	Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion	03.0
Tool	TOOL05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	TOOL06	Project emissions from flaring	04.0
Tool	TOOL14	Project and leakage emissions from anaerobic digesters	02.0
Methodology	AMS-III.H	AMS-III.H.: Methane recovery in wastewater treatment	19.0
Tool	TOOL03	Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion	03.0
Tool	TOOL04	Emissions from solid waste disposal sites	08.1
Tool	TOOL05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	TOOL06	Project emissions from flaring	04.0
Tool	TOOL32	Positive lists of technologies	04.0
Methodology	AMS-III.AQ	AMS-III.AQ.: Introduction of Bio-CNG in transportation applications	02.0
Tool	TOOL03	Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion	03.0

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Tool	TOOL05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	TOOL15	Upstream leakage emissions associated with fossil fuel use	02.0
Tool	TOOL16	Project and leakage emissions from biomass	05.0
Methodology	AMS-III.O	AMS-III.O: Hydrogen production using methane extracted from biogas	02.0
Tool	TOOL03	Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion	03.0
Tool	TOOL06	Project emissions from flaring	04.0
Tool	TOOL21	Demonstration of additionality of small-scale project activities	13.1
Methodology	AMS-I.D	AMS-I.D: Grid connected renewable electricity generation	18.0
Tool	TOOL03	Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion	03.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	07.0
Tool	TOOL10	Tool to determine the remaining lifetime of an equipment	01
Tool	TOOL16	Project and leakage emissions from biomass	05.0
Tool	TOOL11	Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period	03.0.1

Note that PAIs within the Grouped Project may not require the use of all the above approved methodologies and methodological tools. Relevance and/or applicability should be assessed on a case by-case basis.

PAI1

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.H	AMS-III.H.: Methane recovery in wastewater treatment	19.0
Tool	TOOL05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	TOOL06	Project emissions from flaring	04.0
Tool	TOOL32	Positive lists of technologies	04.0
Methodology	AMS-I.D	AMS-I.D: Grid connected renewable electricity generation	18.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	07.0
Tool	TOOL10	Tool to determine the remaining lifetime of an equipment	01

3.2 Applicability of Methodology

The project includes small-scale activities for which methodologies of type AMS-III.D, AMS-III.H, AMS-III.AQ, AMS-III.O and AMS-I.D are applicable:

Grouped Project

Methodology ID	Applicability condition	Justification of compliance
AMS III D	*As per paragraph 3 of AMS-III.D.* This methodology is only applicable under the following conditions: (a) The livestock population in the farm is managed under confined conditions; (b) Manure or the streams obtained after treatment are not discharged into natural water resources (e.g. river or estuaries), otherwise “AMS-III.H	Potentially Applicable If applicable to a PAI within the Grouped Project PAIs must satisfy the following conditions: (a) All the livestock population in the farms within the project boundary is managed under confined conditions; (b) The manure or the streams obtained after treatment are not

Methodology ID	Applicability condition	Justification of compliance
	Methane recovery in wastewater treatment" shall be applied; (c) The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C; (d) In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m; (e) No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario	discharged into natural water resources (e.g. river or estuaries); (c) The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C[12] (d) The retention time of manure waste in the anaerobic treatment system is greater than one month and that depth of anaerobic treatment system is more than 1 meter depth; (e) No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario
AMS III D	*As per paragraph 4 of AMS-III.D.* The project activity shall satisfy the following conditions: (a) The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of "AMS-III.AO Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions and procedures (not resulting in methane emissions) must be ensured (b) Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared	Potentially Applicable If applicable to a PAI within the Grouped Project, the PAIs must satisfy the following conditions: (a) The residual waste will be handled aerobically as manure, and thus no emissions to be considered as in AMS-III.AO. (b) A proper technical measure will be taken care, all the biogas which is produced in the biogas digester will be used to generate electricity, and remaining will be flared (c) The storage time of the manure after removal is less than 45 days before it is being fed into the anaerobic digester

Methodology ID	Applicability condition	Justification of compliance
	(c) The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when removed from the animal barns is larger than 20%, this time constraint will not apply	
AMS III D	As per paragraph 5 of AMS-III.D. Projects that recover methane from landfills shall use “AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure shall use “AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology “AMS-III.AO Methane recovery through controlled anaerobic digestion".	Not Applicable PAIs to be implemented under this Grouped Project will not recover methane from landfill, treatment of wastewater or involve composting of animal manure or co-digestion of animal manure and other organic matter.
AMS III D	As per paragraph 6 of AMS-III.D. Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the project activity, it should be taken into account accordingly, using zero as its emission factor; however, energy	Potentially Applicable If applicable to a PAI within the Grouped Project, PAIs implemented under grouped project activity may use biogas to generate electricity for auxiliary and own consumption, as per para 4 of AMS-III.H. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the project activity, it should be taken into

Methodology ID	Applicability condition	Justification of compliance
	used for such purposes is not eligible as an SSC CDM Type I project component	account accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component
AMS III D	As per paragraph 7 of AMS-III.D. New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the "General guidelines for SSC CDM methodologies".	Not Applicable For PAIs to be implemented under this Grouped Project, there will be no any new facilities (Greenfield projects) or involve capacity additions compared to the baseline scenario
AMS III D	As per paragraph 8 of AMS-III.D. The requirements concerning demonstration of the remaining lifetime of the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies".	Potentially Applicable If applicable to a PAI within the Grouped Project, PAIs to be implemented under this Grouped Project shall comply with the relevant requirements in the General guidelines to SSC CDM methodologies and the requirements for demonstrating the remaining lifetime of the equipment replaced as described in the general guidelines shall be followed.
AMS III D	As per paragraph 9 of AMS-III.D. Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO2 equivalent annually from all Type III components of the project activity	Potentially Applicable PAIs to be implemented under this Grouped Project will result in less than or equal to 60ktCO_2 equivalent annually.
AMS III H	As per paragraph 2 of AMS-III.H. The methodology comprises measures that recover biogas from	Potentially Applicable If applicable to a PAI within the Grouped Project, PAIs to be

Methodology ID	Applicability condition	Justification of compliance
	biogenic organic matter in wastewater by means of one, or a combination, of the following options: (a) Substitution of aerobic wastewater or sludge treatment systems with anaerobic systems with biogas recovery and combustion; (b) Introduction of anaerobic sludge treatment system with biogas recovery and combustion to a wastewater treatment plant without sludge treatment; (c) Introduction of biogas recovery and combustion to a sludge treatment system; (d) Introduction of biogas recovery and combustion to an anaerobic wastewater treatment system such as anaerobic reactor, lagoon, septic tank or an on-site industrial plant; (e) Introduction of anaerobic wastewater treatment with biogas recovery and combustion, with or without anaerobic sludge treatment to an untreated wastewater stream; (f) Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery).	implemented under this Grouped Project must recover biogas from wastewater treatment system which would have otherwise been vented to the atmosphere.
AMS III H	As per paragraph 3 of AMS-III.H. In cases where baseline system is anaerobic lagoon, the methodology is applicable if:	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, the baseline system for the PAIs to be implemented under this Grouped Project is anaerobic

Methodology ID	Applicability condition	Justification of compliance
	(a) The lagoons are ponds with a depth greater than two meters, without aeration. The value for depth is obtained from engineering design documents, or through direct measurement, or by dividing the surface area by the total volume. If the lagoon filling level varies seasonally, the average of the highest and lowest levels may be taken; (b) Ambient temperature above 15ºC, at least during part of the year on a monthly average basis; (c) The minimum interval between two consecutive sludge removal events shall be 30 days.	lagoon of which must satisfy the following: (a) The lagoons are deeper than two meters and without aeration; (b) Ambient temperature above 15ºC, at least during part of the year on a monthly average basis, for Malaysia as having tropical climate[12]; The minimum interval between two consecutive sludge removal events is more than 30 days.
AMS III H	As per paragraph 4 of AMS-III.H. The recovered biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case, additional guidance provided in Annex 1 shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case, additional guidance provided in Annex 1 shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented under this Grouped Project must involve facilities to utilize biogas generated for renewable energy and/or upgrade hydrogen production or used as fuel in transportation, and that excess gas is burned via flaring.

[12] Malaysia climate: https://climateknowledgeportal.worldbank.org/country/malaysia/climate-data-historical#:~:text=Malaysia%20has%20a%20tropical%20climate,25.9%C2%B0C%20in%20May.

Methodology ID	Applicability condition	Justification of compliance
	network to a group of end users; or (iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users. (d) Hydrogen production; (e) Use as fuel in transportation applications after upgrading.	
AMS III H	As per paragraph 5 of AMS-III.H. If the recovered biogas is used for project activities covered under paragraph 4 (a), that component of the project activity can use a corresponding methodology under Type I.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented under this Grouped Project must have methane captured utilized for renewable electricity generation of which the approved methodologies AMS-I.D is used.
AMS III H	As per paragraph 6 of AMS-III.H. For project activities covered under paragraph 4 (b), if bottles with upgraded biogas are sold outside the project boundary, the end use of the biogas shall be ensured via a contract between the bottled biogas vendor and the end-user. No emission reductions may be claimed from the displacement of fuels from the end use of bottled biogas in such situations. If, however, the end use of the bottled biogas is included in the project boundary and is monitored during the crediting period, CO_2 emissions avoided by the displacement of fossil fuel can be claimed under the corresponding Type I methodology, e.g. AMS-I.C Thermal energy production with or without electricity.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, bottled upgraded biogas is sold outside of the project boundary, end-use of the biogas will be ensured via a contract between bottled biogas vendor and end user. End users will not claim emission reduction for fuel displacement. For end-use within project boundary, emission from displacement of fuel can be claimed.

Methodology ID	Applicability condition	Justification of compliance
AMS III H	As per paragraph 7 of AMS-III.H. For project activities covered under paragraph 4 (c) (i), emission reductions from the displacement of the use of natural gas are eligible under this methodology, provided the geographical extent of the natural gas distribution grid is within the host country boundaries.	Potentially Applicable If applicable to a PAI within the Grouped Project, emission reduction displacement of use of natural gas are eligible as it is located within the host country boundary distribution grid.
AMS III H	As per paragraph 8 of AMS-III.H. For project activities covered under paragraph 4 (c) (ii), emission reductions for the displacement of the use of fuels can be claimed following the provision in the corresponding Type I methodology, e.g. AMS-I.C.	Not Applicable For PAIs to be implemented under this Grouped Project, there will be no activities of transportation of natural gas distribution via a dedicated piped network to a group of end user(s), for the PAIs. Thus is not claimed as in AMS-I.C.
AMS III H	As per paragraph 9 of AMS-III.H. In particular, for the case of 4 (b) and (c) (iii), the physical leakage during storage and transportation of upgraded biogas as well as the emissions from fossil fuel consumed by vehicles for transporting biogas shall be considered. Relevant procedures in paragraph 18 of the appendix of AMS-III.H Methane recovery in wastewater treatment shall be followed in this regard.	Potentially Applicable If applicable to a PAI within the Grouped Project, for bottling, upgrading and transportation to distribution points, physical leakage will be considered in accordance with paragraph 18 or AMS-III.H.
AMS III H	As per paragraph 10 of AMS-III.H. For project activities covered under paragraph 4 (b) and (c), this methodology is applicable if the upgraded methane content of the	Potentially Applicable If applicable to a PAI within the Grouped Project, for PAIs to be implemented under this Grouped Project, as relevant national regulations do not exist, the minimum

Methodology ID	Applicability condition	Justification of compliance
	biogas is in accordance with relevant national regulations (where these exist) or in the absence of national regulations, a minimum of 96% (by volume).	requirement of 96% (by volume) applies.
AMS III H	As per paragraph 11 of AMS-III.H. If the recovered biogas is utilized for the production of hydrogen (project activities covered under paragraph 4 (d)), that component of the project activity shall use the corresponding methodology AMS-III.O Hydrogen production using methane extracted from biogas.	Potentially Applicable If applicable to a PAI within the Grouped Project, , the recovered biogas is utilized for the production of hydrogen that component of the project activity shall use the corresponding methodology AMS-III.O Hydrogen production using methane extracted from biogas
AMS III H	As per paragraph 12 of AMS-III.H. If the recovered biogas is used for project activities covered under paragraph 4 (e), that component of the project activity shall use the corresponding methodology AMS-III.AQ Introduction of Bio-CNG in road transportation.	Potentially Applicable If applicable to a PAI within the Grouped Project, PAIs to be implemented under this Grouped Project will use the recovered biogas as fuel in transportation applications after upgrading.
AMS III H	As per paragraph 13 of AMS-III.H. New facilities (Greenfield projects) and project activities involving a change of equipment resulting in a capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this methodology if they comply with the relevant requirements in the General guidelines to SSC CDM methodologies. In addition, the	Not Applicable PAIs to be implemented under this Grouped Project is not a Greenfield project and the project activities do not result in capacity addition of the treatment system.

Methodology ID	Applicability condition	Justification of compliance
	requirements for demonstrating the remaining lifetime of the equipment replaced as described in the general guidelines shall be followed.	
	As per paragraph 2.2.14 of AMS-III.H. The location of the wastewater treatment plant as well as the source generating the wastewater shall be uniquely defined and described in the Project Design Document (PDD).	Potentially Applicable If applicable to a PAI within the Grouped Project, the location of each PAIs will be identified by specific reference code, address, map and GPS coordinates. The unique location of the wastewater generating source will be defined and described in Section 1.12. of Project Description (PD), and also shown in the baseline and project activity diagram
AMS III H	As per paragraph 2.2.15 of AMS-III.H Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 ktCO_2 equivalent annually from all Type III components of the project activity.	Potentially Applicable The project activity instance under grouped project activity will result in less than or equal to 60ktCO_2 equivalent annually from all Type III, as demonstrated in section 1.10 of PD.
AMS III AQ	As per paragraph 2 of AMS-III.AQ This methodology comprises activities for production of Biogenic Compressed Natural Gas (Bio-CNG) from biomass including biomass residues and cultivated biomass to be used in transportation applications. Biomass cultivated for production of the Bio-CNG should be sourced from dedicated plantations	Potentially Applicable If applicable to a PAI within the Grouped Project, the PAI will utilize biomass waste residues from industrial and/or dairy farm to produce Bio-CNG.
AMS III AQ	As per paragraph 3 of AMS-III.AQ The project activity involves installation and operation of Bio-CNG plant that includes:	Potentially Applicable If applicable to a PAI within the Grouped Project, PAI will install and operate Bio-CNG plant that includes:

Methodology ID	Applicability condition	Justification of compliance
	(a) Anaerobic digester(s) to produce and recover biogas; (b) Biogas treatment system that includes processing, purification and compression of the biogas to obtain up-graded biogas such that methane content, its quality and physical and chemical properties are equivalent to the CNG (c) Filling stations, storage and transportation.	(a) Anaerobic digester to produce and recover biogas, (b) Biogas treatment systems that include processing, purification and compression of the biogas to obtain upgraded biogas such that methane content, its quality and physical and chemical properties are equivalent to CNG, (c) Filling stations, storage and transportation.
AMS III AQ	As per paragraph 4 of AMS-III.AQ This methodology covers the use of Bio-CNG in various types of transportation applications such as Compressed Natural Gas (CNG) vehicles, modified vehicles. Examples include buses, trucks, three-wheeler, cars, jeeps etc.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAI will use bio-CNG for various types of transportation applications such as CNG vehicles, modified vehicles.
AMS III AQ	As per paragraph 5 of AMS-III.AQ If the part of the recovered biogas is injected into a natural gas distribution grid, emission reduction for that component of the project activity can be claimed following the provisions in annex 1of “AMS-III.H: Methane recovery in wastewater treatment”	<u>Not Applicable</u> Project activity instances under grouped project activity do not have part of the recovered biogas injected into a natural gas distribution grid.
AMS III AQ	As per paragraph 6 of AMS-III.AQ The methodology is applicable if the methane content of the upgraded biogas is in accordance with relevant national regulations and in their absence a minimum of 96 percent (by volume).	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAI will have methane content of the upgraded biogas of minimum of 96 percent (by volume), there is no relevant national regulations.

Methodology ID	Applicability condition	Justification of compliance
AMS III AQ	As per paragraph 7 of AMS-III.AQ If the project activity utilizes biomass sourced from dedicated plantations, the applicability conditions prescribed in the methodological tool 'Project emissions from cultivation of biomass" shall apply.	Not Applicable PAI under the grouped project activity do not source biomass from dedicated plantations.
AMS III AQ	As per paragraph 8 of AMS-III.AQ The retailers, final users (where applicable) and the producer of the Bio-CNG are bound by a contract that states that the final consumers and retailers shall not claim emission reductions resulting from its consumption. Only the producer of the Bio-CNG can claim emission reductions under this methodology.	Potentially Applicable If applicable to a PAI within the Grouped Project, PAI under the grouped project will sign a contract that states that final consumers and retailers shall not claim emission reductions resulting from their consumption of Bio-CNG.
AMS III AQ	As per paragraph 9 of AMS-III.AQ The export of Bio-CNG produced under this methodology is not allowed.	Potentially Applicable If applicable to a PAI within the Grouped Project, PAI under the grouped project activity will not be exporting Bio-CNG.
AMS III AQ	As per paragraph 10 of AMS-III.AQ The digested residue waste leaving the reactor shall be handled aerobically and submitted to soil application, the proper procedure and conditions not resulting in the methane emissions shall be ensured; otherwise the emissions shall be taken into account as per relevant procedures "AMS-III.AQ: Methane recovery through controlled anaerobic digestion".	Potentially Applicable If applicable to a PAI within the Grouped Project, PAI under the grouped project activity will handles digested residue wastes aerobically and used for soil application.
AMS III AQ	As per paragraph 10 of AMS-III.AQ	Potentially Applicable

Methodology ID	Applicability condition	Justification of compliance
	Measures are limited to those that result in emission reduction of less than or equal to 60 kt CO_2 equivalent annually. Where applicable the sum of the emission reductions from all Type III components of a project activity should comply with 60 kt CO_2 equivalent annually.	All PAI under the grouped project activity will result in less than or equal to 60ktCO_2 equivalent annually.
AMS III O	As per paragraph 2 of AMS-III.O. The methodology comprises of installation of biogas purification system to isolate methane from biogas to produce hydrogen	Potentially Applicable If applicable to a PAI within the Grouped Project, PAIs to be implemented under this Grouped Project, PAIs satisfy the requirement of use of methane captured to produce biogas and further purified to produce hydrogen.
AMS III O	As per paragraph 3 of AMS-III.O. The project activity installs: (a) A biogas purification system to isolate methane from biogas, which is being flared in the baseline situation for the production of hydrogen displacing liquefied petroleum gas (LPG) as both feedstock and fuel in a hydrogen production unit; or (b) A biogas purification system in combination with installation of new measures that recover methane from biogenic organic matter from waste water treatment plants of landfills, using technologies/ measures covered in AMA-III.H or AMS-III.G.	Potentially Applicable If applicable to a PAI within the Grouped Project, PAIs to be implemented under this Grouped Project satisfy either one of below: (a) Methane that is being flared will be captured and utilized for production of hydrogen displacing LPG as both feedstock and fuel in hydrogen production unit (b) Installation of a new methane recovery from wastewater treatment plants, using AMS-III.H or AMS-III.G.
AMS III O	As per paragraph 4 of AMS-III.O.	Potentially Applicable

Methodology ID	Applicability condition	Justification of compliance
	Emission reductions resulting from the installation of methane recovery system shall be calculated as per AMS-III.H or AMS-III.G.	If applicable to a PAI within the Grouped Project, PAIs to be implemented under this Grouped Project, fulfil the requirement of installation of the new methane recovery will have emission reductions calculated as per AMS-III.H or AMS-III.G.
AMS III O	As per paragraph 5 of AMS-III.O. There is no diversion of biogas that is already being used for thermal or electricity energy generation or utilized in any other (chemical) process in the baseline.	Potentially Applicable If applicable to a PAI within the Grouped Project, PAIs implemented under grouped project activity are not currently using biogas in thermal or electricity generation unit or other (chemical) processes in the baseline.
AMS III O	As per paragraph 6 of AMS-III.O. The project activity complies with all regulations including all safety related measures.	Potentially Applicable If applicable to a PAI within the Grouped Project, Project activity instances under grouped project comply with all related regulations including safety.
AMS III O	As per paragraph 7 of AMS-III.O. Measures are limited to those that result in aggregate emission reductions of less than or equal to 60,000 t CO_2 equivalent annually from all type III components.	Potentially Applicable If applicable to a PAI within the Grouped Project, Project activity instance under grouped project activity fulfil the requirement as aggregate emission reductions result in less than or equal to 60ktCO_2 equivalent annually from all Type III, as demonstrated in section 1.10 of PD.
AMS III O	As per paragraph 8 of AMS-III.O. This methodology is not applicable to technologies displacing the	Not Applicable Project activity instance under grouped project activity is not

Methodology ID	Applicability condition	Justification of compliance
	production of hydrogen from electrolysis.	currently using electrolysis technologies in producing hydrogen.
AMS I D	As per paragraph 2 of AMS-I.D This methodology comprises renewable energy generation units such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, the PAIs comprises renewable energy generation units such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.
AMS I D	As per paragraph 4 of AMS-I.D This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/ unit(s); or (e) Involve a replacement of (an) existing plant(s).	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, the PAIs will satisfy: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/ unit(s); or (e) Involve a replacement of (an) existing plant(s).
AMS I D	As per paragraph 5 of AMS-I.D Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir;	<u>Not Applicable</u> Project activity instances under grouped project activity are not hydro power plant projects.

Methodology ID	Applicability condition	Justification of compliance
	(b) The project activity is implemented in an existing reservoir where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4W/m^2; or (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4W/m^2.	
AMS I D	As per paragraph 6 of AMS-I.D If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15MW.	Potentially Applicable If applicable to a PAI within the Grouped Project, the PAIs will have both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15MW.
AMS I D	As per paragraph 7 of AMS-I.D Combined heat and power (co-generation) systems are not eligible under this category.	Not Applicable PAI under the grouped project activity will not involve combined heat and power (co-generation) systems.
AMS I D	As per paragraph 8 of AMS-I.D In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the	Not Applicable PAI under the grouped project activity will not involve the capacity addition of renewable power generation facility.

Methodology ID	Applicability condition	Justification of compliance
	project should be lower than 15MW and should be physically distinct from the existing units.	
AMS I D	As per paragraph 9 of AMS-I.D In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/ unit shall not exceed the limit of 15MW.	Not Applicable PAI under the grouped project activity will not retrofit, rehabilitation or replacement facility.
AMS I D	As per paragraph 10 of AMS-I.D In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to the grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as 'AMS-I.C: Thermal energy production with or without electricity' shall be explored.	Potentially Applicable If applicable to a PAI within the Grouped Project, Project activity instances under grouped project activity are wastewater treatment projects, and eligible under Type III category and the component is in accordance with type III methodology.
AMS I D	As per paragraph 11 of AMS-I.D In case biomass is sourced from dedicated plantations, the applicability criteria in the tool 'Project emissions from cultivation of biomass' shall apply.	Not Applicable PAI under the grouped project activity will not using biomass sourced from dedicated plantations (i.e. grown specifically for the grouped project utilization).

PAI1

Methodology ID	Applicability condition	Justification of compliance
AMS III H	As per paragraph 2 of AMS-III.H. The methodology comprises measures that recover biogas from biogenic organic matter in wastewater by means of one, or a combination, of the following options: (a) Substitution of aerobic wastewater or sludge treatment systems with anaerobic systems with biogas recovery and combustion; (b) Introduction of anaerobic sludge treatment system with biogas recovery and combustion to a wastewater treatment plant without sludge treatment; (c) Introduction of biogas recovery and combustion to a sludge treatment system; (d) Introduction of biogas recovery and combustion to an anaerobic wastewater treatment system such as anaerobic reactor, lagoon, septic tank or an on-site industrial plant; (e) Introduction of anaerobic wastewater treatment with biogas recovery and combustion, with or without anaerobic sludge treatment to an untreated wastewater stream; (f) Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery).	Applicable Option (f) as in the methodology is applicable to PAI1.

Methodology ID	Applicability condition	Justification of compliance
AMS III H	As per paragraph 3 of AMS-III.H. In cases where baseline system is anaerobic lagoon, the methodology is applicable if: (d) The lagoons are ponds with a depth greater than two meters, without aeration. The value for depth is obtained from engineering design documents, or through direct measurement, or by dividing the surface area by the total volume. If the lagoon filling level varies seasonally, the average of the highest and lowest levels may be taken; (e) Ambient temperature above 15°C, at least during part of the year on a monthly average basis; (f) The minimum interval between two consecutive sludge removal events shall be 30 days.	<u>Applicable</u> The baseline system of PAI1 is an anaerobic lagoon of which: (a) The lagoons are deeper than 2 meters and without aeration[13]. The value is obtained from an engineering design document; (b) The average ambient temperature in Pahang, Malaysia, is reported to be 30°C[14] which is way above 15°C; (c) The sludge removal interval is more than 30 days[15].
AMS III H	As per paragraph 4 of AMS-III.H. The recovered biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case, additional guidance provided in Annex 1 shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case, additional guidance provided in Annex 1 shall be followed:	<u>Applicable</u> PAI1 uses the captured biogas for direct generation of electrical energy.

[13] The values are obtained from an engineering design document provided by PAI1.
[14] https://www.accuweather.com/en/my/temerloh/229429/june-weather/229429
[15] Sludge removal record received from PAI1

Methodology ID	Applicability condition	Justification of compliance
	(i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or (iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users. (d) Hydrogen production; (e) Use as fuel in transportation applications after upgrading.	
AMS III H	As per paragraph 5 of AMS-III.H. If the recovered biogas is used for project activities covered under paragraph 4 (a), that component of the project activity can use a corresponding methodology under Type I.	Applicable PAI1 uses the recovered biogas for direct generation of electrical energy, AMS-I.D is used in the calculation.
AMS III H	As per paragraph 6 of AMS-III.H. For project activities covered under paragraph 4 (b), if bottles with upgraded biogas are sold outside the project boundary, the end use of the biogas shall be ensured via a contract between the bottled biogas vendor and the end-user. No emission reductions may be claimed from the displacement of fuels from the end use of bottled biogas in such situations. If, however, the end use of the bottled biogas is included in the project boundary and is monitored during the crediting period, CO_2 emissions avoided by the displacement of fossil fuel can be	Not applicable PAI1 does not involve bottling of upgraded biogas

Methodology ID	Applicability condition	Justification of compliance
	claimed under the corresponding Type I methodology, e.g. AMS-I.C Thermal energy production with or without electricity.	
AMS III H	As per paragraph 7 of AMS-III.H. For project activities covered under paragraph 4 (c) (i), emission reductions from the displacement of the use of natural gas are eligible under this methodology, provided the geographical extent of the natural gas distribution grid is within the host country boundaries.	Not applicable PAI1 does not involve upgrading and injection of biogas into a natural gas distribution grid.
AMS III H	As per paragraph 8 of AMS-III.H. For project activities covered under paragraph 4 (c) (ii), emission reductions for the displacement of the use of fuels can be claimed following the provision in the corresponding Type I methodology, e.g. AMS-I.C.	Not applicable PAI1 does not involve the transportation of natural gas distribution via a dedicated piped network to a group of end-user (s).
AMS III H	As per paragraph 9 of AMS-III.H. In particular, for the case of 4 (b) and (c) (iii), the physical leakage during storage and transportation of upgraded biogas as well as the emissions from fossil fuel consumed by vehicles for transporting biogas shall be considered. Relevant procedures in paragraph 18 of the appendix of AMS-III.H Methane recovery in wastewater treatment shall be followed in this regard.	Not Applicable PAI1 does not involve thermal or mechanical electrical energy generation after bottling of upgraded biogas or upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users
AMS III H	As per paragraph 10 of AMS-III.H.	Not Applicable

Methodology ID	Applicability condition	Justification of compliance
	For project activities covered under paragraph 4 (b) and (c), this methodology is applicable if the upgraded methane content of the biogas is in accordance with relevant national regulations (where these exist) or in the absence of national regulations, a minimum of 96% (by volume).	PAI1 does not involve bottling, upgrading or transportation of biogas.
AMS III H	As per paragraph 11 of AMS-III.H. If the recovered biogas is utilized for the production of hydrogen (project activities covered under paragraph 4 (d)), that component of the project activity shall use the corresponding methodology AMS-III.O Hydrogen production using methane extracted from biogas.	<u>Not applicable</u> PAI1 does not involve hydrogen production
AMS III H	As per paragraph 12 of AMS-III.H. If the recovered biogas is used for project activities covered under paragraph 4 (e), that component of the project activity shall use the corresponding methodology AMS-III.AQ Introduction of Bio-CNG in road transportation.	<u>Not applicable</u> PAI1 does not involve use of recovered biogas as fuel in transportation applications after upgrading.
AMS III H	As per paragraph 13 of AMS-III.H. New facilities (Greenfield projects) and project activities involving a change of equipment resulting in a capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this	<u>Not Applicable</u> PAI1 is not a Greenfield project. The project activity is implemented in an existing wastewater treatment plant. There is also no capacity addition of wastewater treatment system compared to the designed capacity of the baseline treatment system.

Methodology ID	Applicability condition	Justification of compliance
	methodology if they comply with the relevant requirements in the General guidelines to SSC CDM methodologies. In addition, the requirements for demonstrating the remaining lifetime of the equipment replaced as described in the general guidelines shall be followed.	
	As per paragraph 2.2.14 of AMS-III.H. The location of the wastewater treatment plant as well as the source generating the wastewater shall be uniquely defined and described in the Project Design Document (PDD).	<u>Applicable</u> The location of the wastewater treatment plant as well as the source generating the wastewater is uniquely defined and described in the PD
AMS III H	As per paragraph 2.2.15 of AMS-III.H Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 $ktCO_2$ equivalent annually from all Type III components of the project activity.	<u>Applicable</u> The estimated average emission reductions from Type III component (methane avoidance) over the crediting period is 28,440tCO_2-equivalent/year. This is lower than the 60 $ktCO_2e$ annual threshold limit for Type III component (methane avoidance).
AMS I D	As per paragraph 2 of AMS-I.D This methodology comprises renewable energy generation units such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	<u>Applicable</u> PAI1 utilizes the captured biogas from Palm Oil Mill Effluent for electricity generation for own use and upload to grid.

Methodology ID	Applicability condition	Justification of compliance
AMS I D	As per paragraph 4 of AMS-I.D This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/ unit(s); or (e) Involve a replacement of (an) existing plant(s).	Applicable (f) PAI1 is a new power generation plant that qualifies as greenfield plant.
AMS I D	As per paragraph 5 of AMS-I.D Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than $4W/m^2$; or (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than $4W/m^2$.	Not Applicable PAI1 is not hydro power plant project.
AMS I D	As per paragraph 6 of AMS-I.D	Applicable

Methodology ID	Applicability condition	Justification of compliance
	If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15MW.	The capacity of the renewable component for PAI1 is 1.5MW which is less than 15MW
AMS I D	As per paragraph 7 of AMS-I.D Combined heat and power (co-generation) systems are not eligible under this category.	Not Applicable PAI1 is not co-generation project
AMS I D	As per paragraph 8 of AMS-I.D In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15MW and should be physically distinct from the existing units.	Not Applicable PAI1 is not capacity addition of renewable energy generation units.
AMS I D	As per paragraph 9 of AMS-I.D In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/ unit shall not exceed the limit of 15MW.	Not Applicable PAI1 does not involve retrofit, rehabilitation or replacement of existing facilities.
AMS I D	As per paragraph 10 of AMS-I.D In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible	Applicable PAI1 is wastewater treatment methane recovery project, and eligible under Type III category.

Methodology ID	Applicability condition	Justification of compliance
	under a relevant Type III category. If the recovered methane is used for electricity generation for supply to the grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as 'AMS-I.C: Thermal energy production with or without electricity' shall be explored.	
AMS I D	As per paragraph 11 of AMS-I.D In case biomass is sourced from dedicated plantations, the applicability criteria in the tool 'Project emissions from cultivation of biomass' shall apply.	Not Applicable PAI1 does not_use biomass sourced from dedicated plantations.

3.3 Project Boundary

Grouped Project

The following GHG sources for the baseline and project scenarios should be assessed on relevance for additional project activity instances added to this Grouped Project

Table 2: Emission sources included or excluded for AMS-III.D in the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Baseline Emissions	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	Physical leakage of biogas in the manure management systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source

Source		Gas	Included?	Justification/Explanation
	which includes production, collection and transport of biogas to the point of flaring/combustion or gainful use	N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Emissions from flaring or combustion of the gas stream	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	CO_2 emissions from use of fossil fuels or electricity for the operation of all the installed facilities	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	CO_2 emissions from incremental transportation distances	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Emissions from the storage of manure before being fed into the anaerobic digester	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

Table 3: Emission sources included or excluded for AMS-III.H in the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions on account of electricity or fossil fuel used	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
		CO_2	No	Excluded for simplification

Source		Gas	Included?	Justification/Explanation
	Methane emissions from baseline wastewater treatment systems	CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from baseline sludge treatment systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/lake/sea	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from the decay of the final sludge generated by the baseline treatment systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Emissions from injection of upgraded biogas into a natural gas distribution grid	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO_2 emissions from electricity and fuel used by the project facilities	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from wastewater treatment systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source

Source	Gas	Included?	Justification/Explanation
affected by the project activity, and not equipped with biogas recovery in the project scenario	N_2O	No	Excluded for simplification
	Other	No	Excluded for simplification
Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation	CO_2	No	Excluded for simplification
	CH_4	Yes	Main emission source
	N_2O	No	Excluded for simplification
	Other	No	Excluded for simplification
Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater	CO_2	No	Excluded for simplification
	CH_4	Yes	Main emission source
	N_2O	No	Excluded for simplification
	Other	No	Excluded for simplification
Methane emissions from the decay of the final sludge generated by the project activity treatment systems	CO_2	No	Excluded for simplification
	CH_4	Yes	Main emission source
	N_2O	No	Excluded for simplification
	Other	No	Excluded for simplification
Methane fugitive emissions due to inefficiencies in capture systems	CO_2	No	Excluded for simplification
	CH_4	Yes	Main emission source
	N_2O	No	Excluded for simplification
	Other	No	Excluded for simplification
Methane emissions due to incomplete flaring	CO_2	No	Excluded for simplification
	CH_4	Yes	Main emission source
	N_2O	No	Excluded for simplification
	Other	No	Excluded for simplification
Methane emissions from biomass stored under anaerobic conditions which would not have	CO_2	No	Excluded for simplification
	CH_4	Yes	Main emission source
	N_2O	No	Excluded for simplification

Source		Gas	Included?	Justification/Explanation
	occurred in the baseline situation	Other	No	Excluded for simplification
	Emissions from electricity and fuel used by upgrading facilities	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from the discharge of the upgrading equipment	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Fugitive methane emissions from leaks in compression equipment	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Emissions on account of vent gases from upgrading equipment	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

Table 4: Emission sources included or excluded for AMS- III.AQ in the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	CO_2 emissions from Bio-CNG displaced fossil fuel origin fuel used	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	Emissions include electricity consumption	CO_2	No	Main emission source
		CH_4	Yes	Excluded for simplification
		N_2O	No	Excluded for simplification

Source		Gas	Included?	Justification/Explanation
	associated with operation of Bio-CNG	Other	No	Excluded for simplification
	Emissions include fossil fuel consumption associated with the operation of Bio-CNG plant.	CO_2	No	Main emission source
		CH_4	Yes	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Emissions from transportation of biomass/waste organic matters from places of their origin to the biogas production site/ filling stations	CO_2	No	Emissions from transportation is not accounted for
		CH_4	Yes	Emissions from transportation is not accounted for
		N_2O	No	Emissions from transportation is not accounted for
		Other	No	Emissions from transportation is not accounted for
	Emission from biomass cultivation	CO_2	No	Emissions from biomass cultivation is not accounted for
		CH_4	No	Emissions from biomass cultivation is not accounted for
		N_2O	No	Emissions from biomass cultivation is not accounted for
		Other	No	Emissions from biomass cultivation is not accounted for
	Emissions from physical leakage of methane from systems affected by the project activity	CO_2	No	Main emission source
		CH_4	Yes	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

Table 5: Emission sources included or excluded for AMS- III.O in the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	CO_2 emissions from reactions of LPG as feedstock, and	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification

Source		Gas	Included?	Justification/Explanation
	combustion of LPG as fuel to reactor	Other	No	Excluded for simplification
Project	CO_2 emissions from fossil fuels and/or electricity used	CO_2	Yes/No	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

Table 6: Emission sources included or excluded for AMS- I.D in the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	CO_2 emissions from electricity generation in power plants that are displaced due to the project activity	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO_2 emissions from on-site consumption of fossil fuel due to project activity	CO_2	Yes/No	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

<u>PAI1</u>

The following boundaries apply to PAI1, as per AMS.III-H and AMS ID

Table 7: Emission sources included for AMS-III.H for PAI1

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions on account of electricity or fossil fuel used	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from baseline wastewater treatment systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification

Source		Gas	Included?	Justification/Explanation
		Other	No	Excluded for simplification
	Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/lake/sea	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO_2 emissions from electricity and fuel used by the project facilities	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane fugitive emissions due to inefficiencies in capture systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
		CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source

Source		Gas	Included?	Justification/Explanation
	Methane emissions due to incomplete flaring	N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

Table 8: Emission sources included for AMS- I.D for PAI1

Source		Gas	Included?	Justification/Explanation
Baseline	CO_2 emissions from electricity generation in power plants that are displaced due to the project activity	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO_2 emissions from on-site consumption of fossil fuel due to project activity	CO_2	Yes/No	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

The project boundary is presented below:

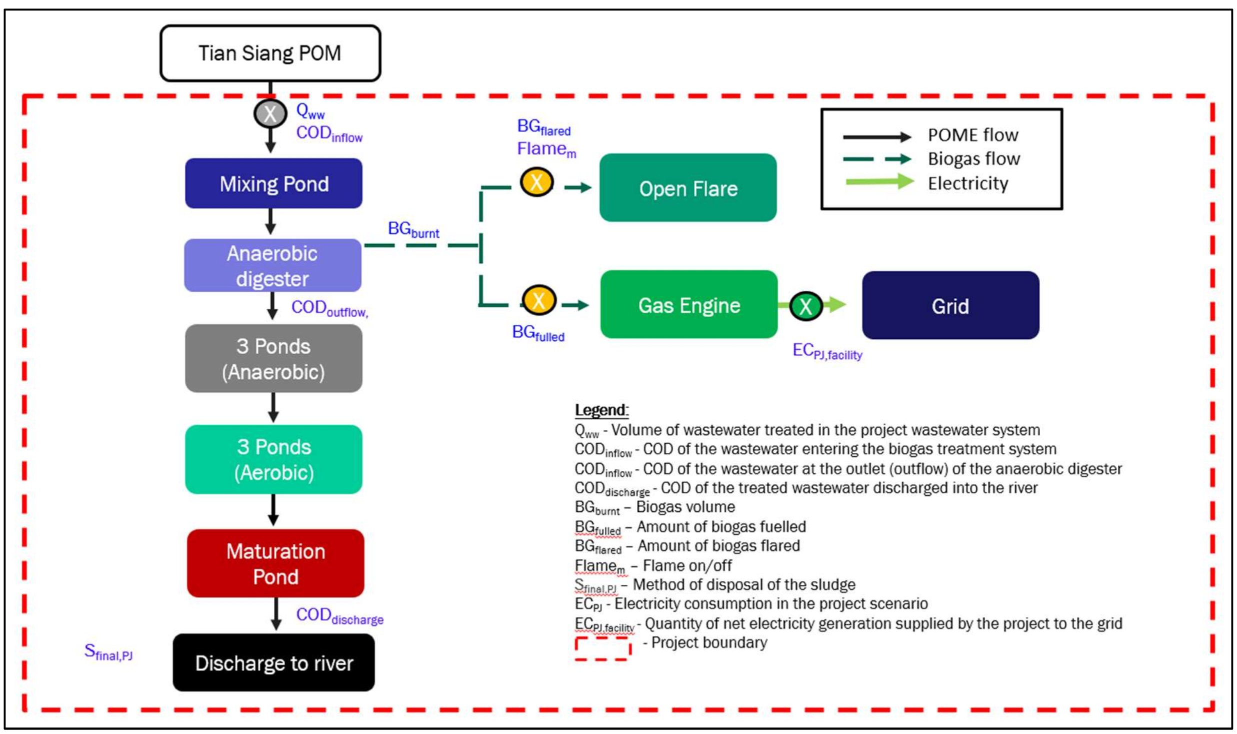

Figure 3: PAI1 Project Scenario

3.4 Baseline Scenario

Grouped Project

According to AMS-III.D version 21.0, the baseline scenario is continuation of existing open anaerobic lagoons which would remain in use, and new open anaerobic lagoons would be constructed for new facilities as they comprise the most common practise in Malaysia to treat animal waste in dairy farm facilities.

According to AMS-III.H version 19.0, the baseline scenario involves a baseline system with anaerobic wastewater treatment system without biogas recovery, which is the case the baseline scenario is determined to be the continuation of the existing open lagoon-based treatment of wastewater without methane recovery as commonly practiced because it only requires small investment and maintenance cost where the treatment system only consume less energy for pumping and mixing.

According to AMS-III.AQ version 02.0, the baseline scenario involves continuation use of GHG intensive derived fossil-fuel or CNG in transportation applications. A baseline system such as this will continue allowing emitting of GHG to the atmosphere. Such practise doesn't require any additional expenditure to modify transport vehicles.

According to AMS-III.O version 02.0, the baseline scenario involves continuation use of LPG as feedstock and/or fuel for hydrogen production and that GHG will continue to be generated in the production process as in the baseline scenario.

According to AMS-I.D version 18.0, the baseline scenario for greenfield power plant, electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Therefore, as a result of the combination of baseline analysis for all the above methodologies, it is concluded that the baseline scenario of the proposed Grouped Project is, in the absence of the project activity:

- For wastewater related to palm oil mill or dairy farm industries, would have been treated in anaerobic open lagoons;
- For transportation applications, use of GHG intensive fuel would continue to be normal practise.
- For hydrogen production, LPG shall continue to be the main feedstock/fuel; and
- For electricity generation, it would have been generated by operation of grid connected power plants and by addition of new generation sources into the grid

PAI1

In the Baseline Scenario, this wastewater is treated in a series of lagoons. Lagoons 1-4 of this system are anaerobic lagoons, with each of the depth deeper than two (2) meters and the average atmospheric temperature in the project location is 31[16] degree Celsius. These conditions create an anaerobic environment in lagoons 1-4, resulting in methane generation. Once the wastewater has completed processing through the baseline treatment system, this wastewater is within the acceptable legal limits applicable to Malaysia and is then discharged into a river.

The sludge removal from lagoons was conducted on an "as-needed" basis. The sludge removed is discarded on the nearby land of the palm oil mill and, as a result, does not produce any emissions of greenhouse gases.

In the absence of PAI1, the baseline scenario will be the continuation of the present wastewater treatment, including ongoing methane emissions to the atmosphere from the anaerobic lagoons.

The baseline scenarios for PAI1 are presented below:

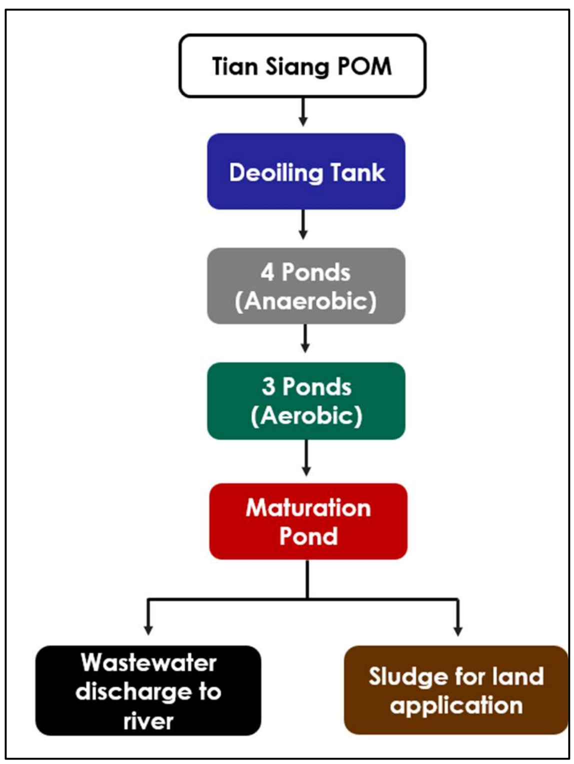

Figure 4: PAI1 Baseline Scenario

[16] https://www.weatheravenue.com/en/asia/my/pahang/triang-weather.html

3.5 Additionality

As per tools Positive lists of technologies - Version 04.0, A positive list of technology and project activity types defined as automatically additional if the PAIs apply such technologies and meet specified conditions.

The positive list of technology requirements for 'waste handling and disposal' is tabulated below:

Table 9: Additionality for PAIs

Requirements for Additionality	PAI Scenario
As per paragraph 12(a): The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation	The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the regulations of Malaysia[17].
As per paragraph 12(b): No regulation in the host country requires the management of biogas from domestic, industrial and agricultural sites	Yes There is no regulation in Malaysia that requires the management of biogas from a site such as the project site[18].
As per paragraph 12(c): There is no capacity increase in the wastewater treatment system	Yes There has been no capacity increase in the wastewater treatment system[19]
As per paragraph 12(d): No other alternative economic activity is expected to be undertaken on the land of the existing lagoon	Yes No other alternative economic activity is expected to be undertaken on the land of the existing lagoon.
As per paragraph 12(e): The biogas is used to generate electricity in one or more power plants, and the total nameplate	Yes The biogas is used to generate electricity in a power plant and the total nameplate capacity is below 5MW[20].

As PAI1 meets all five (5) conditions, PAI1 is automatically additional.

[17] Engineering diagrams for the anaerobic lagoon and documentation relevant to wastewater discharge
[18] MPOB criteria for new licenses and specifics that relate PAI1 (eg. license for mill).
[19] License for mill and supporting documentation.
[20] Technical specifications for the biogas engine

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☐ No

Not applicable. The project is located inside a Non-Annex 1 country, but the project activities are not mandated by any law, statute, or other regulatory framework.

3.5.2 Additionality Methods

Grouped Project

Future PAIs may be able to consider the Positive Lists of Technologies or any other suitable demonstration of additionality. Suitable evidence to demonstrate additionality is required for each PAI. If the PAIs in the Grouped Project need to apply to demonstrate additionality and the procedure in the applied methodology or tool involves several steps. The PAIs will describe how each step is applied and document the outcome of each step.

PAI1

Not applicable to PAI1. As per tools Positive lists of technologies - Version 04.0, A positive list of technology and project activity types is defined as automatically additional if the PAIs apply such technologies and meet specified conditions. PAI1 meets all five (5) conditions. As a result, PAI1 is automatically additional.

3.6 Methodology Deviations

Grouped Project

If a project activity instance, added to this Grouped Project, deviates from the methodology an explanation shall be provided in this section.

PAI1

Not applicable as no deviations from the methodology were made.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Grouped Project

AMS III D, Version 21.0

According to the approved methodology AMS III D, Version 21.0, the baseline emissions for instances in the Grouped Project are determined according to Equations as follows:

The baseline scenario is the situation where, in the absence of the project activity, animal manure is left to decay anaerobically within the project boundary and methane is emitted to the atmosphere. Baseline emissions (BE_y) are calculated by using one of the following two options:

(a) Using the amount of the waste or raw material that would decay anaerobically in the absence of the project activity, with the most recent IPCC Tier 2 approach (please refer to the chapter 'Emissions from Livestock and Manure Management' under the volume 'Agriculture, Forestry and other Land use' of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories). For this calculation, information about the characteristics of the manure and of the management systems in the baseline is required. Manure characteristics include the amount of volatile solids (VS) produced by the livestock and the maximum amount of methane that can be potentially produced from that manure (Bo);

(b) Using the amount of manure that would decay anaerobically in the absence of the project activity based on direct measurement of the quantity of manure treated together with its specific volatile solids (SVS) content

If option (a) is chosen, baseline emissions are determined as follows:

$$BE_y = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j.LT} MCF_j \times B_{o,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bi,j}$$

Where:

Parameter	Description	Unit
BE_y	Baseline emissions in year y	tCO_2e
GWP_{CH4}	Global Warming Potential for methane	tCO_{2e}/tCH_4
D_{CH4}	CH_4 density (0.00067 t/m^3 at room temperature (20 °C) and 1 atm pressure)	t/m^3

Parameter	Description	Unit
UF_b	Model correction factor to account for model uncertainties (0.94)	-
MCF_j	Annual methane conversion factor (MCF) for the baseline animal manure management system j	-
$B_{o,LT}$	Maximum methane producing potential of the volatile solid generated for animal type LT	m³ CH4/kg-dm
$N_{LT,y}$	Annual average number of animals of type LT in year y	numbers
$VS_{LT,y}$	Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis	kg-dm/animal /year
$MS\%_{Bi,j}$	Fraction of manure handled in baseline animal manure management system j	-
LT	Index for all types of livestock	-
j	Index for animal manure management system	-

The maximum methane-producing capacity of the manure (Bo) varies by species and diet. The preferred method to obtain B_o measurement values is to use data from country-specific published sources, measured with a standardised method (B_o shall be based on total as-excreted VS). These values shall be compared to IPCC default values and any significant differences shall be explained. If country specific Bo values are not available, default values from tables 10 A-4 to 10 A-9 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10 can be used, provided that the project participants assess the suitability of those data to the specific situation of the treatment site.

VS are the organic material in livestock manure and consist of both biodegradable and non-biodegradable fractions. For the calculations the total VS excreted by each animal species is required.

(i) The preferred method to obtain VS is to use data from nationally published sources. These values shall be compared with IPCC default values and any significant differences shall be explained

(ii) If data from nationally published sources are not available, country-specific VS excretion rates can be estimated from feed intake levels, via the enhanced characterisation method (tier 2) described in section 10.2 in 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 chapter 10, using the equation below

$$VS_{LT,y} = \left[GE_{LT} \times \left(1 - \frac{DE_{LT}}{100}\right) + (UE \times GE_{LT})\right] \times \left[\left(\frac{1 - ASH}{ED_{LT}}\right)\right] \times nd_y$$

Where:

Parameter	Description	Unit
$VS_{LT,y}$	Annual volatile solid excretions for livestock LT entering all animal waste management systems on a dry matter weight basis	kg-dm/animal/yr
GE_{LT}	Daily average gross energy intake	MJ/animal/day
DE_{LT}	Digestible energy of the feed	per cent
UE	Urinary energy (fraction of GELT)	-
ASH	Ash content of manure (fraction of the dry matter feed intake)	-
ED_{LT}	Energy density of the feed fed to livestock type LT	MJ/kg-dm
nd_y	Number of days treatment plant was operational in year y	-

(iii) If country specific VS values are not available IPCC default values from 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 chapter 10 table 10 A-4 to 10 A-9 can be used provided that the project participants assess the suitability of those data to the specific situation of the treatment site particularly with reference to feed intake levels

Project participants may adjust default IPCC values for VS for a site-specific average animal weight. If so, it shall be well explained and documented. The following equation shall be used:

$$VS_{LT,y} = \left(\frac{W_{site}}{W_{default}}\right) \times VS_{default} \times nd_y$$

Where:

Parameter	Description	Unit
W_{site}	Average animal weight of a defined livestock population at the project site	Kg
$W_{default}$	Default average animal weight of a defined population, this data is sourced from IPCC 2006	kg
$VS_{default}$	Default value for the volatile solid excretion rate per day on a dry-matter basis for a defined livestock population	kg-dm/animal/day
n_{dy}	Number of days treatment plant was operational in year y	-

B_o or VS values applicable to developed countries can be used provided the following four conditions are satisfied:

(i) The genetic source of the livestock originates from an Annex I Party;

(ii) The farm uses formulated feed rations (FFR) which are optimized for the various animal(s), stage of growth, category, weight gain/productivity and/or genetics;

(iii) The use of FFR can be validated (through on-farm record keeping, feed supplier, etc.);

(iv) The project specific animal weights are more similar to developed country IPCC default values

In the case of sequential treatment stages, the reduction of the volatile solids during a treatment stage is estimated based on referenced data for different treatment types. Emissions from the next treatment stage are then calculated following the approach outlined above, but with volatile solids adjusted for the reduction from the previous treatment stages by multiplying by (1 - RVS), where RVS is the relative reduction of volatile solids from the previous stage. The relative reduction of volatile solids (RVS) depends on the treatment technology and should be estimated in a conservative manner. Default values for different treatment technologies can be found in the table in the Appendix.

Methane Conversion Factors (MCF) values are determined for a specific manure management system and represent the degree to which B_o is achieved. Where available country-specific MCF values that reflect the specific management systems used in particular countries or regions shall be used. Alternatively, the IPCC default values provided in table 10.17 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 can be used. The site annual average temperature is taken from official data at the nearest meteorological station, or from data available from historical on site observations

The annual average number of animals ($N_{LT,y}$) is determined as follows:

$$N_{LT,y} = N_{da,y} \times \left(\frac{N_{p,y}}{365}\right)$$

Where:

Parameter	Description	Unit
$N_{da,y}$	Number of days animal is alive in the farm in the year y	numbers
$N_{p,y}$	Number of animals produced annually of type LT for the year y	numbers

If option b is chosen, baseline emissions are determined based on directly measured quantity of manure and its specific volatile solids content, as follows:

$$BE_y = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times Q_{manure,j,LT,y} \times SVS_{j,LT,y}$$

Where:

Parameter	Description	Unit
$Q_{manure,j,LT,y}$	Quantity of manure treated from livestock type LT and animal manure management system j	tonnes/year , dry basis
$SVS_{j,LT,y}$	Specific volatile solids content of animal manure from livestock type LT and animal manure management system j in year y	tonnes/tonnes, dry basis
MCF_j	Annual methane conversion factor (MCF) for the baseline animal manure management system j	-
$B_{0,LT}$	Maximum methane producing potential of the volatile solid generated for animal type LT	m^3 CH4/kg-dm

For each PAI in the Grouped Project, the baseline emissions will be identified according to the procedures set out in *AMS-III.D.: Methane recovery in animal manure management systems --- Version 21.0.* The table below summarises the baseline emissions that are considered for each PAI.

Source	Included/Excluded	Justification
Baseline emissions (BE_y)	Included	Main emissions source in the baseline from animal manure is left to decay anaerobically

AMS III H, Version 19.0

According to the approved methodology AMS III H, Version 19.0, the baseline emissions for instances in the Grouped Project are determined as follows:

Baseline emissions for the systems affected by the project activity may consist of:

(a) Emissions on account of electricity or fossil fuel used ($BE_{power,y}$);

(b) Methane emissions from baseline wastewater treatment systems ($BE_{ww,treatment,y}$);

(c) Methane emissions from baseline sludge treatment systems ($BE_{s,treatment,y}$);

(d) Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/lake/sea ($BE_{ww,discharge,y}$);

(e) Methane emissions from the decay of the final sludge generated by the baseline treatment systems ($BE_{s,final,y}$).

$$BE_y = \{BE_{power,y} + BE_{ww,treatment,y} + BE_{s,treatment,y} + BE_{ww,discharge,y} + BE_{s,final,y}\}$$

Where:

Parameter	Description	Unit
BE_y	Baseline emissions in year y	tCO_2e
$BE_{power,y}$	Emissions on account of electricity or fossil fuel used	tCO_2e
$BE_{ww,treatment,y}$	Methane emissions from baseline wastewater treatment systems	tCO_2e
$BE_{s,treatment,y}$	Methane emissions from baseline sludge treatment systems	tCO_2e
$BE_{ww,discharg\ ,y}$	Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/lake/sea	tCO_2e
$BE_{s,final,y}$	Methane emissions from the decay of the final sludge generated by the baseline treatment systems	tCO_2e

Baseline emissions from electricity and fossil fuel consumption ($BE_{power,y}$) are determined as per the procedures described in the “Tool to calculate baseline, project and/or leakage emissions from electricity consumption” and “Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion”, respectively. The energy consumption shall include all equipment/devices in the baseline wastewater and sludge treatment facility. If recovered biogas in the baseline is used to power auxiliary equipment it should be taken into account, accordingly, using zero as its emission factor

Methane emissions from the baseline wastewater treatment systems affected by the project ($BE_{ww,treatment,y}$) are determined using the COD removal efficiency of the baseline plant:

$$BE_{ww,treatment,y} = \sum_i \left(Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}\right) \times B_{o,ww} \times UF_{BL} \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
$Q_{ww,i,y}$	Volume of wastewater treated in baseline wastewater treatment system i in year y (m^3). For ex ante estimation, forecasted wastewater generation volume or the designed capacity of the wastewater treatment facility can be used. However, the ex post emissions reduction calculation shall be based on the actual monitored volume of treated wastewater	m^3
$COD_{inflow,i,y}$	Chemical oxygen demand of the wastewater inflow to the baseline treatment system *i* in year *y* (t/m^3). Average value may be used through sampling with the confidence/precision level 90/10	t/m^3
$\eta_{COD,BL,i}$	COD removal efficiency of the baseline treatment system *i*, determined as per the paragraphs 35, 36 or 37 of AMS-III.H.	%
$MCF_{ww,treatment,BL,i}$	Methane correction factor for baseline wastewater treatment systems *i* (MCF values as per Table 2 of AMS-III.H.)	-
i	Index for baseline wastewater treatment system	-
$B_{o,ww}$	Methane producing capacity of the wastewater (IPCC value of 0.25 kg CH_4/kg COD)[21]	kg CH_4/kg COD
UF_{BL}	Model correction factor to account for model uncertainties (0.89)[22]	-
GWP_{CH4}	Global Warming Potential for methane	tCO_{2e}/tCH_4

Methane emissions from the baseline sludge treatment systems affected by the project activity are determined using the methane generation potential of the sludge treatment systems:

21 Project activities may use the default value of 0.6 kg CH_4/kg BOD, if the parameter $BOD_{5,20}$ is used to determine the organic content of the wastewater. In this case, baseline and project emissions calculations shall use BOD instead of COD in the equations, and the monitoring of the project activity shall be based in direct measurements of $BOD_{5,20}$, i.e. the estimation of BOD values based on COD measurements is not allowed.

22 Reference: FCCC/SBSTA/2003/10/Add.2, page 25.

$$BE_{treatment,s,y} = \sum_j S_{j,BL,y} \times MCF_{s,treatment,BL,j} \times DOC_s \times UF_{BL} \times DOC_F \times F \times 16/12 \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
$S_{j,BL,y}$	Amount of dry matter in the sludge that would have been treated by the sludge treatment system *j* in the baseline scenario (t). For ex ante estimation, forecasted sludge generation volume or the designed capacity of the sludge treatment facility can be used. However, the ex post emissions reduction calculation shall be based on the actual monitored volume of treated sludge	t
j	Index for baseline sludge treatment system	-
DOC_s	Degradable organic content of the untreated sludge generated in the year *y* (fraction, dry basis). Default values of 0.5 for domestic sludge and 0.257 for industrial sludge[23] shall be used	-
$MCF_{s,treatment,BL,j}$	Methane correction factor for the baseline sludge treatment system *j* (*MCF* values as per Table 2 of AMS-III.H.)	-
UF_{BL}	Model correction factor to account for model uncertainties (0.89)	-
DOC_F	Fraction of DOC dissimilated to biogas (IPCC default value of 0.5)	-
F	Fraction of CH_4 in biogas (IPCC default of 0.5)	-

If the sludge is composted, the following equation shall be applied:

$$BE_{s,treatment,y} = \sum_j S_{j,BL,y} \times EF_{composting} \times GWP_{CH4}$$

[23] The IPCC default values of 0.05 for domestic sludge (wet basis, considering a default dry matter content of 10 per cent) or 0.09 for industrial sludge (wet basis, assuming dry matter content of 35 per cent), were corrected for dry basis.

Where:

Parameter	Description	Unit
$EF_{composting}$	Emission factor for composting organic waste (tCH_4/t waste treated). Emission factors can be based on facility/site-specific measurements, country specific values or IPCC default values (Table 4.1, chapter 4, Volume 5, 2006 IPCC Guidelines for National Greenhouse Gas Inventories). IPCC default value is 0.01 t CH_4/ t sludge treated on a dry weight basis	tCH_4/t waste treated

If the baseline wastewater treatment system is different from the treatment system in the project scenario, the sludge generation rate (amount of sludge generated per unit of COD removed) in the baseline may differ significantly from that of the project scenario. For example, it is known that the amount of sludge generated in aerobic wastewater systems is larger than in anaerobic systems, for the same COD removal efficiency. Therefore, for these cases, the monitored values of the amount of sludge generated during the crediting period will be used to estimate the amount of sludge generated in the baseline, as follows:

$$S_{j,BL,y} = S_{l,PJ,y} \times \frac{SGR_{BL}}{SGR_{PJ}}$$

Where:

Parameter	Description	Unit
$S_{l,PJ,y}$	Amount of dry matter in the sludge treated by the sludge treatment system *i* in year *y* in the project scenario (t)	t
SGR_{BL}	Sludge generation ratio of the wastewater treatment plant in the baseline scenario (tonne of dry matter in sludge/t COD removed). This ratio will be determined as per paragraphs 35, 36 or 37 of AMS-III.H.	tonne of dry matter in sludge/t COD removed
SGR_{PJ}	Sludge generation ratio of the wastewater treatment plant in the project scenario (tonne of dry matter in sludge/t COD removed). Calculated using the monitored values of COD removal (i.e. $COD_{inflow,i}$ minus $COD_{outflow,i}$) and sludge generation in the project scenario	tonne of dry matter in sludge/t COD removed

Methane emissions from degradable organic carbon in treated wastewater discharged in e.g. a river, sea or lake in the baseline situation are determined as follows:

$$BE_{ww,discharge,y} = Q_{ww,y} \times GWP_{CH4} \times B_{o,ww} \times UF_{BL} \times COD_{ww,discharge,BL,y} \times MCF_{ww,BL,discharge}$$

Where:

Parameter	Description	Unit
$Q_{ww,y}$	Volume of treated wastewater discharged in year y (m^3)	m^3
UF_{BL}	Model correction factor to account for model uncertainties (0.89)	-
$COD_{ww,discharge,BL,y}$	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake in the baseline situation in the year y (t/m^3). If the baseline scenario is the discharge of untreated wastewater, the COD of untreated wastewater shall be used	t/m^3
$MCF_{ww,BL,discharge}$	Methane correction factor based on discharge pathway in the baseline situation (e.g. into sea, river or lake) of the wastewater (fraction) (MCF values as per Table 2 of AMS-III.H.)	Equation (1)-

To determine $COD_{ww,discharge,BL,y}$: if the baseline treatment system(s) is different from the treatment system(s) in the project scenario, the monitored values of the COD inflow during crediting period will be used to calculate the baseline emissions ex post. The outflow COD of the baseline systems will be estimated using the removal efficiency of the baseline treatment systems, estimated as per paragraphs 35, 36 or 37 of AMS-III.H.

Methane emissions from anaerobic decay of the final sludge produced are determined as follows:

$$BE_{s,final,y} = S_{final,BL,y} \times DOC_s \times UF_{BL} \times MCF_{s,BL,final} \times DOC_F \times F \times 16/12 \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
$S_{final,BL,y}$	Amount of dry matter in the final sludge generated by the baseline wastewater treatment systems in the year *y* (t). If the baseline wastewater treatment system is different from the project system, it will be estimated using the monitored amount of dry matter in the final sludge generated by the project activity ($S_{final,PJ,y}$) corrected for the sludge generation ratios of the project and baseline systems as per equation (5) above	t
$MCF_{s,BL,final}$	Methane correction factor of the disposal site that receives the final sludge in the baseline situation, estimated as per the procedures described in the methodological tool "Emissions from solid waste disposal sites"	-
UF_{BL}	Model correction factor to account for model uncertainties (0.89)	-

For each PAI in the Grouped Project, the baseline emissions will be identified according to the procedures set out in *AMS-III.H.: Methane recovery in wastewater treatment --- Version 19.0.* The table below summarises the baseline emissions that are considered for each PAI.

Source	Included/Excluded	Justification
Baseline emissions from electricity or fuel consumption in year *y* ($BE_{power,y}$)	Included/ Excluded	Electricity or fuel may or may not be used in the baseline scenario
Baseline emissions of the wastewater treatment systems affected by the project activity in year *y* ($BE_{ww,treatment,y}$)	Included	Main emissions source in the baseline from open lagoons
Baseline emissions of the sludge treatment systems affected by the project activity in year *y* ($BE_{s,treatment,y}$)	Excluded	No sludge treatment activity involved in each PAI
Baseline methane emissions from degradable organic carbon in treated wastewater discharged into	Included/ Excluded	The treated effluent from open anaerobic lagoons may or may not be discharged into sea/river/lake

Source	Included/Excluded	Justification
sea/river/lake in year y ($BE_{ww,discharge,y}$)		
Baseline methane emissions from anaerobic decay of the final sludge produced in year *y* ($BE_{s,final,y}$)	Excluded	No sludge treatment activity involved in each PAI

<u>*AMS III AQ, Version 02.0*</u>

According to the approved methodology AMS III AQ, Version 02.0, the baseline emissions for instances in the Grouped Project are determined according to Equations as follows:

Baseline emissions are calculated by using one of the two available approaches. Under approach 1 baseline emissions are calculated based on the amount of Bio-CNG produced and distributed, and it is applicable to project activities those are:

(a) Use of Bio-CNG in modified diesel vehicles; and/or;

(b) Use of Bio-CNG in modified gasoline vehicles when such vehicles are not included in the boundary

Under approach 2 baseline emissions are calculated based on the quantity of Bio-CNG filled into converted gasoline vehicles and it is applicable to the project activities that are the production and use of Bio-CNG in modified gasoline vehicles when such vehicles are included in the boundary and are monitored. Approach 2 is not applicable to the modified diesel vehicles:

<u>Approach 1</u>

$$BE_y = FS_{Bio-CNG,y} \times NCV_{Bio-CNG} \times EF_{Co2,CNG}$$

Where:

Parameter	Description	Unit
BE_y	Baseline emissions in year y	tCO_2e
$FS_{Bio-CNG,y}$	Amount of Bio-CNG distributed/sold directly to retailers, filling stations by the project activity in year y	tonnes
$NCV_{Bio-CNG}$	CO_2 emission factor of CNG	tCO_2e/GJ
$EF_{CO2,CNG}$	Net calorific value of Bio-CNG	GJ/tonne

Under the condition of:

$$FS_{Bio-CNG,y} \leq FP_{Bio-CNG,y}$$

Where:

Parameter	Description	Unit
$FS_{Bio-CNG,y}$	Quantity of the Bio-CNG produced by the project activity in the year y	tonnes

Approach 2

$$FC_{gasoline,k,y} = FC_{Bio-CNG,k,y} \times \frac{NCV_{Bio-CNG}}{NCV_i} \times n \times f_{FO,gasoline}$$

Where:

Parameter	Description	Unit
$FC_{gasoline,k,y}$	Amount of gasoline of fossil origin which would have been consumed in the baseline by vehicle k in the year y	tonnes
$FS_{Bio-CNG,k,y}$	Bio-CNG consumed by the project vehicle k in the year y	tonnes
$NCV_{Bio-CNG}$	Net calorific value of Bio-CNG	GJ/tonne
NCV_i	Net calorific value of gasoline that was used by project vehicle k	GJ/tonne
n	Discount factor to account for the possible drop in the fuel efficiency of the retrofitted Bio-CNG vehicles. A default value of 0.95 shall be used for converted vehicles that previously used gasoline	-
$f_{FO,gasoline}$	Fraction of gasoline fossil fuel origin. 1.0 if pure gasoline has been displaced. In cases where national regulations require mandatory blending of the fuels with biofuels then the fraction of gasoline (on mass basis) in the blend should be applied.	-

Total baseline emissions for approach 2 are calculated on an annual basis as below:

$$BE_y = \sum_k FC_{gasoline,k,y} \times NCV_{gasoline} \times EF_{CO2,gasoline}$$

Where:

Parameter	Description	Unit
BE_y	Total baseline emission in year y	tCO_2e
$NCV_{gasoline}$	Net calorific value of gasoline	GJ/tonne
$EF_{CO2,gasoline}$	CO_2 emission factor of gasoline	tCO_2e/GJ

Under the condition of:

$$\sum FC_{Bio-CNG,k,y} \leq FP_{Bio-CNG,y}$$

Where:

Parameter	Description	Unit
$FC_{Bio-CNG,k,y}$	Total consumed Bio-CNG by all project vehicles in year y	tonnes

In the cases where project proponents apply both approach 1 and 2, project proponents shall describe in the VCS PD how the double counting of emission reductions has been avoided.

For each PAI in the Grouped Project, the baseline emissions will be identified according to the procedures set out in *AMS-III.AQ.: Introduction of Bio-CNG in transportation applications --- Version 02.0.* The table below summarises the baseline emissions that are considered for each PAI.

Source	Included/ Excluded	Justification
Baseline emissions (BE_y)	Included	Main emissions source in the baseline

<u>AMS III O, Version</u>

According to the approved methodology AMS III O, Version 02.0, the baseline emissions for instances in the Grouped Project are determined according to equations as follows:

The baseline scenario involves continuation use of LPG as feedstock and/or fuel for hydrogen production and that GHG will continue to be generated in the production process as in the baseline scenario.

Baseline emissions (BE_y) are calculated as the summation of the following uses of LPG displaced:

(a) CO_2 generated in reactions of LPG (displaced by methane extracted from biogas in the project scenario) **as feedstock** during the steam-reforming/shift-reaction (BE_{LPG_FEED});

(b) CO_2 generated in the combustion process of LPG (displaced by methane extracted from biogas in the project scenario) **as fuel** to the reactors (BE_{LPG_FUEL}).

$$BE_y = BE_{LPG\_FEED} + BE_{LPG\_FUEL}$$

For LPG used **as feedstock** for the associated CO_2 generated, the baseline emissions are determined as follows:

$$BE_{LPG\_FEED} = R_{CO2/H2} \times m_{H2,BIO} \times MW_{CO2} \times C_1$$

Where:

Parameter	Description	Unit
BE_{LPG_FEED}	Annual baseline CO_2 emissions from the displaced LPG feedstock in the hydrogen production unit	tCO_2e
$R_{CO2/H2}$	CO_2 generation potential per mol of hydrogen produced with LPG as feedstock as defined in paragraph 25	kmol-CO_2/kmol-H_2
$m_{H2,BIO}$	Molar quantity of hydrogen produced annually from methane extracted from biogas as defined in paragraph 29	kmol-H_2
MW_{CO2}	Molecular weight of CO_2 (44)	kg/kmol
C_1	Conversion factor kilograms to tonnes (0.001)	-

For LPG mixture containing m_1 mol of propane and m_2 mol of butane, the reactions during hydrogen production in accordance with para 22 of AMS-III.O version 02.0 are below:

Source Gas	Reaction Type	Ref. Eq.	Reaction
Propane	Steam Reforming	(A)	$C_3H_8 + 3H_2O \leftrightarrow 3CO + 7H_2$
	Shift-Conversion	(B)	$3CO + 3H_2O \leftrightarrow 3CO_2 + 3H_2$
	Sub-total	(C)=(A)+(B)	$C_3H_8 + 6H_2O \leftrightarrow 3CO_2 + 10H_2$
Butane	Steam Reforming	(D)	$C_4H_{10} + 4H_2O \leftrightarrow 4CO + 9H_2$
	Shift-Conversion	(E)	$4CO + 4H_2O \leftrightarrow 4CO_2 + 4H_2$
	Sub-total	(F)=(D)+(E)	$C_4H_{10} + 8H_2O \leftrightarrow 4CO_2 + 13H_2$

Where, for 100 mol of LPG mixture containing m1 mol of propane and m^2 mol of butane, the reactions are as below:

Source Gas	Composition in 100 mol	Ref. Reaction from	Reactions
Propane	m1	(C)	$[m_1]C_3H_8 + [6m_1]H_2O \leftrightarrow [3m_1]CO_2 + [10m_1]H_2$
Butane	m2	(F)	$[m_2]C_4H_{10} + [8m_2]H_2O \leftrightarrow [4m_2]CO_2 + [13m_2]H_2$
Total	m1 + m2	(G)	As $100molLPG = [m_1]C_3H_8 + [m_2]C_4H_{10}$, (1) + (2) is $100molLPG + [6m_1 + 8m_2]H_2O \leftrightarrow [3m_1 + 4m_2]CO_2 + [10m_1 + 13m_2]H_2$

Based on methodology AMS-III.O Version 02.0 para 25, reaction G from table above, CO_2 generation potential per mol of hydrogen produced is defined as:

$$R_{CO2/H2} = \frac{\lfloor 3m_1 + 4m_2 \rfloor}{[10m_1 + 13m_2]}$$

For LPG used as fuel (BE_{LPG_FUEL}), for the associated CO_2 emissions from combustion process of LPG (displaced by methane extracted from biogas in the project scenario) shall be calculated based on:

(a) The specific fuel consumption of the hydrogen production unit using LPG as fuel as described under monitoring methodology, and

(b) The amount of hydrogen produced using methane extracted from biogas as fuel as calculated in paragraphs 16 and 17 of methodology.

$$BE_{LPG\_FUEL} = SFC_{LPG} \times V_{H2,BIO} \times EF_{LPG} \times C_2$$

Where:

Parameter	Description	Unit
BE_{LPG_FUEL}	Annual baseline CO_2 emission from LPG used as fuel in the reactors that is displaced by methane extracted from biogas in the project scenario	tCO_2e
SFC_{LPG}	Specific fuel consumption of the hydrogen production unit using LPG as fuel	kg-LPG/Nm^3-H_2
$V_{H2,BIO}$	Volume of hydrogen produced from methane extracted from biogas under normal condition, annually as defined in paragraphs 30 and 31	Nm^3-H_2
EF_{LPG}	Emission factor of LPG	kg-CO_2/kg LPG
C_2	Conversion factor kilograms to tones (0.001)	-

Emission factor of LPG (EF_{LPG})is calculated based on the weighted average carbon content of LPG, which is evaluated based on its composition as shown below:

Component	% mol	Molecular Weight	Mass 1 mol LPG	Mass of LPG[24]	% carbon	Mass of carbon per mass of LPG
Indication	(A)	(B)	(C)	(D)	(E)	(F)
Formula			(A)*(B)			(D)*(E)
Propane (C_3H_8)	0.30	44	13.2	0.25	0.818	0.201
Butane (C_4H_{10})	0.70	58	40.6	0.75	1.182	0.893
			53.8			1.093

Based on carbon content of LPG, the emission factor is calculated as follows:

EF_{LPG} = 1.093 kgC/kgLPG X 44/12 $kgCO_2$/kgC

= 4.008 kg CO_2/kg LPG

$V_{H2,BIO}$ can be calculated using para 29, from the molar amount of hydrogen (mH_2) produced using the molar amount of hydrogen converted into its equivalent volume of low-pressure gas or vice-versa as defined by 'ideal gas' relationship shown in equation:

$$\boldsymbol{P_N . V_N = m_{H2} . R \; . T_N . C_3}$$

Where:

Parameter	Description	Unit
V_N	Normalized volume of hydrogen produced annually	Nm^3
P_N	Pressure in Pascal at normal condition	Pa
T_N	Temperature in Kelvin at normal condition (273)	K
R	Gas constant in SI Unit (8.314)	$Pa.m^3.mol^{-1}.K^{-1}$
C_3	Conversion factor kmol to mol (1000)	-
m_{H2}	Molar amount of hydrogen produced	kmol

[24] Normalized mass fraction based on Column C

Therefore the value ($V_{N,H2}$) converted (to Nm^3-H_2)) is calculated using formula below,

$$V_{H2,BIO} = \frac{m_{H2,BIO} \text{ x } 1000 \text{ mol/k mol x } 8.314 \text{ Pa m}^3\text{/mol K x } 273.15 \text{ K}}{1 \text{ X } 10^5}$$

Where $m_{H2,BIO}$ is calculated from formula below:

$$m_{H2,BIO} = m_{H2,T} - m_{H2,LPG}$$

Where:

Parameter	Description	Unit
$m_{H2,BIO}$	Molar quantity of hydrogen produced from methane extracted from biogas annually	kmol-H_2
$m_{H2,T}$	Total molar amount of hydrogen produced annually. This parameter shall be based on monitoring of volume of hydrogen produced by the hydrogen production unit. If the volume is reported as normal volume, the equivalent molar amount can be calculated using ideal gas relationship described in paragraph 29	kmol-H_2
$m_{H2,LPG}$	Molar amount of hydrogen produced from LPG annually as calculated in paragraph 28	kmol-H_2

$m_{H2,LPG}$, is calculated through monitored amount of LPG used as feedstock to the reaction (M_{LPG}) multiplied by the hydrogen production potential, calculated from below formula:

$$\boldsymbol{m_{H2,LPG} = R_{H2/LPG} \times \frac{M_{LPG}}{(m_1 \times MW_{C3H8} + m_2 \times MW_{C4H10})}}$$

Where:

Parameter	Description	Unit
$m_{H2,LPG}$	Molar amount of hydrogen produced from LPG annually as calculated in paragraph 28	kmol-H_2
$R_{H2/LPG}$	Molar amount of hydrogen produced from LPG annually	kmol-H_2
M_{LPG}	Mass of LPG used as reaction feedstock annually	Kg-LPG
MW_{LPG}	Molecular weight of LPG	kg-LPG/kmol-LPG
m_1	% mol of propane in LPG	mol/mol
MW_{C3H8}	Molecular weight of propane (44)	kg/kmol
m_2	% mol of butane in LPG	mol/mol
MW_{C4H10}	Molecular weight of butane (66)	kg/kmol

Based on methodology AMS-III.O Version 02.0 para 24, reaction G, generation potential per mol of hydrogen produced is defined and calculated as:

$$R_{H2/LPG} = \frac{[10m_1 + 13m_2]}{100}$$

The composition of LPG for the purpose of baseline emission calculations shall be determined based on the composition analysis of stand-by LPG[25] stock. This shall be based on:

a) Information provided by the supplier; or

Compositional analysis conducted by an independent certified laboratory; or

Product specification statement provided by the national gas supplier of the host-country.

For the purpose of baseline emission calculation, information provided by LPG supplier in Malaysia is considered, where it is of 30/70 propane/butane composition[26]. This molar composition however will be monitored quarterly during crediting period.

LPG mixtures of 30/70 propane butane gives molecular weight of LPG, MW_{LPG} calculated according to below calculation:

Components	Formula	% mol	Molecular Weight (MW) in g/mol	Number of carbon in molecule (n)	Number of hydrogen in molecule (m)
Propane	C_3H_8	m_1,%	44	3	8
Butane	C_4H_{10}	m_2 %	58	4	10

For LPG mixture of m_1%mol of propane and m_2%mol of butane, molecular weight of the mixture (MW_{LPG}) can be calculated as weighted average of individual gas using the following relationship:

MW_{LPG} = m_1% X 44 +m_2% X 58

= 53.8

The table below summarises the baseline emissions that are considered for each AMS-III.O applied PAI.

25 Stand-by LPG is essential for process reliability. Standby LPG is the LPG stock stored by the operator to cover situations where biogas is not available in sufficient amount or production of hydrogen from biogas has halted for some reasons. For example in prolonged dry season wastewater treatment facility treating wastewaters such as palm oil mill effluent may not be operating in full capacity and therefore producing less biogas. Other possibilities include temporary non-availability of H_2S removal system due to maintenance/ repair.

26 http://eprints.usm.my/13194/1/effect_of_variation.pdf

Source	Included/Excluded	Justification
Baseline emissions (BE_y)	Included	Main emissions source in the baseline from fuel and feedstock switch to reduce consumption of fossil fuel

AMS I D, Version 18.0

According to the approved methodology AMS I D, Version 18.0, the baseline emissions include only CO_2 emissions from electricity generation in power plants that are displaced due to the project activity:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where:

Parameter	Description	Unit
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y	MWh
$EF_{grid,y}$	Combined margin CO2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system"	tCO_2/MWh

The table below summarises the baseline emissions that are considered for each PAI.

Source	Included/ Excluded	Justification
Baseline emissions from electricity generation in power plants that are displaced due to the project activity (BE_y)	Included/ Excluded	Electricity generation in power plants that are displaced due to the project activity may or may not be applied in the baseline scenario

Baseline Emissions for the PAI1

AMS III H, Version 19.0

The table below summarises the baseline emissions that are considered for the PAIs:

Source	Included/Excluded	Justification
$BE_{power,y}$	Excluded	No electricity or fossil fuel used
$BE_{ww,treatment,y}$	Included	Main emissions source from open lagoons
$BE_{s,treatment,y}$	Excluded	No sludge treatment activity involved
$BE_{ww,discharge,y}$	Included	The effluent is discharged into river
$BE_{s,final,y}$	Excluded	No sludge treatment activity involved

AMS I D, Version 18.0

The table below summarises the baseline emissions that are considered for the PAIs:

Source	Included/Excluded	Justification
BE_y	Included	Power generation from the PAI1 will be uploaded to the grid

The summary of baseline emission for the PAI1 is tabulated below:

Year	Total BE_y	AMS IIIH		AMS ID
		$BE_{ww,treatment,y}$	$BE_{ww,discharge,y}$	BE_y
01-Apr-2024 – 31-Dec-2024 (275 days)	31,532	27,930	76	3,526
01-Jan-2025 – 31-Dec-2025	41,852	37,071	101	4,680
01-Jan-2026 – 31-Dec-2026	41,852	37,071	101	4,680
01-Jan-2027 – 31-Dec-2027	41,852	37,071	101	4,680
01-Jan-2028 – 31-Dec-2028	41,852	37,071	101	4,680
01-Jan-2029 – 31-Dec-2029	41,852	37,071	101	4,680
01-Jan-2030 – 31-Dec-2030	41,852	37,071	101	4,680
01-Jan-2031 – 31-Mar-2031 (90 days)	10,320	9,141	25	1,154
Total	**292,965**	**259,498**	**708**	**32,760**

The detailed calculation for baseline emission is tabulated below:

AMS III H

$BE_{ww,treatment,y}$ - Baseline emission of the wastewater treatment systems affected by the project activity

Parameter	01-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 31-Mar-2031
$\Sigma Q_{ww,i,y}$	100,399	133,257	133,257	133,257	133,257	133,257	133,257	32,858
$COD_{inflow,i,y}$	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657
$h_{COD, BL, i}$	85%	85%	85%	85%	85%	85%	85%	85%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
GWP_{CH4}	28	28	28	28	28	28	28	28
$BE_{ww,treatment,y}$	**27,930**	**37,071**	**37,071**	**37,071**	**37,071**	**37,071**	**37,071**	**9,141**

$BE_{ww,discharge,y}$ - Baseline methane emissions from degradable organic carbon in treated wastewater discharged into river

Parameter	01-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 31-Mar-2031
$Q_{ww,y}$	100,399	133,257	133,257	133,257	133,257	133,257	133,257	32,858
GWP_{CH4}	28	28	28	28	28	28	28	28
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
$COD_{ww,discharge,BL,y}$	0.00122	0.00122	0.00122	0.00122	0.00122	0.00122	0.00122	0.00122
$MCF_{ww,BL,discharge}$	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
$BE_{ww,discharge,y}$	**76**	**101**	**101**	**101**	**101**	**101**	**101**	**25**

AMS ID

BE_y - Baseline emissions from electricity generation in power plants that are displaced due to the project activity

Parameter	01-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 31-Mar-2031
EG_{BLy}	6,027	8,000	8,000	8,000	8,000	8,000	8,000	1,973
EF_{CO2}	0.585	0.585	0.585	0.585	0.585	0.585	0.585	0.585
$BE_{,y}$	**3,526**	**4,680**	**4,680**	**4,680**	**4,680**	**4,680**	**4,680**	**1,154**

4.2 Project Emissions

Grouped Project

AMS III D, Version 21.0

According to the approved methodology AMS III D, Version 21.0, the project emissions for instances in the Grouped Project are determined as follows:

(a) Physical leakage of biogas in the manure management systems which includes production, collection and transport of biogas to the point of flaring/combustion or gainful use ($PE_{PL,y}$);

(b) Emissions from flaring or combustion of the gas stream ($PE_{flare,y}$);

(c) CO_2 emissions from use of fossil fuels or electricity for the operation of all the installed facilities ($PE_{power,y}$);

(d) CO_2 emissions from incremental transportation distances;

(e) Emissions from the storage of manure before being fed into the anaerobic digester ($PE_{storage,y}$).

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y}$$

Where:

Parameter	Description	Unit
PE_y	Project emissions in year y	tCO_2e
$PE_{PL,y}$	Emissions due to physical leakage of biogas in year y	tCO_2e
$PE_{flare,y}$	Emissions from flaring or combustion of the biogas stream in the year y	tCO_2e
$PE_{power,y}$	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y	tCO_2e
$PE_{transport,y}$	Emissions from incremental transportation in the year y, as per relevant paragraph in AMS-III.AO	tCO_2e
$PE_{storage,y}$	Emissions from the storage of manure	tCO_2e

Project emissions due to physical leakage of biogas from the animal manure management systems used to produce, collect and transport the biogas to the point of flaring or gainful use are estimated as:

(a) 10% of the maximum methane producing potential of the manure fed into the management systems implemented by the project activity

(i) If the option (a) in the baseline emission is chosen, it is determined as:

$$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{i,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y}$$

Where:

Parameter	Description	Unit
$MS\%_{i,y}$	Fraction of manure handled in system i in year y If the project activity involves sequential manure management systems, the procedure specified in paragraph 18(e) shall be used to estimate the project emissions due to physical leakage of biogas in each stage	%

(ii) If the option (b) in the baseline emission is chosen, it is determined as:

$$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{i,LT} B_{0,LT} \times Q_{manure,LT,y} \times SVS_{LT,y} \times MS\%_{i,y}$$

(b) Optionally, the relevant procedure in the methodological tool "Project and leakage emissions from anaerobic digesters" may be followed. In such a case, $PE_{PL,y}$ is equivalent to PECH4,y in the tool

In the case of flaring of the recovered biogas, project emissions are estimated using the procedures described in the methodological tool "Project emissions from flaring". If the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. However, this use of the recovered biogas shall be included in the project boundary and its output shall be monitored in order to ensure that the recovered biogas is actually destroyed, even if the emission reductions from this component are not claimed.

Project emissions from electricity and fossil fuel consumption are determined by following the methodological tool "Project and leakage emissions from anaerobic digesters", where $PE_{Power,y}$ is the sum of $PE_{EC,y}$ and $PE_{FC,y}$ in the tool.

Project emission from incremental transportation of the manure to the site ($PE_{transp,y}$) is not expected to happen, as the digester is located within the project boundary and does not contribute to incremental transportation. Therefore, the project emissions from the incremental transportation of manure to the site is taken as zero.

Project emissions on account of storage of manure before being fed into the anaerobic digester shall be accounted for if both condition (a) and condition (b) below are satisfied:

(a) The storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester;

(b) The dry matter content of the manure when removed from the animal barns is less than 20%.

The following method shall be used to calculate project emissions from manure storage:

$$PE_{storage} = GWP_{CH4} \times D_{CH4} \times \sum_{LT,l}\left[\frac{365}{AI_l} \times \sum_{d=1}^{AI}\left(N_{LT} \times VS_{LT,d} \times MS\%_l \times \left(1 - e^{-k(AI_l - d)}\right) \times MCF_i \times B_{0,LT}\right)\right]$$

Where:

Parameter	Description	Unit
$PE_{storage}$	Project emissions on account of manure storage in year y	tCO_2e
AI_l	Annual average interval between manure collection and delivery for treatment at a given storage device l	days
$VS_{LT,d}$	Amount of volatile solid production by type of animal LT in a day	kg VS/head/d
$MS\%_l$	Fraction of volatile solids (%) handled by storage device l	-
k	Degradation rate constant (0.069)	-
d	Days for which cumulative methane emissions are calculated; d can vary from 1 to 45 and to be run from 1 up to AI_l	-
MCF_l	Annual methane conversion factor for the project manure storage device l from Table 10.17, Chapter 10, Volume 4	-

Project emission due to physical leakage ($PE_{PL,y}$) also allows for option to the tool *'Project and leakage emissions from anaerobic digesters*.

Project emission from storage of manure before being fed into the anaerobic digester ($PE_{storage,y}$) is expected to be zero, as the The project activity has been designed such that the manure is stored less than 24 hours before it is fed into the anaerobic digester.

For each project activity instance in the Grouped Project, the project emissions will be identified according to the procedures set out in *AMS-III.D.: Methane recovery in animal manure management systems --- Version 21.0.* The table below summarises the project emissions that will be considered for each project activity instance in the Grouped Project.

Parameter	Included/Excluded	Justification
Emissions due to physical leakage of biogas ($PE_{PL,y}$)	Included/Excluded	The project may or may not emit GHG due to physical leakage of biogas.
Emissions from flaring or combustion of the biogas stream ($PE_{flare,y}$)	Included/Excluded	The project may or may not emit GHG from incomplete flaring.
Emissions from the use of fossil fuel or electricity for the operation of the installed facilities ($PE_{power,y}$)	Included/Excluded	Electricity or fuel may or may not be used in the project scenario
Emissions from incremental transportation ($PE_{transport,y}$)	Included/Excluded	The project may or may not emit GHG from incremental transportation.
Emissions from the storage of manure ($PE_{storage,y}$)	Included/Excluded	The project may or may not emit GHG from storage of manure.

AMS III H, Version 19.0

According to the approved methodology AMS III H, Version 19.0, the project emissions for instances in the Grouped Project are determined as follows:

Project emissions consist of:

a) CO_2 emissions from electricity and fuel used by the project facilities ($PE_{power,y}$);

b) Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario ($PE_{ww,treatment,y}$);

c) Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation ($PE_{s,treatment,y}$)

d) Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater ($PE_{ww,discharge,y}$);

e) Methane emissions from the decay of the final sludge generated by the project activity treatment systems ($PE_{s,final,y}$);

f) Methane fugitive emissions due to inefficiencies in capture systems ($PE_{fugitive,y}$);

g) Methane emissions due to incomplete flaring ($PE_{flaring,y}$);

h) Methane emissions from biomass stored under anaerobic conditions which would not have occurred in the baseline situation ($PE_{biomass,y}$);

$$PE_y = \begin{Bmatrix} PE_{power,y} + PE_{ww,treatment,y} + PE_{s,treatment,y} + PE_{ww,discharge,y} + PE_{s,final,y} + \\ PE_{fugitive,y} + PE_{biomass,y} + PE_{flaring,y} \end{Bmatrix}$$

Where:

Parameter	Description	Unit
PE_y	Project emissions in year y	tCO_2e
$PE_{power,y}$	Emissions from electricity or fuel consumption in the year y (t CO2e). These emissions shall be calculated as per paragraph 26 of AMS III H, for the situation of the project scenario, using energy consumption data of all equipment/devices used in the project activity wastewater and sludge treatment systems and systems for biogas recovery and flaring/gainful use	tCO_2e
$PE_{ww,treatment,y}$	Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery, in year y (t CO_2e). These emissions shall be calculated as per equation (2) in paragraph 27 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project situation ($MCF_{ww,treatment,PJ,k}$ and $\eta_{,PJ,k,y}$) and with the following changed definition of parameters: $MCF_{ww,treatment,PJ,k}$ — Methane correction factor for project wastewater treatment system k (MCF values as per Table 2) $\eta_{,PJ,k,y}$ — Chemical oxygen demand removal efficiency of the project wastewater treatment system k in year y (t/m^3), measured based on inflow COD and outflow COD in system k	tCO_2e
$PE_{s,treatment,y}$	Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery, in year y (t CO_2e). These emissions shall be calculated as per equations (3) and (4) in paragraphs 30 and 31 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project situation ($S_{l,PJ,y}$, $MCF_{s,treatment,l}$) and with the following changed definition of parameters:	tCO_2e

Parameter	Description		Unit
	$S_{l,PJ,y}$	Amount of dry matter in the sludge treated by the sludge treatment system I in the project scenario in year y (t)	
	$MCF_{s,treatment,l}$	Methane correction factor for the project sludge treatment system I (MCF values as per Table 2 of AMS-III.H)	
$PE_{ww,discharg\ ,y}$	Methane emissions from degradable organic carbon in treated wastewater in year y (tCO_2e). These emissions shall be calculated as per equation (6) in paragraph 33 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project conditions ($COD_{ww,discharge,PJ,y}$, $MCF_{ww,PJ,discharge}$) and with the following changed definition of parameters:		tCO_2e
	$COD_{ww,discharge,PJ,y}$	Chemical oxygen demand of the treated wastewater discharged into the sea, river or lake in the project scenario in year y (t/m^3)	
	$MCF_{ww,PJ,discharge}$	Methane correction factor based on the discharge pathway of the wastewater in the project scenario (e.g. into sea, river or lake) (MCF values as per Table 2)	
$PE_{s,final,y}$	Methane emissions from anaerobic decay of the final sludge produced in year y (t CO_2e). These emissions shall be calculated as per equation (7) in paragraph 35 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project conditions ($MCF_{s,PJ,final}$, $S_{final,PJ,y}$). If the sludge is controlled combusted, disposed in a landfill with biogas recovery, or used for soil application in aerobic conditions in the project activity, this term shall be neglected, and the sludge treatment and/or use and/or final disposal shall be monitored during the crediting period with the following revised definition of the parameters:		tCO_2e
	$MCF_{s,PJ,final}$	Methane correction factor of the disposal site that receives the final sludge in the project situation, estimated as per the procedures described in the	

Parameter	Description	Unit
	methodological tool "Emissions from solid waste disposal sites" $S_{final,PJ,y}$ Amount of dry matter in final sludge generated by the project wastewater treatment systems in the year y (t)	
$PE_{fugitive,y}$	Methane emissions from biogas release in capture systems in year y, calculated as per paragraph 4 (tCO_2e)	tCO_2e
$PE_{biomass,y}$	Methane emissions due to incomplete flaring in year y (t CO_2e). For ex ante estimation, baseline emission calculation for wastewater and/or sludge treatment (i.e. equation (2) and/or equation (3)) can be used but without the consideration of GWP for CH_4. However, the ex post emission reduction shall be calculated as per methodological tool "Project emissions from flaring"	tCO_2e
$PE_{flaring,y}$	Methane emissions from biomass stored under anaerobic conditions. If storage of biomass under anaerobic conditions takes place in the project and does not occur in the baseline, methane emissions due to anaerobic decay of this biomass shall be considered and be determined as per the procedure in the methodological tool "Emissions from solid waste disposal sites" (t CO_2e)	tCO_2e

Project emissions from methane release in capture systems are determined as follows:

Based on the methane emission potential of wastewater and/or sludge:

$$PE_{fugitive,y} = PE_{fugitive,ww,y} + PE_{fugitive,s,y}$$

Where:

Parameter	Description	Unit
$PE_{fugitive,ww,y}$	Fugitive emissions through capture inefficiencies in the anaerobic wastewater treatment systems in the year y	tCO_2e
$PE_{fugitive,s,y}$	Fugitive emissions through capture inefficiencies in the anaerobic sludge treatment systems in the year y	tCO_2e

$$PE_{fugitive,ww,y} = (1 - CFE_{ww}) \times MEP_{ww,treatment,y} \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
CFE_{ww}	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems (a default value of 0.9 shall be used)	-
$MEP_{ww,treatment,y}$	Methane emission potential of wastewater treatment systems equipped with biogas recovery system in year y (t)	-

$$MEP_{ww,treatment,y} = Q_{ww,y} \times B_{o,ww} \times UF_{PJ} \times \sum_{k} COD_{removed,PJ,k,y} \times MCF_{ww,treatment,PJ,k}$$

Where:

Parameter	Description	Unit
$COD_{removed,PJ,k,y}$	The chemical oxygen demand removed[27] by the treatment system k of the project activity equipped with biogas recovery in the year y	t/m^3
$MCF_{ww,treatment,PJ,k}$	Methane correction factor for the project wastewater treatment system k equipped with biogas recovery equipment (MCF values as per Table 2 above)	-
UF_{PJ}	Model correction factor to account for model uncertainties (1.12)	-

$$PE_{fugitive,s,y} = (1 - CFE_s) \times MEP_{s,treatment,y} \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
CFE_s	Capture efficiency of the biogas recovery equipment in the sludge treatment systems (a default value of 0.9 shall be used)	-
$MEP_{s,treatment,y}$	Methane emission potential of the sludge treatment systems equipped with a biogas recovery system in year y	t

$$MEP_{s,treatment,y} = \sum_{l} \left(S_{l,PJ,y} \times MCF_{s,treatment,PJ,l}\right) \times DOC_s \times UF_{PJ} \times DOC_F \times F \times 16/12$$

Where:

[27] Difference between the inflow COD and the outflow COD.

Parameter	Description	Unit
$S_{l,PJ,y}$	Amount of sludge treated in the project sludge treatment system l equipped with a biogas recovery system (on a dry basis) in year y	t
$MCF_{s,treatment,PJ,l}$	Methane correction factor for the sludge treatment system equipped with biogas recovery equipment (MCF values as per Table 2 above)	-
UF_{PJ}	Model correction factor to account for model uncertainties (1.12)	-

For each project activity instance in the Grouped Project, the project emissions will be identified according to the procedures set out in *AMS-III.H.: Methane recovery in wastewater treatment --- Version 19.0.* The table below summarises the project emissions that will be considered for each project activity instance in the Grouped Project.

Source	Included/ Excluded	Justification
CO_2 emissions from electricity and fuel used by the project facilities ($PE_{power,y}$)	Included/ Excluded	Electricity or fuel may or may not be used in the project scenario
Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario ($PE_{ww,treatment,y}$)	Included/ Excluded	Each project activity instance to determine based on the project design
Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation ($PE_{s,treatment,y}$)	Excluded	No sludge treatment activity involved in each PAI
Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater ($PE_{ww,discharge,y}$)	Included/ Excluded	The treatment effluent from open anaerobic lagoons may or may not be discharged into sea/river/lake
Methane emissions from the decay of the final sludge generated by the project activity treatment systems ($PE_{s,final,y}$)	Excluded	No sludge treatment activity involved in each PAI
Methane fugitive emissions due to inefficiencies in capture systems ($PE_{fugitive,y}$)	Included	Project emissions due to potential inefficiencies of biogas capture system.

Source	Included/ Excluded	Justification
Methane emissions due to incomplete flaring ($PE_{flaring,y}$)	Included/ Excluded	The project may or may not emit GHG from incomplete flaring.
Methane emissions from biomass stored under anaerobic conditions which would not have occurred in the baseline situation ($PE_{biomass,y}$)	Excluded	There is no storage of additional biomass under anaerobic conditions in project activity instances.

AMS III AQ, Version 02.0

The project emissions should be calculated as follows

$$PE_y = PE_{elec,y} + PE_{fuel,y} + PE_{transport,y} + PE_{cultivation,y} + PE_{CH4,y}$$

Where:

Parameter	Description	Unit
PE_y	Project emissions in year y	tCO_2e
$PE_{elec,y}$	Project emissions due to electricity consumption in year y	tCO_2e
$PE_{fuel,y}$	Project emissions due to fossil fuels consumption in year y	tCO_2e
$PE_{transport,y}$	Project emissions from transportation of the biomass from the places of their origin to the biogas production site and where applicable, transportation Bio-CNG from biogas plant to filling stations where it is used by final consumers in year y	tCO_2e
$PE_{cultivation,y}$	Project emissions from biomass cultivation in a dedicated plantation in year y	tCO_2e
$PE_{CH4,y}$	Project emissions due to the physical leakage of methane from the systems affected by the project activity for production, processing, purification, compression; storage and filling of the Bio-CNG in year y	tCO_2e

Calculation of $PE_{elec,y}$

The emissions include electricity consumption (including auxiliary use) $PE_{elec,y}$ associated with the operation of Bio-CNG plant, calculated as per the parameter $PE_{EC,y}$ in the "Tool to calculate baseline, project and/or leakage emissions from electricity consumption".

Calculation of $PE_{fuel,y}$

The emissions include fossil fuel consumption (including auxiliary use) $PE_{fuel,y}$ associated with the operation of Bio-CNG plant, calculated as per the parameter $PE_{FC,j,y}$ in the "Tool to calculate project or leakage CO2 emissions from fossil fuel combustion", where each combustion processes j in the tool should correspond to one of the fossil fuel consumption sources of the plant

In cases where it is demonstrated that the energy requirements of the biogas production and treatment system and Bio-CNG plant are met only by renewable energy source the values of $PE_{elec,y}$ and $PE_{fuel,y}$ are considered as zero.

Calculation of $PE_{transport,y}$

Project emissions from transportation of the biomass and/or waste organic matters from the places of their origin to the biogas production site and where applicable, transportation Bio-CNG from biogas plant to filling stations where it is used by final consumers have to be accounted following the procedures in "AMS-III.AK: Biodiesel production and use for transport applications" if the transportation distance is more than 200 km, otherwise they can be neglected.

Calculation of $PE_{cultivationH4,y}$

If the project activity utilizes biomass sourced from dedicated plantations, project emissions from biomass cultivation shall be calculated as per the methodological tool "Project emissions from cultivation of biomass".

Calculation of $PE_{CH4,y}$

Project emissions associated with the physical leakage of methane from the systems affected by the project activity are calculated as follows:

$$PE_{CH4,y} = PE_{AD,y} + PE_{Bio-CNG,y}$$

Where:

Parameter	Description	Unit
$PE_{AD,y}$	CH_4 leakage emissions from the anaerobic digesters in year y	tCO_2e
$PE_{Bio-C\quad,y}$	Project emissions of CH4 from biogas and Bio-CNG processing, upgrading, purification, compression, storage and transportation (leaks and dissolved in wastewater) in year y	tCO_2e

Methane emissions from physical leakage emissions from the anaerobic digesters ($PE_{AD,y}$)

Methane emissions due to physical leakages from the digester and recovery system ($PE_{AD,y}$) shall be estimated using a default factor of 0.05 m^3 biogas leaked/m^3 biogas produced. For ex ante estimation the expected biogas production of the digester may be used, for ex post calculations the effectively recovered biogas amount shall be used for the calculation

Methane emissions from physical leakage due to the biogas treatment system ($PE_{Bio-CNG,y}$)

The following project emission sources shall be determined as per the relevant procedures in annex 1 of "AMS-III.H: Methane recovery in wastewater treatment":

(a) Methane emissions from the discharge of the upgrading equipment are determined;
(b) Fugitive methane emissions from leaks in compression equipment;
(c) Methane emissions due to the vent gases from upgrade equipment;
(d) Methane emissions related to physical leakage from filling operations shall be computed as per the procedures for calculating emissions from compressor leaks;
(e) Where applicable methane emissions associated with the physical leakage of the upgraded biogas from the dedicated pipelines;
(f) Where applicable methane emissions due to physical leakage from Bio-CNG/biogas filled bottles (e.g. mobile cascades) which are used for the storage and transportation of Bio-CNG/biogas.

Methane emissions from physical leakage due to the biogas treatment system ($PE_{Bio-CNG,y}$)

Parameters for calculating methane emissions from physical leakage of methane from the systems affected by the project activity for production, processing, purification, compression, storage and filing of the Bio-CNG shall be monitored as per procedure as described in Appendix of AMS-III.H.

Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, additional guidance in the appendix AMS-II.H shall be followed which shall include:

$$PE_{process,y} = PE_{power,upgrade,y} + PE_{ww,upgrade,y} + PE_{CH4,equip,y} + PE_{ventgas,y}$$

Where:

Parameter	Description	Unit
$PE_{process,y}$	Project emissions related to the upgrading and compression of the biogas in year y	tCO_2e
$PE_{power,upgrade,y}$	CO2 emissions from electricity and fuel used by the upgrading facilities, as per paragraph 39 of AMS-III.H	tCO_2e
$PE_{ww,upgrade,y}$	Emissions from the methane contained in any wastewater discharge of upgrading installation in year y	tCO_2e
$PE_{CH4,equip,y}$	Emissions from compressor leaks in year y	tCO_2e
$PE_{ventgas,y}$	Emissions from venting gases retained in the upgrading equipment in year y	tCO_2e

Calculation of $PE_{power,upgrade,y}$

Project emissions associated with the electricity and fuel used by the upgrading facilities will be calculated according to "Tool to calculate baseline, project and/or leakage emissions from electricity consumption", and "Tool to calculate project or leakage CO_2 emissions from fossil f uel combustion", respectively.

These parameters have already been considered in the monitoring for methodology AMS-III.H and/or AMS-III.D for capture of methane, and therefore will not be further considered.

Calculation of $PE_{ww,upgrade,y}$

Emissions from methane contained in any wastewater discharge of upgrading installation are calculated as follows:

$$PE_{ww,upgrade,y} = Q_{ww,upgrade,y} \times [CH_4]_{ww,upgrade,y} \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
$Q_{ww,upgrade,y}$	Volume of wastewater discharge from upgrading installation in year y	tCO_2e
$[CH4]_{upgrade,y}$	Dissolved methane contained in the wastewater discharge in year y	tCO_2e
GWP_{CH4}	Global warming potential of CH_4	$tCO_2e/$ tCH_4

Calculation of $PE_{CH4\ equip,y}$

Emissions from compressor leaks are calculated as follows:

$$PE_{CH4,equip,y} = GWP_{CH4} \times \left(\frac{1}{1000}\right) \times \sum_{equipment} w_{CH4,stream,y} \times EF_{equipment} \times T_{equipment,y}$$

Where:

Parameter	Description	Unit
$w_{CH4,stream,y}$	Average methane weight fraction of the gas (kg-CH_4/kg) in year y	tCO_2e
$EF_{equipment,y}$	Leakage rate for fugitive emissions from the compression technology as per specification from compressor manufacturer in kg/hour/compressor. If no default value from the technology provider is available, the approach below shall be used	kg/ hour/compressor

Parameter	Description	Unit
$T_{equipment,y}$	Operation time of the equipment in hours in year y (in the absence of detailed information, it can be assumed that the equipment is used continuously , as a conservative approach)	hour

$W_{CH4stream,y}$ refers to the methane content in biogas in year y, and therefore doesn't require further monitoring.

Methane emission factor ($EF_{equipment,y}$) in table in page 34/40 of AMS-III.H is adopted:

Equipment type	Emission factor (kg/hour/source) for methane
Valves	4.5E-0.3
Pump seals	2.4E-0.3
Others[28]	8.8E-0.3
Connectors	2.0E-0.4
Flanges	3.9E-0.4
Open ended lines	2.0E-0.3

Calculation of $PE_{vent\ gas,y}$

Emissions from venting gas retained in the upgrading equipment are calculated as follows:

$$PE_{ventgas,y} = \sum_{h=1}^{8760} TM_{RG,h} \times (1-\eta_{flare,h}) \times \frac{GWP_{CH4}}{1000}$$

Where:

Parameter	Description	Unit
$TM_{RG,h}$	Mass flow rate of methane in the residual gas in hour h	kg/hour
$\eta_{flare,h}$	Flare efficiency in hour h	%

28 The emission factor for "other" equipment type was derived from compressors, diaphragms, drains, dump arms, hatches, instruments, meters, pressure relief valves, polished rods, relief valves and vents. This "other" equipment type should be applied for any equipment type other than connectors, flanges, open-ended lines, pumps or valves.

In case of project activities that are transported via a dedicated piped network to a group of end users, emissions due to physical leakage of upgraded biogas from the dedicated piped network ($PE_{leakage,pipeline,y}$) shall be determined as follows:

$$PE_{leakage,pipeline,y} = Q_{methane,pipeline,y} \times LR_{pipeline} \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
$Q_{methane\ pipeline,y}$	Total quantity of methane transported in the dedicated pipeline network in year y	m³
$LR_{pipeline,}$	Physical leakage rate from the dedicated piped network (if no project-specific values can be identified a conservative default value of 0.0125 Gg per 10^6 m3 of utility sales shall be applied)	kg/ hour/compressor

The digested residue waste leaving the reactor shall be treated aerobically, and disposed in land properly, such as to avoid methane emissions. If disposed under anaerobic conditions (e.g. landfill) the methane emissions shall be estimated and discounted as project emissions following the relevant provisions in "AMS-III.AO: Methane recovery through controlled anaerobic digestion".

For each project activity instance in the Grouped Project, the project emissions will be identified according to the procedures set out in *AMS-III.AQ.: Introduction of Bio-CNG in transportation applications --- Version 02.0.* The table below summarises the project emissions that will be considered for each project activity instance in the Grouped Project.

Source	Included/ Excluded	Justification
Project emissions due to electricity consumption ($PE_{elec,y}$)	Included/ Excluded	Electricity may or may not be used in the project scenario
Project emissions due to fossil fuels consumption ($PE_{fuel,y}$)	Included/ Excluded	Fuel may or may not be used in the project scenario
Project emissions from transportation of the biomass from the places of their origin to the biogas production site and where applicable, transportation Bio-CNG from biogas plant to filling stations where it is used by final consumers ($PE_{transport,y}$)	Included/ Excluded	Transportation of biomass from the places of their origin to the biogas production may or may not be occurred in the project scenario

Source	Included/ Excluded	Justification
Project emissions from biomass cultivation in a dedicated plantation ($PE_{cultivation,y}$)	Included/ Excluded	Biomass cultivation may or may not be happened in the dedicated plantation during project scenario
Project emissions due to the physical leakage of methane from the systems affected by the project activity for production, processing, purification, compression; storage and filling of the Bio-CNG ($PE_{CH4,y}$)	Included/ Excluded	Physical leakage of methane from the system may or may not be happened in the project scenario

AMS III O, Version 02.0

According to the approved methodology AMS III O, Version 02.0, the project emissions for instances in the Grouped Project are calculated as the summation of the emissions from fossil fuels and/or electricity used.

The emissions from fossil fuels and/or electricity used to generate steam for the purpose of regeneration of the biogas purification system for operating the biogas purification system calculated in accordance with the methods specified in:

(a) AMS-I.D - Grid connected renewable electricity generation; and

(b) Tool to calculate project or leakage CO2 emissions from fossil fuel combustion

Project Emissions for the Project Activity Instances 1

AMS III H, Version 19.0

The table below summarises the project emissions that are considered for the PAI1

Source	Included/Excluded	Justification
$PE_{power,y}$	Excluded	No electricity or fossil fuel used in the project scenario
$PE_{ww,treatment,y}$	Included	Main emissions source in the baseline from open lagoons
$PE_{s,treatment,y}$	Excluded	No sludge treatment activity involved in the project activity
$PE_{ww,discharge,y}$	Included	The effluent is discharged into river
$PE_{s,final,y}$	Excluded	No sludge treatment activity involved in the project activity

Source	Included/Excluded	Justification
$PE_{fugitive,y}$	Included	Project emissions due to potential inefficiencies of biogas capture system
$PE_{flaring,y}$	Included	Project emission due to incomplete flaring
$PE_{biomass,y}$	Excluded	No storage of additional biomass under anaerobic conditions in project activity instances

The project emissions calculated for the PAI1 is calculated below:

Year	PE_y	$PE_{ww,treatment,y}$	$PE_{ww,discharge,y}$	$PE_{fugitive,y}$	$PE_{flaring,y}$
01-Apr-2024 – 31-Dec-2024 (275 days)	10,105	5,274	318	3,516	998
01-Jan-2025 – 31-Dec-2025	13,412	7,000	422	4,667	1,324
01-Jan-2026 – 31-Dec-2026	13,412	7,000	422	4,667	1,324
01-Jan-2027 – 31-Dec-2027	13,412	7,000	422	4,667	1,324
01-Jan-2028 – 31-Dec-2028	13,412	7,000	422	4,667	1,324
01-Jan-2029 – 31-Dec-2029	13,412	7,000	422	4,667	1,324
01-Jan-2030 – 31-Dec-2030	13,412	7,000	422	4,667	1,324
01-Jan-2031 – 31-Mar-2031 (90 days)	3,307	1,726	104	1,151	326
Total	**93,887**	**49,001**	**2,951**	**32,667**	**9,268**

The detailed calculation for project emission is tabulated below:

AMS III H

$PE_{ww,treatment,y}$ - Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery

Parameter	01-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 31-Mar-2031
$\Sigma Q_{ww,i,y}$	100,399	133,257	133,257	133,257	133,257	133,257	133,257	32,858
$COD_{removed,PJ,y}$	0.0099	0.0099	0.0099	0.0099	0.0099	0.0099	0.0099	0.0099
$h_{COD, PJ, i}$	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%
$MCF_{ww,treatment,PJ,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
GWP_{CH4}	28	28	28	28	28	28	28	28
$PE_{ww,treatment,y}$	**5,274**	**7,000**	**7,000**	**7,000**	**7,000**	**7,000**	**7,000**	**1,726**

PE $_{ww,discharge,y}$ - Methane emissions from degradable organic carbon in treated wastewater

Parameter	01-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 - 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 31-Mar-2031
$\Sigma Q_{ww,i,y}$	100,399	133,257	133,257	133,257	133,257	133,257	133,257	32,858
COD $_{ww,discharge,PJ,y}$	0.004035	0.004035	0.004035	0.004035	0.004035	0.004035	0.004035	0.004035
$MCF_{ww,discharge,PJ,k}$	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF $_{PJ}$	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
GWP_{CH4}	28	28	28	28	28	28	28	28
$PE_{ww,discharge,y}$	**318**	**422**	**422**	**422**	**422**	**422**	**422**	**104**

$PE_{fugitive,y}$ - Emissions on account of inefficiencies in capture systems

Parameter	01-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 31-Mar-2031
CFE_{ww}	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
GWP_{CH4}	28	28	28	28	28	28	28	28
$Q_{ww,y}$	100,399	133,257	133,257	133,257	133,257	133,257	133,257	32,858
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
$COD_{removed,PJ,k,y}$	0.0558	0.0558	0.0558	0.0558	0.0558	0.0558	0.0558	0.0558
$MCF_{ww,treatment, PJ,k}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$MEP_{ww,treatment,y}$	1,256	1,667	1,667	1,667	1,667	1,667	1,667	411
$PE_{,fugitive,y}$	**3,516**	**4,667**	**4,667**	**4,667**	**4,667**	**4,667**	**4,667**	**1,151**

$PE_{flaring,y}$ - Methane emissions due to incomplete flaring

Parameter	01-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 31-Mar-2031
$BE_{ww,treatment,y}$	27,930	37,071	37,071	37,071	37,071	37,071	37,071	9,141
GWP_{CH4}	28	28	28	28	28	28	28	28
$PE_{flaring,y}$	**998**	**1,324**	**1,324**	**1,324**	**1,324**	**1,324**	**1,324**	**326**

AMS I D, Version 18.0

According to AMS-I.D paragraph 39, the project emission is 0, and CO_2 emissions from on-site consumption of fossil fuel due to project activity shall be calculated using the latest version of "Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion.

4.3 Leakage Emissions

Grouped Project

AMS-III.D version 21.0

According to AMS III-D Version 21.0, paragraph 26, the leakage is determined by methodological tool "Project and leakage emissions from anaerobic digesters".

Leakage emissions from anaerobic digesters are determined as follows:

$$LE_{AD,y}=LE_{storage,y}+LE_{comp,y}$$

Where:

Parameter	Description	Unit
$LE_{AD,y}$	Leakage emissions associated with the anaerobic digester in year y (t CO_2e)	tCO_2/yr
$LE_{storage,y}$	Leakage emissions associated with storage of digestate in year y (t CO_2e)	tCO_2/yr
$LE_{comp,y}$	Leakage emissions associated with composting digestate in year y (t CO_2e)	tCO_2/yr

For leakage emissions associated with digestate from digester is stored ($LE_{storage,y}$) under the following anaerobic conditions: (a) In an un-aerated lagoon that has a depth of more than one meter; or (b) In a SWDS, including stockpiles that are considered a SWDS as per the definitions section.

For the grouped project, the digestate are not stored in anaerobic lagoon but is land applied. It is also not used for composting.

Leakage emissions associated with composting of digestate ($LE_{comp,y}$) is considered zero since for the grouped project, the digestate will not be composted, but land applied.

Therefore, there is no leakage emission associated with anaerobic digesters.

AMS-III.H Version 19.0

According to AMS III-H Version 19.0, paragraph 41, if the technology is using equipment transferred from another activity, leakage effects at the site of the other activity are to be considered and estimated (LE_y). There are no relevant leakage effects to be considered for all the PAIs.

AMS-III.AQ Version 02..0

Leakage emission for AMS III A.Q, paragraph 35, leakage emissions $LE_{BIOMASS,y}$ due to competing use of biomass shall be accounted for as per the approved "General guidance on leakage in biomass project.

The substitution of Bio-CNG for CNG from fossil origin reduces indirect ("upstream") emissions associated with the production of fossil CNG and is treated as negative leakage $LE_{PROCESS,y,CNG}$ that can be calculated as per the latest approved version of the tool "Upstream leakage emissions associated with fossil fuel use".

The substitution of Bio-CNG for gasoline reduces indirect ("upstream") emissions associated with the production of gasoline and is treated as negative leakage $LE_{PROCESS,y,GAS}$ (leakage emissions related to production and refining of the gasoline) that can be calculated using the latest approved version of the tool "Upstream leakage emissions associated with fossil fuel use".

Negative leakage emissions related to the avoided production of fossil fuel (CNG, gasoline) (tCO_2/yr) shall be calculated as per the equation below:

$$LE_{PROCESS,y,FF} = LE_{PROCESS,y,CNG} + LE_{PROCESS,y,GAS}$$

Where:

Parameter	Description	Unit
$LE_{PROCESS,y,FF}$	Leakage related to the avoided production of fossil fuel	tCO_2/yr

According to AMS III-O Version 02.0, paragraph 32, if the technology is using equipment transferred from another activity, or if the displaced equipment is transferred from another activity, leakage effects is to be considered for all the PAIs.

There will be no equipment transfer from another activity for all PAIs under this grouped activity

PAI1

No leakage emission for PAI1

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Grouped Project

As per the AMS-III.D. methodology, Version 21.0, emission reductions for the Grouped Project will be calculated as follows.

The emission reductions achieved by the project activity will be determined ex post through direct measurement of the amount of methane fuelled, flared or gainfully used. It is likely that the project activity involves manure treatment steps with higher methane conversion factors (MCF) than the MCF for the manure treatment systems used in the baseline situation, therefore the emission reductions achieved by the project activity are limited to the ex post calculated baseline emissions minus the project emissions using the actual monitored data for the project activity (i.e. $N_{LT,y}$, $MS\%_{i,y}$, $MS\%_{l}$, AI_l, as well as $VS_{LT,y}$ in cases where adjusted values for animal weight are used). The emission reductions achieved in any year are the lowest value of the following:

$$ER_{y,ex\ post} = min[(BE_{y,ex\ post} - PE_{y,ex\ post}), (MD_y - PE_{power,y,ex\ post})]$$

Where:

Parameter	Description	Unit
$ER_{y,ex\ post}$	Emission reductions achieved by the project activity based on monitored values for year y	tCO_2e
$BE_{y,ex\ post}$	Baseline emissions calculated for projects using option (a) using ex post monitored values of $N_{LT,y}$ and if applicable $VS_{LT,y}$. For projects using option in (b), the ex post monitored values for $Q_{manure,j,LT,y}$ and $SVS_{j,LT,y}$ are used	tCO_2e
$PE_{y,ex\ post}$	Project emissions calculated using ex post monitored values of $N_{LT,y}$, $MS\%_{i,y}$, $MS\%_{l}$, AI_l, $Q_{res\ waste,y}$ and if applicable $VS_{LT,y}$	tCO_2e
MD_y	Methane captured and destroyed or used gainfully by the project activity in year y	tCO_2e
$PE_{power,y,ex\ post}$	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities based on monitored values in the year y	tCO_2e

Biogas flared or combusted, (MD_y) shall be determined using the flare efficiency and methane content of biogas:

$$MD_y = BG_{burnt,y} \times w_{CH4,y} \times D_{CH4} \times FE \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
$BG_{burnt,y}$	Biogas flared/combusted in year y	m^3
$w_{CH4,y}$	Methane content of the biogas in the year y	Volume fraction
D_{CH4}	Density of methane at the temperature and pressure of the biogas in the year y	t/m^3
FE	Flare efficiency in year y (fraction). If the biogas is combusted for gainful purposes, e.g. fed to an engine, an efficiency of 100 per cent may be applied	-

As per the AMS-III.H. methodology, Version 19.0, emission reductions for the Grouped Project will be calculated as follows.

For all options in paragraph 2 of AMS-III.H, emission reductions shall be estimated ex ante in the PD using the equations provided in the baseline, project and leakage emissions sections above. Emission reductions shall be estimated ex ante as follows:

$$ER_{y,ex\ ante} = BE_{y,ex\ ante} - \left(PE_{y,ex\ ante} + LE_{y,ex\ ante}\right)$$

Where:

Parameter	Description	Unit
$ER_{y,ex\ ante}$	Ex-ante emission reduction in year y	tCO_2e
$LE_{y,ex\ ante}$	Ex-ante leakage emissions in year y	tCO_2e
$PE_{y,ex\ ante}$	Ex-ante project emissions in year y calculated as paragraph 39 of AMS-III.H	tCO_2e
$BE_{y,ex\ ante}$	Ex-ante baseline emissions in year y calculated as per paragraph 25 of AMS-III.H	tCO_2e

For cases 2(b), 2(c), 2(d) and 2(f), ex post emission reductions shall be based on the lowest value of the following, as per paragraph 44:

(a) The amount of biogas recovered and fuelled or flared (MD_y) during the crediting period, that is monitored ex post;

(b) Ex post calculated baseline, project and leakage emissions based on actual monitored data for the project activity.

For cases 2(b), 2(c), 2(d) and 2(f): it is possible that the project activity involves wastewater and sludge treatment systems with higher methane conversion factors (*MCF*) or with higher efficiency than the treatment systems used in the baseline situation. Therefore, the emission reductions achieved by the project activity is limited to the ex post calculated baseline emissions minus project emissions using the actual monitored data for the project activity. The emission reductions achieved in any year are the lowest value of the following:

$$ER_{y,ex\ post} = min\left(\left(BE_{y,ex\ post} - PE_{y,ex\ post} - LE_{y,ex\ post}\right), \left(MD_y - PE_{power,y} - PE_{biomass,y} - LE_{y,ex\ post}\right)\right)$$

Where:

Parameter	Description	Unit
$ER_{y,ex\ post}$	Emission reductions achieved by the project activity based on monitored values for year y	tCO_2e
$BE_{y,ex\ post}$	Baseline emissions calculated as per paragraph 25 of AMS-III.H using ex post monitored values	tCO_2e

Parameter	Description	Unit
$PE_{y,ex\ post}$	Project emissions calculated as per paragraph 39 using ex post monitored values	tCO_2e
MD_y	Methane captured and destroyed/gainfully used by the project activity in the year y	tCO_2e

In the case of flaring/combustion MD_y will be measured using the conditions of the flaring process:

$$MD_y = BG_{burnt,y} \times w_{CH4,y} \times D_{CH4} \times FE \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
$BG_{burnt,y}$	Biogas flared/combusted in year y	m^3
$w_{CH4,y}$	Methane content of the biogas in the year y	Volume fraction
D_{CH4}	Density of methane at the temperature and pressure of the biogas in the year y	t/m^3
FE	Flare efficiency in year y (fraction). If the biogas is combusted for gainful purposes, e.g. fed to an engine, an efficiency of 100 per cent may be applied	-

For the cases 2 (a) and (e) the emission reduction achieved by the project activity (ex post) will be the difference between the baseline emissions and the sum of the project emissions and leakage.

$$ER_y = BE_{y,ex\ post} - \left(PE_{y,ex\ post} + LE_{y,ex\ post}\right)$$

The historical records of electricity and fuel consumption, the COD content of untreated and treated wastewater, and the quantity of sludge produced by the replaced units will be used for the baseline calculation.

In case (a), if the volumetric flow and the characteristic properties (e.g. COD) of the inflow and outflow of the wastewater are equivalent in the project and the baseline scenarios (i.e. the project and baseline systems have the same efficiency for COD removal for wastewater treatment), then the higher energy consumption and sludge generation in the baseline scenario are the only significant differences contributing to emissions reductions in the project case. In this case, the emission reductions can be calculated as the difference between the historical energy consumption of the replaced unit and the recorded energy consumption of the new system, plus the difference in emissions from sludge treatment and/or disposal. Project emissions from fugitive emissions and incomplete flaring ($PE_{fugitive,y}$, $PE_{flaring,y}$) shall also be considered in the calculation of the emission reductions, however the emissions from the wastewater outflow and sludge ($PE_{ww,discharge,y}$, $PE_{s,final,y}$) may be disregarded, if they are equivalent in the baseline and project scenarios.

As per the AMS-III.AQ. methodology, Version 02.0, emission reductions for the Grouped Project will be calculated as follows.

$$ER_y = BE_y - PE_y - LE_{BIOMASS,y} + LE_{PROCESS,y,FF}$$

Where:

Parameter	Description	Unit
ER_y	Emission reductions in the year y	tCO_2e

As per the AMS-III.O. methodology, Version 02.0, emission reductions for the Grouped Project will be calculated as follows.

$$ER_y = BE_y - PE_y - LE_y$$

Where:

Parameter	Description	Unit
ER_y	Emission reductions in year y	tCO_2e
BE_y	Baseline emissions in year y	tCO_2e
PE_y	Project activity emissions in year y	tCO_2e
LE_y	Leakage in year y	tCO_2e

<u>Emissions Reduction for the PAI1</u>

The estimated emissions reduction for all the PAIs are calculated as per AMS-III.H, Version 19.0, which is tabulated as below:

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
01-Apr-2024 – 31-Dec-2024	31,532	10,105	0	0	21,427
01-Jan-2025 – 31-Dec-2025	41,852	13,412	0	0	28,440

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
01-Jan-2026 – 31-Dec-2026	41,852	13,412	0	0	28,440
01-Jan-2027 – 31-Dec-2027	41,852	13,412	0	0	28,440
01-Jan-2028 – 31-Dec-2028	41,852	13,412	0	0	28,440
01-Jan-2029 – 31-Dec-2029	41,852	13,412	0	0	28,440
01-Jan-2030 – 31-Dec-2030	41,852	13,412	0	0	28,440
01-Jan-2031 – 31-Mar-2031	10,320	3,307	0	0	7,013
Total	**292,965**	**93,887**	**0**	**0**	**199,079**

5 MONITORING

5.1 Data and Parameters Available at Validation

Grouped Project

AMS-III.D, version 21.0

Data / Parameter	GWP_{CH4}
Data unit	tCO_2/tCH_4
Description	Global warning potential for methane
Source of data	IPCC
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value, latest as per IPCC
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	D_{CH4}
Data unit	t/m^3
Description	Density of CH_4
Source of data	AMS-III.D
Value applied	0.00067 (at room temperature)
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of baseline and project emissions.
Comments	-

Data / Parameter	UF_b
Data unit	Fraction
Description	Correction factor to account for model uncertainties
Source of data	AMS IIID, Version 21
Value applied	0.94
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS IIID Version 21
Purpose of data	Calculation of baseline and project emissions.
Comments	-

Data / Parameter	MCF_j
Data unit	-
Description	Annual methane conversion factor (MCF) for the baseline animal manure management system j
Source of data	2006 IPCC Guidelines for National Greenhouse Gas inventories
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	No country or regional specific value available. Default value from Table 10.17 of 2006 IPCC guidelines for National Greenhouse Gas Inventories Volume 4 of Chapter 10 is applied.
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	$B_{o,LT}$
Data unit	m^3 CH_4/kg-dm
Description	Maximum methane producing potential of the volatile solid generated for animal type LT
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories

Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	Default value from Table 10 A-7 and Table 10 A-8 of 2006 IPCC guidelines for National Greenhouse Gas Inventories Volume 4 of Chapter 10 is applied
Purpose of data	Calculation of baseline and project emissions.
Comments	-

Data / Parameter	$VS_{default}$
Data unit	kg-dm/animal/year
Description	Default value for the volatile solid excretion rate per day on a dry matter basis for a defined livestock population
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	Default value from Table 10 A-7 and Table 10 A-8 of 2006 IPCC guidelines for National Greenhouse Gas Inventories Volume 4 of Chapter 10 is applied. Region: Asia
Purpose of data	Calculation of baseline and project emissions.
Comments	-

Data / Parameter	$MS\%_{BL,j}$
Data unit	-
Description	Fraction of manure handled in baseline animal manure management system *j*
Source of data	-
Value applied	100%
Justification of choice of data or description of measurement methods and procedures applied	Region: Asia Manure Management Systems: Lagoon

Purpose of data	Calculation of baseline and project emissions.
Comments	-

Data / Parameter	$f_{CH4,default}$
Data unit	-
Description	Default value for the fraction of methane in the biogas.
Source of data	Tool to Project and leakage emissions from anaerobic digesters.
Value applied	0.6
Justification of choice of data or description of measurement methods and procedures applied	Default value as per tool
Purpose of data	Calculate the quantity of methane produced in the digester.
Comments	-

Data / Parameter	$W_{default}$
Data unit	Kg
Description	Default average animal weight of a defined population.
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	Default value from Table 10 A-7 and Table 10 A-8 of 2006 IPCC guidelines for National Greenhouse Gas Inventories Volume 4 of Chapter 10 is applied.
Purpose of data	Calculation of baseline emissions.
Comments	-

AMS-III.H, version 19.0

Data / Parameter	GWP_{CH4}
Data unit	tCO_2/tCH_4
Description	Global warning potential for methane
Source of data	IPCC
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value, latest as per IPCC
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	$B_{O,WW}$
Data unit	kg CH_4/kg COD
Description	Methane producing capacity of the wastewater
Source of data	IPCC default value as per methodology AMS III-H version 19
Value applied	0.25
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	UF_{BL}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.H version 19

Value applied	0.89
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	UF_{PJ}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.H version 19
Value applied	1.12
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of project emissions
Comments	-

Data / Parameter	$MCF_{ww, treatment,BL,i}$
Data unit	-
Description	Methane correction factor for the baseline anaerobic wastewater treatment system j
Source of data	Table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)

Value applied	<table><tr><th>Type of wastewater treatment and discharge pathway or system</th><th>MCF value</th></tr><tr><td>Discharge of wastewater to sea, river or lake</td><td>0.1</td></tr><tr><td>Land application</td><td>0.1</td></tr><tr><td>Aerobic treatment, well managed</td><td>0.0</td></tr><tr><td>Aerobic treatment, poorly managed or overloaded</td><td>0.3</td></tr><tr><td>Anaerobic digester for sludge without methane recovery</td><td>0.8</td></tr><tr><td>Anaerobic reactor without methane recovery</td><td>0.8</td></tr><tr><td>Anaerobic shallow lagoon (depth less than 2 metres)</td><td>0.2</td></tr><tr><td>Anaerobic deep lagoon (depth more than 2 metres)</td><td>0.8</td></tr><tr><td>Septic system</td><td>0.5</td></tr><tr><td>Land application</td><td>0.1</td></tr></table>
Justification of choice of data or description of measurement methods and procedures applied	Default value according to table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$MCF_{ww, treatment,PJ,k}$
Data unit	-
Description	Methane correction factor for the project anaerobic wastewater treatment system k
Source of data	Table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)
Value applied	<table><tr><th>Type of wastewater treatment and discharge pathway or system</th><th>MCF value</th></tr></table>

	Discharge of wastewater to sea, river or lake	0.1
	Land application	0.1
	Aerobic treatment, well managed	0.0
	Aerobic treatment, poorly managed or overloaded	0.3
	Anaerobic digester for sludge without methane recovery	0.8
	Anaerobic reactor without methane recovery	0.8
	Anaerobic shallow lagoon (depth less than 2 metres)	0.2
	Anaerobic deep lagoon (depth more than 2 metres)	0.8
	Septic system	0.5
	Land application	0.1
Justification of choice of data or description of measurement methods and procedures applied	Default value according to table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)	
Purpose of data	Calculation of project emissions	
Comments	-	

Data / Parameter	CFE_{ww}
Data unit	-
Description	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems
Source of data	AMS-III.H, Version 19
Value applied	0.9
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of project emissions
Comments	-

Data / Parameter	$\eta_{COD,BL,i}$
Data unit	%
Description	COD removal efficiency of the baseline system
Source of data	Historical records of at least one (1) year prior to the project implementation or 10-day baseline measurement campaign for the existing wastewater system
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	COD removal efficiency is based on either historical records of at least one (1) year prior to the project implementation or 10-day baseline measurement campaign for the existing wastewater system.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$\eta_{COD,PJ,i}$
Data unit	%
Description	COD removal efficiency of the project system
Source of data	Technology provider
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	COD removal efficiency is based on the expertise of the technology provider.
Purpose of data	Ex ante estimation of project emissions.
Comments	Note that ex-post calculation of project estimations is based on the monitored data for actual COD reductions, as relevant to the PAI.

Data / Parameter	$EF_{equipment}$
Data unit	Kg/hour/source
Description	Methane emission factors for equipment[29]
Source of data	AMS-III.H Appendix para 12
Value applied	Equipment type: Emission factor for methane Valves: 4.5E-0.3 Pump seals: 2.4E-0.3 Others[30]: 8.8E-0.3 Connectors: 2.0E-0.4 Flanges: 3.9E-0.4 Open ended lines: 2.0E-0.3
Justification of choice of data or description of measurement methods and procedures applied	Taken from Protocol from Equipment Leak estimates, published by EPA (United States Environmental Protection Agency)
Purpose of data	For the purpose of estimation of methane emissions in Project emissions
Comments	For PAIs that apply AMS-III.H (para 4(b) and 4(c))

Equipment type	Emission factor for methane
Valves	4.5E-0.3
Pump seals	2.4E-0.3
Others[30]	8.8E-0.3
Connectors	2.0E-0.4
Flanges	3.9E-0.4
Open ended lines	2.0E-0.3

Data / Parameter	$LR_{pipeline}$
Data unit	$Gg/10^6m^3$
Description	Physical leakage rate from the dedicated piped network
Source of data	AMS-III.H Appendix para 17
Value applied	0.0125

[29] Please refer to the document US EPA-453/R-95-017 Table 2.4, page 2-15, accessed on 23/10/2007

[30] The emission factor for "other" equipment type was derived from compressors, diaphragms, drains, dump arms, hatches, instruments, meters, pressure relief valves, polished rods, relief valves and vents. This "other" equipment type should be applied for any equipment type other than connectors, flanges, open-ended lines, pumps or valves.

Justification of choice of data or description of measurement methods and procedures applied	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, chapter 4, Table 4.2.5 provides default values for fugitive emissions from gas operations in developing countries. The default values provided for fugitive emissions for the distribution of natural gas to end users range from 1.1 E-3 to 2.5 E-3 Gg per 106 m^3 of utility sales. The uncertainty in this value is -20 per cent to 500 per cent. A conservative value of 2.5 E-3 * 500% = 0.0125 Gg per 106 m3 of utility sales shall be taken.
Purpose of data	To calculate physical leakage of upgraded biogas from dedicated piped network
Comments	For PAIs that apply AMS-III.H (para 4(b) and 4(c))

Data / Parameter	$SPEC_{flare}$
Data unit	Temperature - °C Flow rate or heat flux - kg/h or m^3 /h Maintenance schedule - number of days
Description	Manufacturer’s flare operating specifications for temperature, flow rate and maintenance schedule
Source of data	Flare manufacturer
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of project emissions
Comments	Only applicable in case of enclosed flares. The maintenance schedule is not required if Option A is selected to determine flare efficiency of an enclosed flare.

AMS-III.AQ, version 02.0

Data / Parameter	GWP_{CH4}
Data unit	tCO_2/tCH_4
Description	Global warning potential for methane
Source of data	IPCC
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value, latest as per IPCC
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	$EF_{CO2,CNG}$
Data unit	tCO_{2e}/GJ
Description	CO_2 Emission factor of CNG
Source of data	IPCC Volume 2 Chapter 1 Table 1.4
Value applied	0.0543
Justification of choice of data or description of measurement methods and procedures applied	In the absence of national data, IPCC default value of 95% confidence interval shall be used only when country or project specific data are not available or demonstrably difficult to obtain. Values shall be updated if national values or IPCC values change.
Purpose of data	Calculation of baseline emission
Comments	For PAIs that apply AMS-III.AQ approach 1 and approach 2

Data / Parameter	$EF_{CO2,gasoline}$
Data unit	tCO_{2e}/GJ
Description	CO_2 Emission factor of CNG
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories *Volume 2 Chapter 1 Table 1.4*

Value applied	*0.0675*
Justification of choice of data or description of measurement methods and procedures applied	In the absence of national data, IPCC default value of 95% confidence interval shall be used only when country or project specific data are not available or demonstrably difficult to obtain. Values shall be updated if national values or IPCC values change.
Purpose of data	Calculation of baseline emission
Comments	For PAIs that apply AMS-III.AQ approach 2

Data / Parameter	n
Data unit	-
Description	Discount factor to account for the possible drop in the fuel efficiency of the retrofitted Bio-CNG vehicles
Source of data	AMS-III.AQ
Value applied	0.95
Justification of choice of data or description of measurement methods and procedures applied	A default value of 0.95 shall be used for converted vehicles that previously used gasoline
Purpose of data	Calculation of baseline emission
Comments	For PAIs that apply AMS-III.AQ approach 2

Data / Parameter	$LR_{pipeline}$
Data unit	$Gg/10^6m^3$
Description	Physical leakage rate from the dedicated piped network
Source of data	AMS-III.H Appendix para 17
Value applied	0.0125
Justification of choice of data or description of measurement methods and procedures applied	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, chapter 4, Table 4.2.5 provides default values for fugitive emissions from gas operations in developing countries. The default values provided for fugitive emissions for the distribution of natural gas to end users range from 1.1 E-3 to 2.5 E-3 Gg per 106 m^3 of utility sales. The uncertainty in this value is

	-20 per cent to 500 per cent. A conservative value of 2.5 E-3 * 500% = 0.0125 Gg per 106 m3 of utility sales shall be taken.
Purpose of data	To calculate physical leakage of upgraded biogas from dedicated piped network
Comments	-

Data / Parameter	$EF_{CO2,FF}$
Data unit	tCO_{2e}/GJ
Description	Emission factor of fossil fuel used in the operation of the Bio-CNG plant
Source of data	IPCC chapter 1 volume 2, table 1.4
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	In the absence of national data, the upper limit IPCC default value of 95% confidence interval shall be used only when country or project specific data are not available or demonstrably difficult to obtain. Values shall be updated if national values or IPCC values change.
Purpose of data	Calculation of project emission from the use of fossil fuel
Comments	-

Data / Parameter	$\rho_{i,y}$
Data unit	Mass unit/volume unit
Description	Weighted average density of fossil fuel used in the operation of the Bio-CNG plant
Source of data	Value provided by fuel supplier
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of project emission from the use of fossil fuel

Comments	-

AMS-III.O, version 02.0

Data / Parameter	EF_{LPG}
Data unit	kg-CO_2/kg LPG
Description	Emission factor of LPG
Source of data	Calculated based on evaluation of carbon content of Malaysia LPG: 30% propane and 70% butane[31]
Value applied	4.008
Justification of choice of data or description of measurement methods and procedures applied	Calculated based on composition of LPG
Purpose of data	For baseline emission calculations
Comments	-

Data / Parameter	SFC_{LPG}
Data unit	kg-LPG/Nm^3-H_2
Description	Specific fuel consumption of baseline process
Source of data	Specific fuel consumption of LPG is should be based on one of the following options: 1. Measurements during crediting period when the hydrogen plant is operated with LPG as fuel 2. Minimum 1-year historical data, 3. Manufacturer's specification
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	Specific fuel consumption of LPG is based on one of the following options: 1. Measurements during crediting period when the hydrogen plant is operated with LPG as fuel,

[31] http://eprints.usm.my/13194/1/effect_of_variation.pdf

	2. Minimum 1 year historical data, 3. Manufacturer's specification Option 2 and 3 can only be used if option 1 is not the case, i.e. LPG is not used as fuel during the crediting period.
Purpose of data	For calculation of baseline emissions.
Comments	-

AMS-I.D, version 18.0

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2/MWh
Description	Grid Emission Factor for Peninsula Malaysia, Sabah, and Sarawak
Source of data	Ministry of Energy, Science, Technology, Environment and Climate Change
Value applied	<table><tr><th>Region</th><th>Combined Margin (CM) (tCO_2/MWh)</th></tr><tr><td>Peninsula Malaysia</td><td>0.585</td></tr><tr><td>Sabah</td><td>0.525</td></tr><tr><td>Sarawak</td><td>0.330</td></tr></table>
Justification of choice of data or description of measurement methods and procedures applied	Calculated by official sources according to the "Tool to calculate the emission factor for an electricity system"[32]
Purpose of data	Calculation of baseline emissions
Comments	Year 2017

[32] https://www.mgtc.gov.my/wp-content/uploads/2019/12/2017-CDM-Electricity-Baseline-Final-Report-Publication-Version.pdf

PAI1

AMS-III.H, version 19.0

Data / Parameter	GWP_{CH4}
Data unit	tCO_2/tCH_4
Description	Global warning potential for methane
Source of data	IPCC
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value, latest as per IPCC
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	$B_{0,WW}$
Data unit	kg CH_4/kg COD
Description	Methane producing capacity of the wastewater
Source of data	IPCC default value as per methodology AMS III-H version 19
Value applied	0.25
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	UF_{BL}
Data unit	-
Description	Model correction factor to account for model uncertainties

Source of data	AMS-III.H version 19
Value applied	0.89
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	UF_{PJ}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.H version 19
Value applied	1.12
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of project emissions
Comments	-

Data / Parameter	$MCF_{ww, treatment,BL,i}$
Data unit	-
Description	Methane correction factor for the baseline anaerobic wastewater treatment system j
Source of data	Table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)
Value applied	0.8
Justification of choice of data or description of	The baseline wastewater treatment system consists of a succession of anaerobic deep lagoons (depth more than 2 metres) therefore the MCF value is chosen as 0.8

measurement methods and procedures applied	
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$MCF_{ww,\ treatment,PJ,k}$
Data unit	-
Description	Methane correction factor for the project anaerobic wastewater treatment system k
Source of data	Table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	The baseline wastewater treatment system consists of a succession of anaerobic deep lagoons (depth more than 2 metres) therefore the MCF value is chosen as 0.8
Purpose of data	Calculation of project emissions
Comments	-

Data / Parameter	CFE_{ww}
Data unit	-
Description	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems
Source of data	AMS-III.H, Version 19
Value applied	0.9
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of project emissions
Comments	-

<table>
<tr><td>Data / Parameter</td><td>$\eta_{COD,BL,i}$</td></tr>
<tr><td>Data unit</td><td>%</td></tr>
<tr><td>Description</td><td>COD removal efficiency of the baseline system</td></tr>
<tr><td>Source of data</td><td>10-day baseline measurement campaign for the existing wastewater system</td></tr>
<tr><td>Value applied</td><td>95.5%</td></tr>
<tr><td>Justification of choice of data or description of measurement methods and procedures applied</td><td>COD removal efficiency is based on COD values for the baseline system before implementation of project activity.
<table>
<tr><th>DATE</th><th>Point A - Raw POME</th><th>Point C - Treated POME</th><th>Point D - Final Discharge</th></tr>
<tr><td>5/7/2023</td><td>57,400</td><td>4,550</td><td>1,100</td></tr>
<tr><td>6/7/2023</td><td>86,700</td><td>5,300</td><td>1,070</td></tr>
<tr><td>7/7/2023</td><td>60,700</td><td>3,670</td><td>1,090</td></tr>
<tr><td>11/7/2023</td><td>59,800</td><td>2,150</td><td>1,400</td></tr>
<tr><td>12/7/2023</td><td>57,400</td><td>1,920</td><td>1,100</td></tr>
<tr><td>15/7/2023</td><td>65,000</td><td>2,200</td><td>1,180</td></tr>
<tr><td>18/7/2023</td><td>32,500</td><td>2,520</td><td>1,160</td></tr>
<tr><td>21/7/2023</td><td>93,700</td><td>2,210</td><td>1,200</td></tr>
<tr><td>22/7/2023</td><td>71,600</td><td>2,180</td><td>1,420</td></tr>
<tr><td>25/7/2023</td><td>72,100</td><td>3,040</td><td>1,460</td></tr>
<tr><td>Average</td><td>65,690</td><td>2,974</td><td>1,218</td></tr>
<tr><td>tCOD/m^3</td><td>0.0657</td><td>0.0030</td><td>0.0012</td></tr>
<tr><td>COD Removal (tCOD/m^3)</td><td>0.0627</td><td>1,756</td><td>64,472</td></tr>
<tr><td>Removal Rate</td><td>95.5%</td><td>59.0%</td><td>98.1%</td></tr>
</table>
</td></tr>
<tr><td>Purpose of data</td><td>Calculation of baseline emissions</td></tr>
<tr><td>Comments</td><td>-</td></tr>
</table>

Data / Parameter	$\eta_{COD,PJ,i}$
Data unit	%
Description	COD removal efficiency of the project system
Source of data	Technology provider
Value applied	90%

Justification of choice of data or description of measurement methods and procedures applied	COD removal efficiency is based on the expertise of the technology provider.
Purpose of data	Ex ante estimation of project emissions.
Comments	Note that ex-post calculation of project estimations is based on the monitored data for actual COD reductions, as relevant to the PAI.

AMS-I.D, version 18.0

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2/MWh
Description	Grid Emission Factor for Peninsula Malaysia
Source of data	Ministry of Energy, Science, Technology, Environment and Climate Change
Value applied	0.585
Justification of choice of data or description of measurement methods and procedures applied	Calculated by official sources according to the "Tool to calculate the emission factor for an electricity system"
Purpose of data	Calculation of baseline emissions
Comments	Year 2017

5.2 Data and Parameters Monitored

<u>Grouped Project</u>

AMS-III.D, version 21.0

Data / Parameter	$VS_{LT,y}$
Data unit	kg dry matter/animal/year
Description	Volatile solids for livestock LT entering the animal manure management system in year y
Source of data	-
Description of measurement methods and procedures to be applied	Only required when data from national published source are not available or IPCC default value from 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 chapter 10 table 10 A-4 to 10 A-9 are not used. When country-specific excretion rates is to be estimated from feed intake levels as indicated in the paragraph 18(b), via the enhanced characterisation method (Tier 2) described in section 10.2 in 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 chapter 10, parameters of GE_{LT}, DE_{LT}, UE, ASH and ED_{LT} shall be monitored as detailed below to derive this value. When developed country values are to be used in the project, relevant parameters specified in the paragraph 18(d) and 36(b) shall be monitored/documented. If IPCC default values are to be adjusted for a site-specific average animal weight as specified in paragraph 18(c), the average animal weight of a defined livestock population at the project site (Wsite) shall be monitored as detailed in the table 4 below.
Frequency of monitoring/recording	Annually
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-

Comments	-

Data / Parameter	$N_{da,y}$
Data unit	Number
Description	Number of days animals alive in the farm in the year y
Source of data	Specific by PAIs
Description of measurement methods and procedures to be applied	Describe the system for monitoring the number of livestock population. The consistency between the value and indirect information (records of sales, records of food purchases) should be assessed sales, records of food purchases) should be assessed
Frequency of monitoring/recording	Annually, based on monthly records
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	$N_{p,y}$
Data unit	Number
Description	Number of animals produced annually of type LT for the year y
Source of data	Specific by PAIs

Description of measurement methods and procedures to be applied	Describe the system on monitoring the number of livestock population. The consistency between the value and indirect information (records of sales, records of food purchases) should be assessed
Frequency of monitoring/recording	Annually, based on monthly records
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	W_{site}
Data unit	kg
Description	Average animal weight of a defined livestock population at the project site
Source of data	Specific by PAIs
Description of measurement methods and procedures to be applied	The average animal weight of a defined livestock population will be recorded monthly in the operation records.
Frequency of monitoring/recording	Annually
Value applied	-
Monitoring equipment	-

QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	$BG_{burnt,y}$
Data unit	m^3
Description	Biogas volume in year y
Source of data	Onsite measurement for $BG_{fuelled,y}$ .and $BG_{flared,y}$
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas recovered, fuelled, flared or otherwise utilized (e.g. injected into a natural gas distribution grid or distributed via a dedicated piped network) shall be monitored ex post, using continuous flow meters. If the biogas streams flared and fuelled (or utilized) are monitored separately, the two fractions can be added together to determine the total biogas recovered, without the need to monitor the recovered biogas before the separation. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place, and on the same basis (wet or dry).
Frequency of monitoring/recording	Annually, monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value is not used for ex ante estimates. Value to be monitored ex-post at $BG_{fuelled,y}$ and $BG_{flared,y}$
Monitoring equipment	Gas flow meter(s)
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach

Purpose of data	Calculation of project emissions
Calculation method	Add the values for $BG_{flared,y}$ and $BG_{fuelled,y}$ (if applicable)
Comments	-

Data / Parameter	$BG_{fuelled,y}$
Data unit	m^3
Description	Amount of biogas fuelled in year y
Source of data	Gas flowmeter
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas fuelled shall be monitored ex post, using continuous flow meters. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$BG_{flared,y}$
Data unit	m^3
Description	Amount of biogas flared in year y

Source of data	Gas flowmeter
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas flared shall be monitored ex post, using continuous flow meters. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	W_{CH4}
Data unit	%
Description	Methane content in biogas in the year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The fraction of methane in the gas should be measured with a continuous analyser or, alternatively, with periodical measurements at a 90/10 confidence/precision level, or, alternatively a default value of 60% methane content can be used. It shall be measured using equipment that can directly measure methane content in the biogas - the estimation of methane content of biogas based on measurement of other constituents of biogas such as CO_2 is not permitted. The methane content measurement shall be carried out close to a location in the system

	where a biogas flow measurement takes place, and on the same basis (wet or dry)
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Methane analyser
QA/QC procedures to be applied	Methane analyser to be maintained according to manufacturer's specifications. In the case of temporary situation where the methane analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	T
Data unit	^oC
Description	Temperature of the biogas at the flow measurement site
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The temperature of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas

Frequency of monitoring/recording	Shall be measured at the same time when methane content in biogas (w_{CH4}) is measured
Value applied	Value to be monitored ex-post
Monitoring equipment	Temperature transmitter
QA/QC procedures to be applied	Temperature transmitter to be maintained according to manufacturer's specifications. In the case of temporary situation where the temperature transmitter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	P
Data unit	Pa
Description	Pressure of the biogas at the flow measurement site
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The pressure of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	Shall be measured at the same time when methane content in biogas (w_{CH4}) is measured
Value applied	Value to be monitored ex-post
Monitoring equipment	Pressure transmitter
QA/QC procedures to be applied	Pressure transmitter to be maintained according to manufacturer's specifications. In the case of temporary situation where the pressure transmitter malfunctions leading to no readings captured, the historical

	record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	FE
Data unit	%
Description	The flare efficiency
Source of data	Default value
Description of measurement methods and procedures to be applied	Enclosed flare: Default value of 90% or measured efficiency Open flare: Default value of 50% or 0%
Frequency of monitoring/recording	-
Value applied	90% or 50%/0%
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

Data / Parameter	$Q_{manure,LT,y}$
Data unit	Tonnes-dm/year
Description	Quantity of manure treated from livestock type LT at animal manure management system j
Source of data	Specific PAI.
Description of measurement methods and procedures to be applied	Manure weight shall be directly measured or alternatively manure volume can be measured together with the density determined from representative sample (90/10 precision). The quantity of animal manure from different farms and different animal types shall be recorded separately for crosscheck. Recording of the baseline animal manure management system where the animal manure would have been treated anaerobically is also required
Frequency of monitoring/recording	Annually, based on daily measurement and monthly aggregation
Value applied	-
Monitoring equipment	Weighbridge
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	$SVS_{i,LT,y}$
Data unit	Tonnes VS/tonnes--dm
Description	Specific volatile solids content of animal manure from livestock type LT and animal manure management system j in year y
Source of data	Specific PAI.
Description of measurement methods	If animal manure is treated in a centralized plant, testing shall be performed according to the guideline in annex 2 of AM0073. It can be a sample basis by following the “Standard for sampling and

and procedures to be applied	surveys for CDM project activities and programme of activities", with a maximum margin of error of 10% at a 90% confidence level
Frequency of monitoring/recording	Annually
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	$MS\%_{i,y}$
Data unit	%
Description	Fraction of manure handled in system i in the project activity in year y
Source of data	Specific PAI.
Description of measurement methods and procedures to be applied	If animal manure is treated in different treatment systems, manure weight delivered to each system shall be directly measured or alternatively manure volume can be measured together with the density determined from representative sample (90/10 precision). The quantity of animal manure from different farms and different animal types shall be recorded separately for cross-check. Recording of the baseline animal manure management system where the animal manure would have been treated anaerobically is also required
Frequency of monitoring/recording	Annually, based on daily measurement and monthly aggregate
Value applied	-
Monitoring equipment	-

QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	AL_l
Data unit	days
Description	Annual average interval between manure collection and delivery for treatment at a given storage device l
Source of data	Specific PAI.
Description of measurement methods and procedures to be applied	The interval between manure collection and delivery will be recorded on a monthly basis to calculate for an annual average
Frequency of monitoring/recording	Annually, based on monthly records
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

Data / Parameter	nd_y
Data unit	days
Description	Number of days that the animal manure management system was operational
Source of data	Specific PAI.
Description of measurement methods and procedures to be applied	If any farm has no operations on a given day, it will be documented and taken into account for the calculation of $BE_{ex\ post}$.
Frequency of monitoring/recording	Annually, based on daily records and monthly aggregation
Value applied	365 days for ex-ante estimation.
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	To estimate the annual volatile solid excretions for livestock LT entering all animal waste management systems on a dry matter weight basis ($V_{SLT,y}$). Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	$MS\%_l$
Data unit	%
Description	Fraction of volatile solids handled by storage device l
Source of data	Specific PAI.
Description of measurement methods	Default value from storage device technical specification

and procedures to be applied	
Frequency of monitoring/recording	Monthly
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

Data / Parameter	GE_{LT}
Data unit	MJ/day
Description	Daily average gross energy intake
Source of data	Specific PAI.
Description of measurement methods and procedures to be applied	Only when country-specific excretion rates are to be estimated from feed intake levels via the enhanced characterization method (Tier 2) described in section 10.2 in *2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 chapter 10*
Frequency of monitoring/recording	Annually
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-

Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	DE_{LT}
Data unit	%
Description	Digestible energy of the feed
Source of data	
Description of measurement methods and procedures to be applied	If IPCC Tier 2 is used for VS determination. IPCC 2006 Table 10.2, Chapter 10, Volume 4
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	UE
Data unit	Fraction of GE
Description	Urinary energy, expressed as fraction of GE

Source of data	-
Description of measurement methods and procedures to be applied	If IPCC Tier 2 is used for VS determination. Typically, 0.04GE can be considered urinary energy excretion by most ruminants (reduce to 0.02 for ruminants fed with 85% or more grain in the diet or for swine). Use country-specific values where available
Frequency of monitoring/recording	Annually
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	ASH
Data unit	Fraction of the dry matter feed intake
Description	Ash content of the manure calculated as a fraction of the dry matter feed intake
Source of data	-
Description of measurement methods and procedures to be applied	If IPCC Tier 2 is used for VS determination. Use country-specific values where available
Frequency of monitoring/recording	-
Value applied	-

Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	ED_{LT}
Data unit	MJ/kg-dm
Description	Energy density of the feed in MJ/kg fed to livestock type LT
Source of data	-
Description of measurement methods and procedures to be applied	If IPCC Tier 2 is used for VS determination. IPCC notes the energy density of feed, ED, is typically 18.45 MJ/kg-dm, which is relatively constant across a wide variety of grain-based feeds. The project proponent will record the composition of the feed to enable the DOE to verify the energy density of the feed
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	EE_y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Source of data	Project equipment technical specification
Description of measurement methods and procedures to be applied	Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and the efficiency specification is for this fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	-
Calculation method	-
Comments	-

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh/year
Description	Electricity consumption in the project scenario in the year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measurement by electricity meters on the electricity entering the Biogas Plant (from the Mill and/or directly from the national electricity grid).

Frequency of monitoring/recording	Continuous
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice. The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity providers. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

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Data / Parameter	$Q_{ww,i,y}$
Data unit	m^3 /month
Description	Volume of wastewater treated in the project wastewater system in year y
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured continuously using flow meter
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be

	attained). Parameter aggregated monthly and annually for calculations.
Value applied	-
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Flow meters will undergo maintenance/calibration subject to appropriate industry standards. If local/national standards or the manufacturer's specifications are not available, international standards may be used. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	$COD_{ww,untreated,y}$ or $COD_{inflow,y}$
Data unit	$tCOD/m^3$
Description	Chemical oxygen demand of the wastewater entering the biogas treatment system in year y ($tonnes/m^3$)
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured the COD according to national or international standards. COD is measured through international sampling
Frequency of monitoring/recording	Periodically, at least monthly
Value applied	-
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD

	samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	
Comments	-

Data / Parameter	$COD_{ww,treated,y}$ or $COD_{outflow,y}$
Data unit	$tCOD/m^3$
Description	Chemical Oxygen Demand of the wastewater at the outlet (outflow) of the anaerobic digester (tonnes/m^3)
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured the COD according to national or international standards. COD is measured through international sampling
Frequency of monitoring/recording	Periodically, at least monthly
Value applied	-
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	$COD_{ww,discharge,PJ,y}$
Data unit	$tCOD/m^3$
Description	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake (tonnes/m^3)
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured the COD according to national or international standards. COD is measured through international sampling
Frequency of monitoring/recording	Periodically, at least monthly
Value applied	-
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	$S_{final,PJ,y}$
Data unit	t
Description	Method of disposal of the sludge
Source of data	Record of site and date of sludge disposal
Description of measurement methods and procedures to be applied	If the methane emissions from anaerobic decay of the final sludge are to be neglected because the sludge is controlled combusted, disposed of in a landfill with methane recovery, or used for soil application, then the end-use of the final sludge will be monitored during the crediting period. If the baseline emissions include the anaerobic decay of final sludge generated by the baseline treatment systems in a landfill

	without methane recovery, the baseline disposal site shall be clearly defined, and verified by the DOE.
Frequency of monitoring/recording	At each incidence of sludge disposal.
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	Plant manager's signature is required on the record.
Purpose of data	Monitoring of sludge disposal.
Calculation method	-
Comments	-

Data / Parameter	$BG_{burnt,y}$
Data unit	m^3
Description	Biogas volume in year y
Source of data	Onsite measurement for $BG_{fuelled,y}$ .and $BG_{flared,y}$
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas recovered, fuelled, flared or otherwise utilized (e.g. injected into a natural gas distribution grid or distributed via a dedicated piped network) shall be monitored ex post, using continuous flow meters. If the biogas streams flared and fuelled (or utilized) are monitored separately, the two fractions can be added together to determine the total biogas recovered, without the need to monitor the recovered biogas before the separation. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place.
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value is not used for ex ante estimates. Value to be monitored ex-post at $BG_{fuelled,y}$ and $BG_{flared,y}$

Monitoring equipment	Gas flow meter(s)
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	Add the values for $BG_{flared,y}$ and $BG_{fuelled,y}$ (if applicable)
Comments	-

Data / Parameter	$BG_{fuelled,y}$
Data unit	m^3
Description	Amount of biogas fuelled in year y
Source of data	Gas flowmeter
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas fuelled shall be monitored ex post, using continuous flow meters. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions

Calculation method	-
Comments	-

Data / Parameter	$BG_{flared,y}$
Data unit	m^3
Description	Amount of biogas flared in year y
Source of data	Gas flowmeter
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas flared shall be monitored ex post, using continuous flow meters. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$W_{CH4,y}$
Data unit	%
Description	Methane content in biogas in the year y
Source of data	On site measurement

Description of measurement methods and procedures to be applied	The fraction of methane in the gas should be measured with a continuous analyser or, alternatively, with periodical measurements at a 90/10 confidence/precision level. It shall be measured using equipment that can directly measure methane content in the biogas - the estimation of methane content of biogas based on measurement of other constituents of biogas such as CO_2 is not permitted. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Methane analyser
QA/QC procedures to be applied	Methane analyser to be maintained according to manufacturer's specifications. In the case of temporary situation where the methane analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	T
Data unit	oC
Description	Temperature of the biogas

Source of data	On site measurement
Description of measurement methods and procedures to be applied	The temperature of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	Shall be measured at the same time when methane content in biogas ($w_{CH4,y}$) is measured
Value applied	Value to be monitored ex-post
Monitoring equipment	Temperature transmitter
QA/QC procedures to be applied	Temperature transmitter to be maintained according to manufacturer's specifications. In the case of temporary situation where the temperature transmitter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	P
Data unit	Pa
Description	Pressure of the biogas
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The pressure of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	Shall be measured at the same time when methane content in biogas ($w_{CH4,y}$) is measured

Value applied	Value to be monitored ex-post
Monitoring equipment	Pressure transmitter
QA/QC procedures to be applied	Pressure transmitter to be maintained according to manufacturer's specifications. In the case of temporary situation where the pressure transmitter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$Flame_m$
Data unit	Flame on or Flame off
Description	Flame detection of flare in the minute m
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measure using a fixed installation optical flame detector
Frequency of monitoring/recording	Once per minute. Detection of flame recorded as a minute that the flame was on, otherwise recorded as a minute that the flame was off
Value applied	Value to be monitored ex-post
Monitoring equipment	Optical flame detector: ultra violet detector or infra-red or both
QA/QC procedures to be applied	The flame detection will be monitored and cross checked with the amount of biogas sent to flare. If there is data for biogas sent to flare means the flame is on. Equipment will be maintained and calibrated in accordance with manufacturer's recommendations.

Purpose of data	Calculation of project emissions
Calculation method	-
Comments	For all PAIs where biogas is fueled, utilized and excess flared.

Data / Parameter	$T_{EG,m}$
Data unit	°C
Description	Temperature in the exhaust gas of the enclosed flare in minute m
Source of data	Measurements by project participants
Description of measurement methods and procedures to be applied	Measure the temperature of the exhaust gas stream in the flare by a Type N thermocouple. A temperature above 500 ºC indicates that a significant amount of gases are still being burnt and that the flare is operating.
Frequency of monitoring/recording	Continuously.
Value applied	Value to be monitored ex-post
Monitoring equipment	Temperature transmitter
QA/QC procedures to be applied	Flow meters are to be periodically calibrated according to the manufacturer's recommendation or industry standard.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh/year
Description	Electricity consumption in the project scenario in the year y

Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measurement by electricity meters on the electricity entering the Biogas Plant (from the Mill and/or directly from the national electricity grid).
Frequency of monitoring/recording	Continuous
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice. The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity providers. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

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Data / Parameter	$FC_{Bio\text{-}CNG,k,y}$
Data unit	tonne
Description	Bio-CNG consumed by the project vehicle k in the year y
Source of data	From PAI.

Description of measurement methods and procedures to be applied	Measurements of the amount of Bio-CNG filled into vehicles of the end users are undertaken using calibrated meters at the filling station site.
Frequency of monitoring/recording	Continuous
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Measurement results shall be cross-checked with production and sales data.
Purpose of data	Calculation of baseline emissions.
Calculation method	
Comments	For AMS-III.AQ approach 2 PAIs.

Data / Parameter	$FS_{Bio\text{-}CNG,y}$
Data unit	tonnes
Description	Amount of Bio-CNG distributed/sold to retailers, filling stations by the project activity in year y
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measurements of the amount of Bio-CNG distributed/sold to retailers/filling stations are undertaken using calibrated meters at the delivery section of Bio-CNG production site.
Frequency of monitoring/recording	Continuously or in batches
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter

QA/QC procedures to be applied	Measurements results shall be cross checked with records for sold amount (e.g. invoices/receipts) and with the amount of biogas produced. The meter will be maintained and calibrated according to industrial standards
Purpose of data	Calculation of baseline emissions.
Calculation method	
Comments	For AMS-III.AQ approach 1 PAIs.

Data / Parameter	$FP_{Bio\text{-}CNG,y}$
Data unit	tonne
Description	Quantity of the Bio-CNG produced by the project activity in the year y
Source of data	Measurement data by a gas flowmeter
Description of measurement methods and procedures to be applied	Measurements are undertaken using calibrated meters at the outlet of the biogas upgrading section, i.e. outlet of the Purification system of the Bio-CNG production site.
Frequency of monitoring/recording	Continuously or in batches
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	The meter will be maintained and calibrated according to industrial standards
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	For AMS-III.AQ applied PAIs.

Data / Parameter	$NCV_{Bio-CNG}$
Data unit	GJ/tonne
Description	Net calorific value of Bio-CNG
Source of data	Measurement
Description of measurement methods and procedures to be applied	Measured according to relevant national/international standards through sampling. Analysis has to be carried out by accredited laboratory
Frequency of monitoring/recording	Monthly or as prescribed by the applied national/ international standard
Value applied	46.5
Monitoring equipment	Measured according to relevant national/international standards through sampling. Analysis has to be carried out by accredited laboratory.
QA/QC procedures to be applied	In the absence of national data, IPCC default value of 95% confidence interval shall be used only when country or project specific data are not available or demonstrably difficult to obtain. Values shall be updated if national values or IPCC values change.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	AMS-III.AQ Approach 2 applied PAIs.

Data / Parameter	NCV_i
Data unit	GJ/tonne
Description	Net calorific value of gasoline/ blended gasoline that was used by project vehicle k
Source of data	Measurement
Description of measurement methods	Measured according to relevant national/international standards through sampling. Systematic sampling method in "Standard for

and procedures to be applied	sampling and surveys for CDM project activities and programme of activities" will be used to determine the sampling frequency. Analysis has to be carried out by accredited laboratory
Frequency of monitoring/recording	At the validation, and annually during the crediting period
Value applied	Value to be monitored ex-post
Monitoring equipment	N/A
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	AMS-III.AQ Approach 2 applied PAIs.

Data / Parameter	$w_{CH4,y}$
Data unit	%
Description	Methane content in the Bio-CNG
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	The fraction of methane in the gas should be measured with a continuous analyzer, or alternatively with periodical measurements at a 90/10 sampling confidence/precision level. It shall be measured using equipment that can directly measure methane content in the biogas – the estimation of methane content of biogas based on measurement of other constituents of biogas such as CO_2 is not permitted. The methane content measurement shall be carried out at the location where $FP_{Bio\text{-}CNG}$ is measured.
Frequency of monitoring/recording	Continuous/periodic
Value applied	Value to be monitored ex-post
Monitoring equipment	Continuous analyzer

QA/QC procedures to be applied	The analyzer will be maintained and calibrated according to industrial standards.
Purpose of data	Calculation of baseline emissions.
Calculation method	-
Comments	For AMS-III.AQ applied PAIs.

Data / Parameter	$f_{FO,gasoline}$
Data unit	%
Description	Fraction of gasoline from fossil fuel origin in the displaced gasoline
Source of data	Measurement
Description of measurement methods and procedures to be applied	As per the following options (in preferential order): 1. Data from supplier of the gasoline; 2. If it accrues to national regulations requiring mandatory blending of biofuels, the regulatory blend fraction may be used; 3. If measured, it shall be according to relevant national/international standards through sampling
Frequency of monitoring/recording	Continuously or in batches
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions.
Calculation method	-
Comments	For AMS-III.AQ applied approach 2 PAI.

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Data / Parameter	$V_{H2,BIO}$
Data unit	Nm^3-H_2
Description	Volume of hydrogen produced from methane extracted from biogas under normal conditions
Source of data	-
Description of measurement methods and procedures to be applied	Continuous metering on volumetric basis
Frequency of monitoring/recording	Continuous
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	The analyzer will be maintained and calibrated according to industrial standards.
Purpose of data	For baseline emissions calculation.
Calculation method	The volume of hydrogen derived from biogas is equivalent to total hydrogen production
Comments	For AMS-III.O applied PAIs.

Data / Parameter	M_{LPG}
Data unit	kg-LPG
Description	LPG used as feedstock to hydrogen production unit
Source of data	Flowmeter
Description of measurement methods	-

and procedures to be applied	
Frequency of monitoring/recording	Continuous
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	The analyzer will be maintained and calibrated according to industrial standards.
Purpose of data	For baseline emissions calculation.
Calculation method	-
Comments	For AMS-III.O applied PAIs.

Data / Parameter	m_1, m_2
Data unit	%
Description	$m_{1:}$ % mol of Propane in LPG m_2 % mol of butane in LPG
Source of data	Material property analysis from LPG supplier or if any available based on chromatography analysis of sample performed every quarter.
Description of measurement methods and procedures to be applied	Accredited laboratory methods.
Frequency of monitoring/recording	Quarterly
Value applied	30% propane; and 70% butane
Monitoring equipment	Accredited laboratory

QA/QC procedures to be applied	If values are based on sampling, analysis must be performed by an accredited laboratory.
Purpose of data	For baseline emissions calculation.
Calculation method	-
Comments	For AMS-III.O applied PAIs.

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh/year
Description	Electricity consumption in the project scenario in the year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measurement by electricity meters on the electricity entering the Biogas Plant (from the Mill and/or directly from the national electricity grid).
Frequency of monitoring/recording	Continuous
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice. The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity providers. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach

Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	PE_{Elec}
Data unit	tCO_2/year
Description	Project emissions from electricity consumption in year y
Source of data	As per AMS-I.D: Grid connected renewable electricity generation
Description of measurement methods and procedures to be applied	AMS-I.D: Grid connected renewable electricity generation
Frequency of monitoring/recording	AMS-I.D: Grid connected renewable electricity generation
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	As per AMS-I.D: Grid connected renewable electricity generation
Purpose of data	For project emissions calculation.
Calculation method	-
Comments	-

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Data / Parameter	$EG_{PJ,facility,y}$
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter
Frequency of monitoring/recording	Continuous, at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice. The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity providers. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	$EG_{PJ,add,y}$

Data unit	MWh
Description	Quantity of net electricity generation supplied to the grid in year y by the project plant/unit that has been added under the project activity
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter
Frequency of monitoring/recording	Continuous, at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice. The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity providers. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Tool 06: Project emissions from flaring, version 04.0

Data / Parameter	$F_{CH4,EG,t}$
Data unit	kg
Description	Mass flow of methane in the exhaust gas of the flare on a dry basis at reference conditions in the time period t
Source of data	Measurements undertaken by a third party accredited entity
Description of measurement methods and procedures to be applied	Measure the mass flow of methane in the exhaust gas according to an appropriate national or international standard e.g. UKs Technical Guidance LFTGN05. The time period t over which the mass flow is measured must be at least one hour. The average flow rate to the flare during the time period t must be greater than the average flow rate observed for the previous six months
Frequency of monitoring/recording	Biannual
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	According to the standard applied
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is required in the case of enclosed flares and if the project participants decide to determine flare efficiency through biannual measurement

Data / Parameter	$T_{EG,m}$
Data unit	^{o}C
Description	Temperature in the exhaust gas of the enclosed flare in the minute *m*

Source of data	Project participants
Description of measurement methods and procedures to be applied	Measure the temperature of the exhaust gas in the flare by an appropriate temperature measurement equipment, Measurements outside the operational temperature specified by the manufacturer may indicate that the flare is not functioning correctly and may require maintenance. Flare manufacturers must provide suitable monitoring ports for the monitoring of the temperature of the flare. These would normally be expected to be in the middle third of the flare. Where more than one temperature port is fitted to the flare, the flare manufacturer must provide written instructions detailing the conditions under which each location shall be used and the port most suitable for monitoring the operation of the flare according to manufacturer's operating specifications for temperature.
Frequency of monitoring/recording	Once per minute
Value applied	Temperature measurement equipment should be replaced or calibrated in accordance with their maintenance schedule
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Temperature measurement equipment should be replaced or calibrated in accordane with their maintenance schedule.
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Unexpected changes such as a sudden increase/drop in temperature can occur for different reasons. These events should be noted in the site records along with any corrective action that was implemented to correct the issue. Monitoring of this parameter is applicable in case of enclosed flares. Measurements are required to determine if manufacturer's flare operating specifications for operating temperature are met.

Data / Parameter	$v_{i,RG,m}$
Data unit	-
Description	Volumetric fraction of component i in the residual gas on a dry basis in the minute m where i = CH_4, CO, CO_2, O_2, H_2, H_2S, NH_4, N_2

Source of data	Measurements by project participants using a continuous gas analyser (values are recorded with the same frequency as the flow).
Description of measurement methods and procedures to be applied	Measurement may be made on either dry or wet basis. If the value is made on a wet basis, then it shall be converted to dry basis for reporting
Frequency of monitoring/recording	Continuously. Values are to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Analysers must be periodically calibrated according to the manufacturer's recommendation. A zero check and a typical value check should be performed by comparison with a standard certified gas. In the case of temporary situation where the gas analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	As a simplified approach, project participants may only measure the content CH_4, CO and CO_2 of the residual gas and consider the remaining part as N_2. Monitoring of this parameter is only applicable in case of enclosed flares and continuous monitoring of the flare efficiency. The methane content measurement shall be carried out close to a

	location in the system where a biogas flow measurement takes place.

Data / Parameter	$V_{RG,m}$
Data unit	m^3
Description	Volumetric flow of the residual gas on a dry basis at reference conditions in the minute m
Source of data	Measurements by project participants using a flow meter
Description of measurement methods and procedures to be applied	Instruments with recordable electronic signal (analogical or digital)
Frequency of monitoring/recording	Continuously. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Flow meters are to be periodically calibrated according to the manufacturer's recommendation In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is applicable in case of enclosed flares and continuous monitoring of the flare efficiency and if project participant selects to calculate $VRG_{,m}$ instead of monitoring directly. Monitoring of this parameter may also be necessary for confirming that the manufacturer's operating specifications for flow

	rate/heat flux are met. In this case the flow rate should be measured in a m^3/h basis

Data / Parameter	$M_{RG,m}$
Data unit	kg
Description	Mass flow of the residual gas on a dry basis at reference conditions in the minute m
Source of data	-
Description of measurement methods and procedures to be applied	Instruments with recordable electronic signal (analogical or digital)
Frequency of monitoring/recording	Continuous. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory. Calibration and frequency of calibration is according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is applicable in case of enclosed flares and continuous monitoring of the flare efficiency and if project participant selects to monitor $M_{RG,m}$ directly, instead of calculating.

	Monitoring of this parameter may also be necessary for confirming that the manufacturer's specifications for flow rate/heat flux are met. In this case the flow rate should be measured in a kg/h basis

Data / Parameter	$v_{O2,EG,m}$
Data unit	-
Description	Volumetric fraction of O_2 in the exhaust gas on a dry basis at reference conditions in the minute m
Source of data	Measurements by project participants using a continuous gas analyser
Description of measurement methods and procedures to be applied	Extractive sampling analysers with water and particulates removal devices or in situ analysers for wet basis determination. The point of measurement (sampling point) shall be in the upper section of the flare (80% of total flare height). Sampling shall be conducted with appropriate sampling probes adequate to high temperatures level (e.g. inconel probes)
Frequency of monitoring/recording	Continuously. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Analysers must be periodically calibrated according to the manufacturer's recommendation. A zero check and a typical value check should be performed by comparison with a standard gas. In the case of temporary situation where the gas analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach

Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is only applicable in case of enclosed flares and continuous monitoring of the flare efficiency

Data / Parameter	$fc_{CH4,EG,m}$
Data unit	mg/m^3
Description	Concentration of methane in the exhaust gas of the flare on a dry basis at reference conditions in the minute m
Source of data	Measurements by project participants using a continuous gas analyser
Description of measurement methods and procedures to be applied	Extractive sampling analysers with water and particulates removal devices or in situ analyser for wet basis determination. The point of measurement (sampling point) shall be in the upper section of the flare in order that the sampling is of the gas after consumption has taken place (80% of total flare height). Sampling shall be conducted with appropriate sampling probes adequate to high temperatures level (e.g. inconel probes)
Frequency of monitoring/recording	Continuously. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Analysers must be periodically calibrated according to the manufacturer's recommendation. A zero check and a typical value check should be performed by comparison with a standard gas. In the case of temporary situation where the gas analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach.

	2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is only applicable in case of enclosed flares and continuous monitoring of the flare efficiency. Measurement instruments may read ppmv or % values. To convert from ppmv to mg/m^3 simply multiply by 0.716. 1% equals 10 000 ppmv

Data / Parameter	$Flame_m$
Data unit	Flame on or Flame off
Description	Flame detection of flare in the minute m
Source of data	Project participants
Description of measurement methods and procedures to be applied	Measure using a fixed installation optical flame detector: Ultra Violet detector or Infra-Red or both
Frequency of monitoring/recording	Once per minute. Detectionn of flame recorded das a minute that the flame was on, otherwise recoded as a minute that the flame was off.
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Equipment shall be maintained and calibrated in accordance with manufacturer's recommendations.
Purpose of data	Calculation of project emissions.
Calculation method	-

Comments	Applicable to all flares

Data / Parameter	$Maintenance_y$
Data unit	Calendar dates
Description	Maintenance events completed in year y
Source of data	Specific PAI
Description of measurement methods and procedures to be applied	Record the date that maintenance events were completed in year y. Records of maintenance logs must include all aspects of the maintenance including the details of the person(s) undertaking the work, parts replaced, or needing to be replaced, source of replacement parts, serial numbers and calibration certificates
Frequency of monitoring/recording	Annual
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Records must be kept in a maintenance log for two years beyond the life of the flare
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is required for the case of enclosed flares and the project participant selects Option B to determine flare efficiency. These dates are required so that they can be compared to the maintenance schedule to check that maintenance events were completed within the minimum time between maintenance events specified by the manufacturer ($SPEC_{,flare}$)

Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 3.0

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2/MWh
Description	Combined margin emission factor for the grid in year y
Source of data	Calculate the combined margin emission factor, using the procedures in the latest approved version of the "Tool to calculate the emission factor for an electricity system"
Description of measurement methods and procedures to be applied	As per the "Tool to calculate the emission factor for an electricity system"
Frequency of monitoring/recording	As per the "Tool to calculate the emission factor for an electricity system"
Value applied	
Monitoring equipment	-
QA/QC procedures to be applied	As per the "Tool to calculate the emission factor for an electricity system"
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Option A1 is chosen.

Data / Parameter	TDLj,y and TDLk,y and TDLl,y
Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to source j, k or l in year y
Source of data	1. Use annual average value based on the most recent data available within the host country; 2. Use as default values of 20% for:

	(a) project or leakage electricity consumption sources; (b) baseline electricity consumption sources if the electricity consumption by all project and leakage electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies is larger than the electricity consumption of all baseline electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies; 3. Use as default values of 3% for: (a) baseline electricity consumption sources; (b) project and leakage electricity consumption sources if the electricity consumption by all project and leakage electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies is smaller than the electricity consumption of all baseline electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies
Description of measurement methods and procedures to be applied	The distribution losses can either be calculated by the project participants or be based on references from utilities, network operators or other official documentation
Frequency of monitoring/recording	Annually. In the absence of data from the relevant year, most recent figures should be used, but not older than 5 years
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

Data / Parameter	$FC_{n,i,t}$
Data unit	Mass or () volume unit per year (in m^3, ton or l)
Description	Quantity of fossil fuel type i fired in the captive power plant n in the time period t

Source of data	Annual data during the crediting period: Onsite measurements Historical data: Historical records / onsite measurements
Description of measurement methods and procedures to be applied	1. In cases where fuel is supplied from small daily tanks, rulers can be used to determine mass or volume of the fuel consumed, with the following conditions: The ruler gauge must be part of the daily tank and calibrated at least once a year and have a book of control for recording the measurements (on a daily basis or per shift); 2. Accessories such as transducers, sonar and piezoelectronic devices are accepted if they are properly calibrated with the ruler gauge and receiving a maintenance per supplier specifications; 3. In case of daily tanks with pre-heaters for heavy oil, the calibration will be made with the system at typical operational conditions
Frequency of monitoring/recording	Continuously, aggregated at least annually
Value applied	Value to be monitored ex-post
Monitoring equipment	Mass or Volumetric meter
QA/QC procedures to be applied	The consistency of metered fuel consumption quantities should be cross-checked with an annual energy balance that is based on purchased quantities and stock changes.
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

Data / Parameter	$EG_{n,t}$
Data unit	MWh
Description	Quantity of electricity generated in captive power plant n in the time period t
Source of data	Onsite measurements
Description of measurement methods	Use electricity meters

and procedures to be applied	
Frequency of monitoring/recording	Continuously, aggregated at least annually
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity Meter
QA/QC procedures to be applied	Cross check measurement results with records for sold electricity where relevant
Purpose of data	Calculation of baseline emissions.
Calculation method	-
Comments	-

Data / Parameter	$NCV_{i,t}$
Data unit	GJ / mass or volume unit
Description	Average net calorific value of fossil fuel type i used in the period t
Source of data	(a) To be provided by fuel supplier (b) Measurements by the project proponents (c) Regional or national default values (d) IPCC default values at the upper or lower limit – whatever is more conservative – of the uncertainty at a 95% confidence interval as provided in Table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Description of measurement methods and procedures to be applied	Measurements taken will be in line with national or international fuel standards
Frequency of monitoring/recording	(a & b) The NCV should be obtained for each fuel delivery, from which weighted average values for the period t should be calculated (c) Review appropriateness of the values annually

	(d) Any future revision of the IPCC Guidelines should be taken into account
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	Verify if the values are within the uncertainty range of the IPCC default values as provided in Table 1.2, Vol. 2 of the 2006 IPCC Guidelines. If the values fall out this range collect additional information from the testing laboratory to justify the outcome or conduct additional measurements. The laboratories should have ISO17025 accreditation or justify that they can comply with similar quality standards
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	

Data / Parameter	$EF_{CO2,i,t}$
Data unit	t CO_2 / GJ
Description	CO_2 emission factor of fossil fuel type i used in the period t
Source of data	(a) To be provided by fuel supplier (b) Measurements by the project proponents (c) Regional or national default values (d) IPCC default values at the upper or lower limit – whatever is more conservative – of the uncertainty at a 95% confidence interval as provided in Table 1.4 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Description of measurement methods and procedures to be applied	Measurements taken will be in line with national or international fuel standards
Frequency of monitoring/recording	(a & b) The CO_2 emission factor should be obtained for each fuel delivery, from which weighted average values for the period t should be calculated

	(c) Review appropriateness of the values annually (d) Any future revision of the IPCC Guidelines should be taken into account
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh/yr
Description	Quantity of electricity consumed by the project electricity consumption source j in year y Net increase in electricity consumption of source I in year y as a result of leakage
Source of data	Direct measurement or calculated based on measurements from more than one electricity meters
Description of measurement methods and procedures to be applied	Use electricity meters installed at the electricity consumption sources
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Transformer power meter

QA/QC procedures to be applied	In cases where electricity meters are regulated (e.g. the electricity is supplied by the electric grid), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier or requirements set by the grid operators. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. In cases where electricity meters are not regulated (e.g. the electricity is supplied by captive power plants), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier or national requirements. If these standards are not available, and meter supplier does not specify, calibrate the meters every 3 years and use the meters with at least 0.5 accuracy class (e.g. a meter with 0.2 accuracy class is more accurate and thus it is accepted).
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

Data / Parameter	$EC_{BL,k,y}$
Data unit	MWh/yr
Description	Quantity of electricity that would be consumed by the baseline electricity consumption source k in year y
Source of data	Direct measurement or calculated based on measurements from more than one electricity meters
Description of measurement methods and procedures to be applied	Use electricity meters installed at the electricity consumption sources

Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Transformer power meter
QA/QC procedures to be applied	In cases where electricity meters are regulated (e.g. the electricity is supplied by the electric grid), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier or requirements set by the grid operators. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. In cases where electricity meters are not regulated (e.g. the electricity is supplied by captive power plants), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier or national requirements. If these standards are not available, and meter supplier does not specify, calibrate the meters every 3 years and use the meters with at least 0.5 accuracy class (e.g. a meter with 0.2 accuracy class is more accurate and thus it is accepted).
Purpose of data	Calculation of baseline emissions.
Calculation method	-
Comments	-

Data / Parameter	$EG_{PJ,grid,y}$ or $EG_{PJ,facility,I,y}$
Data unit	MWh/yr
Description	Quantity of electricity generated and supplied by the project power plant to the grid in year y

	Quantity of electricity generated and supplied by the project power plant to the consumers/electricity consuming facility i in year y
Source of data	Direct measurement or calculated based on measurements from more than one electricity meters
Description of measurement methods and procedures to be applied	Use electricity meters installed at the grid interface for electricity export to grid and for supply to captive consumers use electricity meters installed at the entrance of the electricity consuming facility. In case of grid and net electricity generation: This parameter should be either monitored using bi-directional energy meter or calculated as difference between (a) the quantity of electricity supplied by the project plant/unit to the grid; and (b) the quantity of electricity the project plant/unit from the grid. If it is calculated, then the following parameters shall be measured: (a) The quantity of electricity supplied by the project plant/unit to the grid; and (b) The quantity of electricity delivered to the project plant/unit from the grid
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	In cases where electricity meters are regulated (e.g. the electricity is supplied by the electric grid), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier or requirements set by the grid operators. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. In cases where electricity meters are not regulated (e.g. the electricity is supplied by captive power plants), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national

	standards or requirements set by the meter supplier. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier or national requirements. If these standards are not available, and meter supplier does not specify, calibrate the meters every 3 years and use the meters with at least 0.5 accuracy class (e.g. a meter with 0.2 accuracy class is more accurate and thus it is accepted).
Purpose of data	Calculation of baseline emissions.
Calculation method	-
Comments	-

Tool 03: Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion, version 3.0

Data / Parameter	$FC_{i,j,y}$
Data unit	Mass or volume unit per year (e.g. ton/yr or m^3/yr)
Description	Quantity of fuel type i combusted in process j during the year y
Source of data	Onsite measurements
Description of measurement methods and procedures to be applied	(a) In cases where fuel is supplied from small daily tanks, rulers can be used to determine mass or volume of the fuel consumed, with the following conditions: The ruler gauge must be part of the daily tank and calibrated at least once a year and have a book of control for recording the measurements (on a daily basis or per shift); (b) Accessories such as transducers, sonar and piezoelectronic devices are accepted if they are properly calibrated with the ruler gauge and receiving a reasonable maintenance; (c) In case of daily tanks with pre-heaters for heavy oil, the calibration will be made with the system at typical operational conditions
Frequency of monitoring/recording	Continuous
Value applied	Value to be monitored ex-post
Monitoring equipment	Mass or Volumetric Meter

QA/QC procedures to be applied	The consistency of metered fuel consumption quantities should be cross-checked by an annual energy balance that is based on purchased quantities and stock changes. Where the purchased fuel invoices can be identified specifically for the CDM project, the metered fuel consumption quantities should also be cross-checked with available purchase invoices from the financial records
Purpose of data	Calculation of baseline and project emissions.
Calculation method	
Comments	

Data / Parameter	$w_{C,i,y}$
Data unit	tC/mass unit of the fuel
Description	Weighted average mass fraction of carbon in fuel type i in year y
Source of data	(a) Values provided by the fuel supplier in invoices (b) Measurements by the project participants
Description of measurement methods and procedures to be applied	Measurements should be undertaken in line with national or international fuel standards
Frequency of monitoring/recording	The mass fraction of carbon should be obtained for each fuel delivery, from which weighted average annual values should be calculated
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	Verify if the values are within the uncertainty range of the product of the IPCC default values as provided in Table 1.2 and Table 1.3, Vol. 2 of the 2006 IPCC Guidelines. If the values fall below this range collect additional information from the testing laboratory to justify the outcome or conduct additional measurements. The laboratories in (b) should have ISO17025 accreditation or justify that they can comply with similar quality standards

Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	$\rho_{i,y}$
Data unit	Mass unit/volume unit
Description	Weighted average density of fuel type i in year y
Source of data	(a) Values provided by the fuel supplier in invoices (b) Measurements by the project participants (c) Regional or national default values
Description of measurement methods and procedures to be applied	Measurements should be undertaken in line with national or international fuel standards
Frequency of monitoring/recording	The density of the fuel should be obtained for each fuel delivery, from which weighted average annual values should be calculated
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	$NCV_{i,y}$
Data unit	GJ per mass or volume unit (e.g. GJ/m^3, GJ/ton)

Description	Weighted average net calorific value of fuel type i in year y
Source of data	(a) Values provided by the fuel supplier in invoices (b) Measurements by the project participants (c) Regional or national default values (d) IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in Table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Description of measurement methods and procedures to be applied	Measurements should be undertaken in line with national or international fuel standards
Frequency of monitoring/recording	(a & b) The NCV should be obtained for each fuel delivery, from which weighted average annual values should be calculated. (c) Review appropriateness of the values annually (d) Any future revision of the IPCC Guidelines should be taken into account
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	Verify if the values are within the uncertainty range of the IPCC default values as provided in Table 1.2, Vol. 2 of the 2006 IPCC Guidelines. If the values fall below this range collect additional information from the testing laboratory to justify the outcome or conduct additional measurements. The laboratories should have ISO17025 accreditation or justify that they can comply with similar quality standards
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	$EF_{CO2,(c),y}$
Data unit	tCO_2/GJ

Description	Weighted average CO_2 emission factor of fuel type (c) in year y
Source of data	(a) Values provided by the fuel supplier in invoices (b) Measurements by the project participants (c) Regional or national default values (d) IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in Table 1.4 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Description of measurement methods and procedures to be applied	Measurements should be undertaken in line with national or international fuel standards
Frequency of monitoring/recording	(a & b) The CO_2 emission factor should be obtained for each fuel delivery, from which weighted average annual values should be calculated. (c) Review appropriateness of the values annually (d) Any future revision of the IPCC Guidelines should be taken into account
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	
Comments	

PAI1

AMS-III.H, version 19.0

Data / Parameter	$Q_{ww,i,y}$
Data unit	m^3 /month
Description	Volume of wastewater treated in the project wastewater system in year y
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured continuously using flow meter
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained). Parameter aggregated monthly and annually for calculations.
Value applied	133,257 m^3/year
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Flow meters will undergo maintenance/calibration subject to appropriate industry standards. If local/national standards or the manufacturer's specifications are not available, international standards may be used. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	$COD_{ww,untreated,PJ,y}$ or $COD_{inflow, PJ,y}$
Data unit	$tCOD/m^3$
Description	Chemical oxygen demand of the wastewater entering the biogas treatment system in year y (tonnes/m^3)
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured the COD according to national or international standards. COD is measured through international sampling
Frequency of monitoring/recording	Periodically, at least monthly
Value applied	0.0657 t/m^3
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	$COD_{ww,treated,y}$ or $COD_{ww,outflow,y}$
Data unit	$tCOD/m^3$
Description	Chemical oxygen demand of the wastewater after the treatment system affected by the project activity.
Source of data	On site measurement
Description of measurement methods	Measure the COD according to national or international standards. COD is measured through representative sampling

and procedures to be applied	
Frequency of monitoring/recording	Periodically, at least monthly.
Value applied	0.0030 t/m^3
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	$COD_{ww,discharge,y}$
Data unit	tCOD/m^3
Description	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake (tonnes/m^3)
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measure the COD according to national or international standards. COD is measured through representative sampling.
Frequency of monitoring/recording	Periodically, at least monthly.
Value applied	0.0012 t/m^3
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external

	laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	$S_{final,PJ,y}$
Data unit	t
Description	Method of disposal of the sludge
Source of data	Record of site and date of sludge disposal
Description of measurement methods and procedures to be applied	The methane emissions from anaerobic decay of the final sludge are to be neglected because the sludge is used for soil application, then the end-use of the final sludge will be monitored during the crediting period.
Frequency of monitoring/recording	At each incidence of sludge disposal.
Value applied	No sludge treatment involved in the PAI1
Monitoring equipment	N/A
QA/QC procedures to be applied	Plant manager's signature is required on the record.
Purpose of data	Monitoring of sludge disposal.
Calculation method	-
Comments	-

Data / Parameter	$BG_{burnt,y}$
Data unit	m^3
Description	Biogas volume in year y

Source of data	Onsite measurement for $BG_{fuelled,y}$ .and $BG_{flared,y}$
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas recovered, fuelled, flared or otherwise utilized (e.g. injected into a natural gas distribution grid or distributed via a dedicated piped network) shall be monitored ex post, using continuous flow meters. If the biogas streams flared and fuelled (or utilized) are monitored separately, the two fractions can be added together to determine the total biogas recovered, without the need to monitor the recovered biogas before the separation. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place.
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value is not used for ex ante estimates. Value to be monitored ex-post at $BG_{fuelled,y}$ and $BG_{flared,y}$
Monitoring equipment	Gas flow meter(s)
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	Add the values for $BG_{flared,y}$ and $BG_{fuelled,y}$
Comments	-

Data / Parameter	$BG_{fuelled,y}$
Data unit	m^3
Description	Amount of biogas fuelled in year y
Source of data	Gas flowmeter

Description of measurement methods and procedures to be applied	In all cases, the amount of biogas fuelled shall be monitored ex post, using continuous flow meters. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$BG_{flared,y}$
Data unit	m^3
Description	Amount of biogas flared in year y
Source of data	Gas flowmeter
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas flared shall be monitored ex post, using continuous flow meters. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)

Value applied	Value to be monitored ex-post
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$W_{CH4,y}$
Data unit	%
Description	Methane content in biogas in the year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The fraction of methane in the gas should be measured with a continuous analyser or, alternatively, with periodical measurements at a 90/10 confidence/precision level. It shall be measured using equipment that can directly measure methane content in the biogas - the estimation of methane content of biogas based on measurement of other constituents of biogas such as CO_2 is not permitted. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Methane analyser

QA/QC procedures to be applied	Methane analyser to be maintained according to manufacturer's specifications. In the case of temporary situation where the methane analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	T
Data unit	^{o}C
Description	Temperature of the biogas
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The temperature of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	Shall be measured at the same time when methane content in biogas ($w_{CH4,y}$) is measured
Value applied	Value to be monitored ex-post
Monitoring equipment	Temperature transmitter

QA/QC procedures to be applied	Temperature transmitter to be maintained according to manufacturer's specifications. In the case of temporary situation where the temperature transmitter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	P
Data unit	Pa
Description	Pressure of the biogas
LI Source of data	On site measurement
Description of measurement methods and procedures to be applied	The pressure of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	Shall be measured at the same time when methane content in biogas ($w_{CH4,y}$) is measured
Value applied	Value to be monitored ex-post
Monitoring equipment	Pressure transmitter
QA/QC procedures to be applied	Pressure transmitter to be maintained according to manufacturer's specifications. In the case of temporary situation where the pressure transmitter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach

Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$Flame_m$
Data unit	Flame on or Flame off
Description	Flame detection of flare in the minute m
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measure using a fixed installation optical flame detector
Frequency of monitoring/recording	Once per minute. Detection of flame recorded as a minute that the flame was on, otherwise recorded as a minute that the flame was off
Value applied	Value to be monitored ex-post
Monitoring equipment	Optical flame detector: ultra violet detector or infra-red or both
QA/QC procedures to be applied	The flame detection will be monitored and cross checked with the amount of biogas sent to flare. If there is data for biogas sent to flare means the flame is on. Equipment will be maintained and calibrated in accordance with manufacturer's recommendations.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	For all PAIs where biogas is fueled, utilized and excess flared.

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh/year

Description	Electricity consumption in the project scenario in the year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measurement by electricity meters on the electricity entering the Biogas Plant (from the Mill and/or directly from the national electricity grid).
Frequency of monitoring/recording	Continuous, ongoing
Value applied	0
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice. The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity providers. For PAI1, the electricity will be supplied from mill. The electricity consumption will be recorded from the electricity meter owned by the mil. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$EG_{PJ,facility,y}$
Data unit	MWh/year

Description	Quantity of net electricity generation supplied by the project to the grid in year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter
Frequency of monitoring/recording	Continuous,
Value applied	8,000 MWh/year
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice. The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity providers. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	$F_{CH4,EG,t}$
Data unit	kg
Description	Mass flow of methane in the exhaust gas of the flare on a dry basis at reference conditions in the time period t
Source of data	Measurements are undertaken by a third-party accredited entity.
Description of measurement methods and procedures to be applied	Measure the mass flow of methane in the exhaust gas according to an appropriate national or international standard e.g. UKs Technical Guidance LFTGN05. The time period t over which the mass flow is measured must be at least one hour. The average flow rate to the flare during the time period t must be greater than the average flow rate observed for the previous six months
Frequency of monitoring/recording	Biannual
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	According to the standard applied
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Monitoring of this parameter is required in the case of enclosed flares and if the project participants decide to determine flare efficiency through biannual measurement

Data / Parameter	$T_{EG,m}$
Data unit	^{o}C
Description	Temperature in the exhaust gas of the enclosed flare in the minute *m*
Source of data	Project participants

Description of measurement methods and procedures to be applied	Measure the temperature of the exhaust gas in the flare by an appropriate temperature measurement equipment, Measurements outside the operational temperature specified by the manufacturer may indicate that the flare is not functioning correctly and may require maintenance. Flare manufacturers must provide suitable monitoring ports for the monitoring of the temperature of the flare. These would normally be expected to be in the middle third of the flare. Where more than one temperature port is fitted to the flare, the flare manufacturer must provide written instructions detailing the conditions under which each location shall be used and the port most suitable for monitoring the operation of the flare according to manufacturer's operating specifications for temperature.
Frequency of monitoring/recording	Once per minute
Value applied	Temperature measurement equipment should be replaced or calibrated in accordance with their maintenance schedule
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Temperature measurement equipment should be replaced or calibrated in accordane with their maintenance schedule.
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Unexpected changes such as a sudden increase/drop in temperature can occur for different reasons. These events should be noted in the site records along with any corrective action that was implemented to correct the issue. Monitoring of this parameter is applicable in case of enclosed flares. Measurements are required to determine if manufacturer's flare operating specifications for operating temperature are met.

Data / Parameter	$v_{i,RG,m}$
Data unit	-
Description	Volumetric fraction of component i in the residual gas on a dry basis in the minute m where i = CH_4, CO, CO_2, O_2, H_2, H_2S, NH_4, N_2

Source of data	Measurements by project participants using a continuous gas analyser (values are recorded with the same frequency as the flow).
Description of measurement methods and procedures to be applied	Measurement may be made on either dry or wet basis. If the value is made on a wet basis, then it shall be converted to dry basis for reporting
Frequency of monitoring/recording	Continuously. Values are to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Analysers must be periodically calibrated according to the manufacturer's recommendation. A zero check and a typical value check should be performed by comparison with a standard certified gas. In the case of temporary situation where the gas analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 3. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 4. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	As a simplified approach, project participants may only measure the content CH_4, CO and CO_2 of the residual gas and consider the remaining part as N_2. Monitoring of this parameter is only applicable in case of enclosed flares and continuous monitoring of the flare efficiency. The methane content measurement shall be carried out close to a

	location in the system where a biogas flow measurement takes place.

Data / Parameter	$V_{RG,m}$
Data unit	m^3
Description	Volumetric flow of the residual gas on a dry basis at reference conditions in the minute m
Source of data	Measurements by project participants using a flow meter
Description of measurement methods and procedures to be applied	Instruments with recordable electronic signal (analogical or digital)
Frequency of monitoring/recording	Continuously. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Flow meters are to be periodically calibrated according to the manufacturer's recommendation In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is applicable in case of enclosed flares and continuous monitoring of the flare efficiency and if project participant selects to calculate $VRG_{,m}$ instead of monitoring directly. Monitoring of this parameter may also be necessary for confirming that the manufacturer's operating specifications for flow

	rate/heat flux are met. In this case the flow rate should be measured in a m^3/h basis

Data / Parameter	$M_{RG,m}$
Data unit	kg
Description	Mass flow of the residual gas on a dry basis at reference conditions in the minute m
Source of data	-
Description of measurement methods and procedures to be applied	Instruments with recordable electronic signal (analogical or digital)
Frequency of monitoring/recording	Continuous. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory. Calibration and frequency of calibration is according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is applicable in case of enclosed flares and continuous monitoring of the flare efficiency and if project participant selects to monitor $M_{RG,m}$ directly, instead of calculating.

	Monitoring of this parameter may also be necessary for confirming that the manufacturer's specifications for flow rate/heat flux are met. In this case the flow rate should be measured in a kg/h basis

Data / Parameter	$v_{O2,EG,m}$
Data unit	-
Description	Volumetric fraction of O_2 in the exhaust gas on a dry basis at reference conditions in the minute m
Source of data	Measurements by project participants using a continuous gas analyser
Description of measurement methods and procedures to be applied	Extractive sampling analysers with water and particulates removal devices or in situ analysers for wet basis determination. The point of measurement (sampling point) shall be in the upper section of the flare (80% of total flare height). Sampling shall be conducted with appropriate sampling probes adequate to high temperatures level (e.g. inconel probes)
Frequency of monitoring/recording	Continuously. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Analysers must be periodically calibrated according to the manufacturer's recommendation. A zero check and a typical value check should be performed by comparison with a standard gas. In the case of temporary situation where the gas analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 3. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 4. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach

Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is only applicable in case of enclosed flares and continuous monitoring of the flare efficiency

Data / Parameter	$fc_{CH4,EG,m}$
Data unit	mg/m^3
Description	Concentration of methane in the exhaust gas of the flare on a dry basis at reference conditions in the minute m
Source of data	Measurements by project participants using a continuous gas analyser
Description of measurement methods and procedures to be applied	Extractive sampling analysers with water and particulates removal devices or in situ analyser for wet basis determination. The point of measurement (sampling point) shall be in the upper section of the flare in order that the sampling is of the gas after consumption has taken place (80% of total flare height). Sampling shall be conducted with appropriate sampling probes adequate to high temperatures level (e.g. inconel probes)
Frequency of monitoring/recording	Continuously. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Analysers must be periodically calibrated according to the manufacturer's recommendation. A zero check and a typical value check should be performed by comparison with a standard gas. In the case of temporary situation where the gas analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 3. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach.

	4. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Monitoring of this parameter is only applicable in case of enclosed flares and continuous monitoring of the flare efficiency. Measurement instruments may read ppmv or % values. To convert from ppmv to mg/m^3 simply multiply by 0.716. 1% equals 10 000 ppmv

Data / Parameter	$Maintenance_y$
Data unit	Calendar dates
Description	Maintenance events completed in year *y*
Source of data	Specific PAI
Description of measurement methods and procedures to be applied	Record the date that maintenance events were completed in year *y*. Records of maintenance logs must include all aspects of the maintenance including the details of the person(s) undertaking the work, parts replaced, or needing to be replaced, source of replacement parts, serial numbers and calibration certificates
Frequency of monitoring/recording	Annual
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Records must be kept in a maintenance log for two years beyond the life of the flare
Purpose of data	Calculation of project emissions.
Calculation method	

Comments	Monitoring of this parameter is required for the case of enclosed flares and the project participant selects Option B to determine flare efficiency. These dates are required so that they can be compared to the maintenance schedule to check that maintenance events were completed within the minimum time between maintenance events specified by the manufacturer ($SPEC_{,flare}$)

5.3 Monitoring Plan

Grouped Project

a) Organizational structure and responsibilities

The project owner organises a specific team in assigning responsibility for data collection, supervision, and witness the whole process of data measuring and recording. The organizational structure, responsibilities and competencies of the personnel that will carry out the monitoring and management of the project activities are as follows:

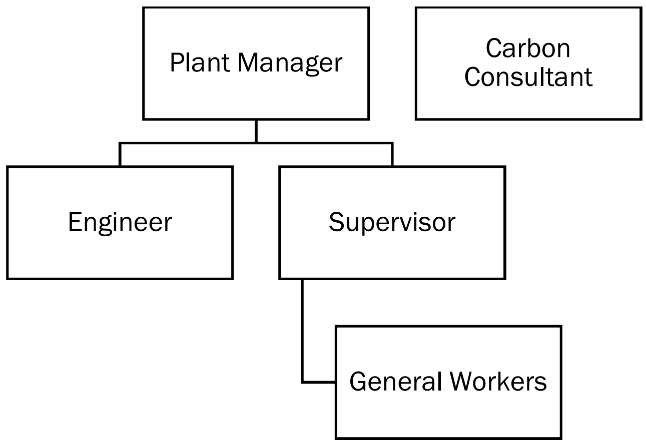

Figure 5: Grouped PAI VCS Organization Chart

The followings are the position and overall responsibility of the operation and monitoring team:

Table 9: Description of roles for PAI VCS team

Personnel	Description of Roles in Monitoring Plan
Plant Manager	• Report and seek approval from the higher management on VCS- related matters • Take full responsibility in the overall monitoring of the project • Conduct internal meetings on VCS matters • Oversees and coordinate the VCS monitoring plan data gathering • Monitor & ensure that data QA/QC is carried out by appointing (internal) independent verifiers to do cross-checking on irregular basis • Ensure all documentations were prepared accordingly based on VCS-PD requirement. • Coordinate with VCS consultant for matters (data/documents/records etc) that require rectification from plant • Point of contact for the mill, to respond to grievance from external parties.
Engineer	• Assists the VCS Manager in performing the VCS monitoring works • Assisting in the daily and monthly monitoring records for all the VCS related equipment.

Personnel	Description of Roles in Monitoring Plan
	• Prepare monthly summary record of the operations of plants/farms. • Ensure maintenance and calibration is carried out for proper running of machinery and equipment
Supervisors	• Report to the Engineer on any VCS monitoring issues • Conduct daily checking of data that have been identified for collection. • Ensuring on the functionality of the data recording devices and calibrated as planned (including performing QA/QC) • Responsible with the security of locked Programmable Logic Controller (PLC) control room.
General Workers	• Perform regular operational and maintenance tasks. • Conduct the daily recordings in daily monitoring log sheets and request verification from the supervisors on the log sheets. • Report any fault to supervisor-in-charge when necessary
VCS Consultant	• Provides advice on all VCS - related matters • Prepare monitoring reports for verification audit. • Liaises with verifier on verification processes • Conducts data audits on VCS monitoring

b) Data and parameters to be monitored

AMS-III.D, version 21.0

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
$VS_{LT,y}$	Volatile solids for livestock LT entering the animal manure management system in year y
$N_{da,y}$	Number of days animals alive in the farm in the year y
$N_{p,y}$	Number of animals produced annually of type LT for the year y
W_{site}	Average animal weight of a defined livestock population at the project site (kg)
$Q_{manure,LT,y}$	Quantity of manure treated from livestock type LT at animal manure management system j
$MS\%_{i,y}$	Fraction of manure handled in system I in the project activity in year y (%)
AL_{l}	Annual average interval between manure collection and delivery for treatment at a given storage device I (days)

Data to be monitored	Description
nd_y	Number of days that the animal manure management system was operational (days)
$MS\%_l$	Fraction of volatile solids handled by storage device l (%)
$SVS_{j,LT,y}$	Specific volatile solids content of animal manure from livestock type LT and animal manure management system j in year y (tonnes VS/tonnes--dm)
EE_y	Energy conversion efficiency of the project equipment (%)
GE_{LT}	Daily average gross energy intake (MJ/day)
DE_{LT}	Digestible energy of the feed (%)
UE	Urinary energy (fraction of GE)
ASH	Ash content of the manure calculated as a fraction of the dry matter feed intake (Fraction of dry matter feed intake)
ED_{LT}	Energy density of the feed fed to livestock type LT (MJ/kg-dm)
$BG_{burnt,y}$	Biogas volume in year y (m^3)
$BG_{fuelled,y}$	Amount of biogas fuelled (m^3)
$BG_{flarred,y}$	Amount of biogas flared (m^3)
w_{CH4}	Methane content in biogas in the year y (%)
T	Temperature of the biogas (^oC)
P	Pressure of the biogas (Pa)
FE	Flare efficiency (%)
$EC_{PJ,j,y}$	Electricity consumption in the project scenario in the year y (MWh/year)

AMS-III.H, version 19.0

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
$Q_{ww,i,y}$	Volume of wastewater treated in the project wastewater system in year y (m^3 /month)
$COD_{ww,untreated,PJ,y}$ or $COD_{inflow,\ PJ,y}$	Chemical oxygen demand of the wastewater entering the biogas treatment system in year y (tonnes/m^3)
$COD_{ww,treated,y}$ or $COD_{ww,outflow,y}$	Chemical oxygen demand of the wastewater after the treatment system affected by the project activity (tonnes/m^3)
$COD_{ww,discharge,y}$	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake (tonnes/m^3)

Data to be monitored	Description
$S_{final,PJ,y}$	Method of disposal of the sludge (t)
$BG_{burnt,y}$	Biogas volume in year y (m^3)
$BG_{fuelled,y}$	Amount of biogas fuelled in year y (m^3)
$BG_{flared,y}$	Amount of biogas flared in year y (m^3)
$w_{CH4,y}$	Methane content in biogas in the year y (%)
T	Temperature of the biogas at the flow measurement site (^oC)
P	Pressure of the biogas at the flow measurement site (Pa)
$Flame_m$	Flame detection of flare in the minute m (flame on and off)
$T_{EG,m}$	Temperature in the exhaust gas of the enclosed flare in minute m (^oC)
$EC_{PJ,j,y}$	Electricity consumption in the project scenario in the year y (MWh/year)

AMS-III.AQ, version 02.0

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
$FS_{Bio\text{-}CNG,y}$	Amount of Bio-CNG distributed/sold to retailers, filling stations bu the project activity in year y (tonnes)
$FC_{Bio\text{-}CNG,k,y}$	Bio-CNG consumed by the project vehicle k in the year y (tonnes)
$NCV_{Bio\text{-}CNG}$	Net calorific value of Bio-CNG (GJ/tonne)
$FP_{Bio\text{-}CNG,y}$	Quantity of the Bio-CNG produced by the project activity in the year y (tonnes)
$f_{FO,gasoline}$	Fraction of gasoline from fossil fuel origin in the displaced gasoline (%)
NCV_i	Net calorific value of gasoline/ blended gasoline that was used by project vehicle k (GJ/tonne)
$w_{CH4,y}$	Methane content in the Bio-CNG (%)

Additional monitoring is required in the event that AMS-III.AQ Approach 2 is applied for the PAIs, where filling stations will be equipped with the following devices systems:

a) Automatic Number Plate Recognition (ANPR); or Electronic Vehicle Identification (EVI);

b) Automatic locking and unlocking function of dispenser directly controlled by equipped device/system responsible for project vehicle identification to ensure that all the Bio-CNG that is produced is only consumed in the project vehicles;

c) System for logging of the data on quantity of Bio-CNG filled into identified vehicles;

d) Natural gas analyzer capable of analysing ethane and propane to ensure that the gas delivered to the vehicle by the dispenser does not contain ethane or/and propane.

AMS-III.O, version 02.0

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
V_{H2BIO}	Volume of hydrogen produced from methane extracted from under normal condition (Nm^3-H_2)
M_{LPG}	LPG used as feedstock to hydrogen production unit (kg-LPG)
m_1	% mol of propane in LPG (%)
m_2	% mol of butane in LPG (%)
$EC_{PJ,j,y}$	Electricity consumption in the project scenario in the year y (MWh/year)
PE_{Elec}	Project emissions from electricity consumption in year y (tCO_2/year)
PE_{FF}	Project emissions from fossil fuel comsumption in year y (tCO_2/year)

AMS-I.D, version 18.0

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
$EC_{PJ,\ facility,j,y}$	Quantity of net electricity generation supplied by the project to the grid in year *y* (MWh)
$EG_{PJ,add,y}$	Quantity of net electricity generation supplied to the grid in year y by the project plant/unit that has been added under the project activity

TOOL06: Project emissions from flaring

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
$F_{CH4,EG,t}$	Mass flow of methane in the exhaust gas of the flare on a dry basis at reference conditions in the time period t (kg)
$T_{EG,m}$	Temperature in the exhaust gas of the enclosed flare in the minute m (°C)

Data to be monitored	Description
$v_{i,RG,m}$	Volumetric fraction of component i in the residual gas on a dry basis in the minute m where i = CH_4, CO, CO_2, O_2, H_2, H_2S, NH_4, N_2
$V_{RG,m}$	Volumetric flow of the residual gas on a dry basis at reference conditions in the minute m (m^3)
$M_{RG,m}$	Mass flow of the residual gas on a dry basis at reference conditions in the minute m (kg)
$v_{O2,EG,m}$	Volumetric fraction of O2 in the exhaust gas on a dry basis at reference conditions in the minute m
$fc_{CH4,EG,m}$	Concentration of methane in the exhaust gas of the flare on a dry basis at reference conditions in the minute m (mg/m^3)
$Flame_m$	Flame detection of flare in the minute m (on or off)
$Maintenance_y$	Maintenance events completed in year y (calendar dates)

<u>TOOL05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation</u>

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
$EF_{grid,CM,y}$	Grid Emission Factor for Peninsula Malaysia, Sabah, and Sarawak (tCO_2/MWh)
$TDL_{j,y}$ and $TDL_{k,y}$ and $TDL_{l,y}$	Average technical transmission and distribution losses for providing electricity to source j, k or l in year y
$FC_{n,i,t}$	Quantity of fossil fuel type i fired in the captive power plant n in the time period t (m^3, ton or l)
$EG_{n,t}$	Quantity of electricity generated in captive power plant n in the time period t (MWh)
$NCV_{i,t}$	Average net calorific value of fossil fuel type i used in the period t (GJ / mass or volume unit)
$EF_{CO2,i,t}$	CO2 emission factor of fossil fuel type i used in the period t ($t CO_2$ / GJ)
$EC_{PJ,j,y}$;	Quantity of electricity consumed by the project electricity consumption source j in year y
$EC_{BL,k,y}$	Quantity of electricity that would be consumed by the baseline electricity consumption source k in year y (MWh/yr)

Data to be monitored	Description
$EG_{PJ,grid,y}$ or $EG_{PJ,facility,I,y}$	Quantity of electricity generated and supplied by the project power plant to the grid in year y Quantity of electricity generated and supplied by the project power plant to the consumers/electricity consuming facility i in year y (MWh/yr)

Tool 03: Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion:

Data to be monitored	Description
$FC_{i,j,y}$	Quantity of fuel type i combusted in process j during the year y (ton/yr or m^3/yr)
$w_{C,i,y}$	Weighted average mass fraction of carbon in fuel type i in year y (tC/mass unit of the fuel)
$\rho_{i,y}$	Weighted average density of fuel type i in year y (Mass unit/volume unit)
$NCV_{i,y}$	Weighted average net calorific value of fuel type i in year y (GJ/m^3, GJ/ton)
$EF_{CO2,i,y}$	CO_2 emission factor of fossil fuel type i used in the period t (t CO_2 / GJ)

c) Training of monitoring personnel

All plant staff will be trained by Green Lagoon Technology Sdn Bhd, prior to the commissioning of the plants. The involved personnel shall receive adequate and regular training on the project performances and the monitoring plan involving methane capture up to utilization of the biogas.

Training of the running of day-to-day operations is under the responsibility of the individual PAIs. This include collection of data and ensuring data integrity is not tolerated.

A VCS monitoring Manual will be drafted and brief training will be provided by the appointed VCS consultant on the monitoring parameters.

d) Procedures for handling non-conformances with the validated monitoring plan

In case of a temporary situations such as the installed flow meter(s)/gas analyser malfunctioned or gave unrepresentative results, the data shall be measured according to the method listed below:

1. Manually with a back-up flow meter/gas analyser. The number of samples needed for the calculation will be calculated by using raosoft calculator, assuming the 95% confidence interval, 10% margin of error. The lower bound of the 95% confidence interval will be applied in the calculation sheet as a conservative approach
2. Alternatively, if a back-up meter is not available, the historical records and future data (at least one (1) month before and after the malfunction period) will be used to estimate the quantity. The 95% confidence interval will be calculated from the data obtained. The lower bound of the 95% confidence interval will be applied in the calculation sheet as a conservative approach

In relation to malfunctions of measurement of COD, the following will be observed. In case of temporary situation such as the COD sampling device/meter malfunctioned, the amount of COD shall be based on the most recent testing result

e) Sampling approaches and precision level

For AMS-III.D related PAIs, the approach used is 'simple random sampling' in accordance with EB86 Annex 4, 'Sampling and surveys for CDM project activities and programme of activities'. The purpose is to ensure 90% confidence that the margin of error in our estimate is not more than ±10% in relative terms.

For AMS-III.H, AMS-III.AQ and AMSI-III.O related PAIs, all data that are collected in accordance with the VCS-PD monitoring frequency and data collection, are included and considered in the emission reduction calculation.

In relation to the measurement of COD: According to raosoft, the total number of months in a year is 12 months and the total number of samples needed to be collected is 11 samples to meet the 90% confidence interval.

f) Quality Assurance and Quality Control & maintenance procedures

Maintenance of the equipment shall be conducted regularly to ensure a good and proper functioning of the equipment.

Calibration of monitoring equipment and instruments will be done periodically according to the standard and procedures to maintain the accuracy of the measuring instrument

Emergency/Contingency preparedness for monitoring system is presented as below:

1. Data collection

   If the person-in-charge for daily raw data logging is unavailable, there will be another back-up person to take over the job.

2. Data storage, transfer and backup

   The monitored parameters that are collected and will be stored in the following ways:

   (a) Hard copy - Log sheets/ logbooks will be kept in proper manner, and available for audit verification purpose,

   (b) Soft copy - Raw data from log sheets/ logbooks will be stored as soft copy; together with monthly CER calculation, the entire data of the project activity will be back-up to an external hard drive or another computer every quarterly.

Immediate action shall be taken immediately if any matters/ uncertainties affecting the data quality.

In case of leakage occurrences in any parts of the biogas plant system, the team shall take immediate action to put the plant operation back on track. The plant operation manual shall include the procedure on safety and always made available on site to ensure the operators are fully aware of emergency procedures.

PAI1

The monitoring plan for this project includes all monitoring requirements following the procedures set by AMS-III.H (version 19.0) and AMS-I.D (version 18.0). The overall monitoring plan can be illustrated in the figure below:

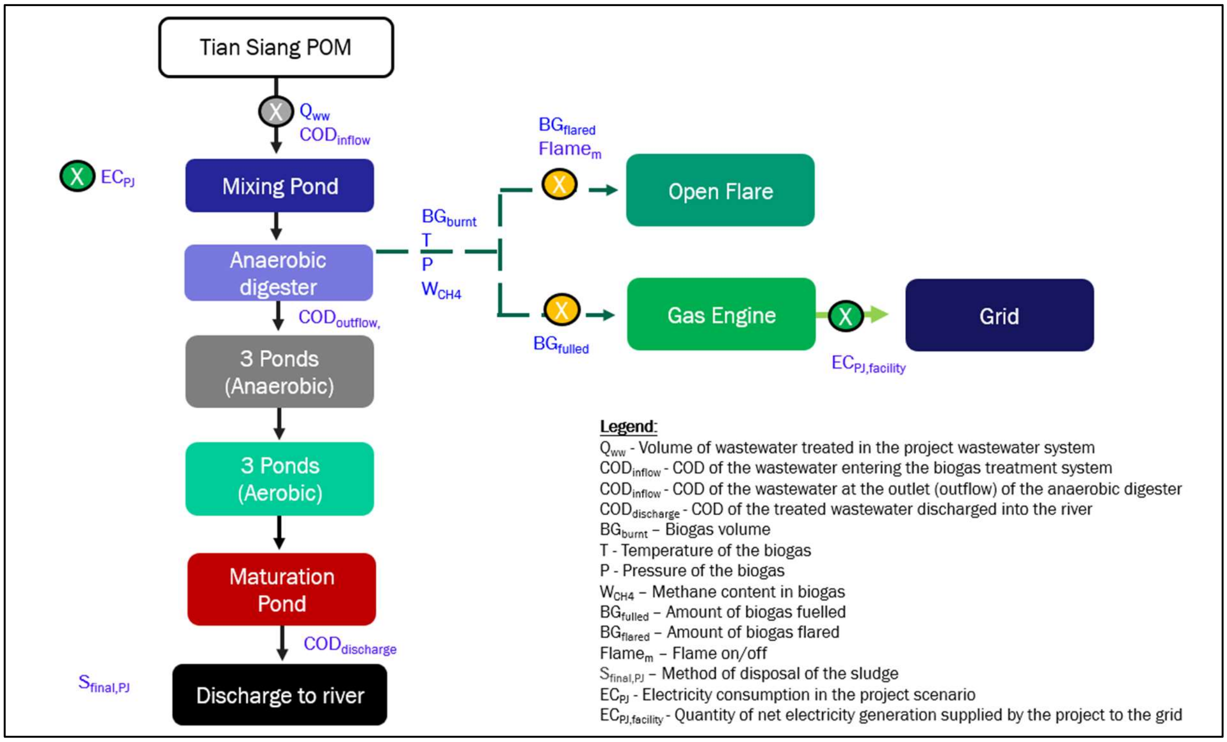

Figure 6: Monitoring Plan for PAI1

Implementation of the Monitoring Plan

The biogas plant operator will adopt the instructions presented in the monitoring plan and ensure all activities related to the biogas plant operation and monitoring are implemented. The main responsibilities of monitoring include:

- Data reading and handling: maintaining an adequate system for collecting, recording and storing data in accordance with the protocols determined by the monitoring plan as well as checking data quality, collection and record-keeping procedures on a regular basis. This also includes making sure that all relevant electronic data is backed-up electronically for at least 2 years after the end of the crediting period in case of an unanticipated loss of data.

- In the case of erroneous data recorded for the monitored parameter occurred during short period, data for the affected period shall be derived based on the approach based on 95% confidence interval principles (source: "IPCC Good Practice Guidance and Uncertainty Management in National Greenhouse Gas Inventories", page 6.6). A total of 10 readings before and after the constant period will be used to calculate the lower bound of 95% confidence interval with reference to the above-mentioned guideline. The lower bound of the interval boundaries calculated will be applied to the period for the constant data as a conservative approach. Alternately, if the 10 readings before and after the erroneous period is unavailable, the record from the daily manual log-sheet will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the period for the constant data as a conservative approach.

- Reporting: preparing periodic summaries and reports that include emission reductions generated and observations regarding monitoring plan procedures;
- Training: assuring assigned personnel received adequate and regular training on project performances and the monitoring plan. Any new staff taking over the role of monitoring and reporting will be fully trained as well; and
- Quality control and quality assurance: complying with quality control and quality assurance procedures to facilitate periodical audits and verification. Any observation of matters affecting data quality will be reported to the management immediately.

Responsibility of Monitoring Plan

A dedicated VCS monitoring team will be assigned to be responsible for undertaking the monitoring as well as the quality control and assurance tasks. The following are the position and responsibility of the VCS monitoring team:

Table 9: Description of roles for PAI1 VCS team

Personnel	Description of Roles in Monitoring Plan
Plant Manager	• Report and seek approval from the higher management on VCS-related matters • Take full responsibility in the overall monitoring of the project • Conduct internal meetings on VCS matters • Oversees and coordinate the VCS monitoring plan data gathering • Monitor & ensure that data QA/QC is carried out by appointing (internal) independent verifiers to do cross-checking on irregular basis • Ensure all documentations were prepared accordingly based on VCS-PD requirement. • Coordinate with VCS consultant for matters (data/documents/records etc) that require rectification from plant
Engineer	• Assists the VCS Manager in performing the VCS monitoring works • Assisting in the daily and monthly monitoring records for all the VCS related equipment. • Prepare monthly summary record of the operations of plants/farms. • Ensure maintenance and calibration is carried out for proper running of machinery and equipment
Supervisors	• Report to the Engineer on any VCS monitoring issues • Conduct daily checking of data that have been identified for collection.

Personnel	Description of Roles in Monitoring Plan
	• Ensuring on the functionality of the data recording devices and calibrated as planned (including performing QA/QC) • Responsible with the security of locked Programmable Logic Controller (PLC) control room.
General Workers	• Perform regular operational and maintenance tasks. • Conduct the daily recordings in daily monitoring log sheets and request verification from the supervisors on the log sheets. • Report any fault to supervisor-in-charge when necessary
VCS Consultant	• Provides advice on all VCS - related matters • Prepare monitoring reports for verification audit. • Liaises with verifier on verification processes • Conducts data audits on VCS monitoring

<u>Quality Control, and Quality Assurance</u>

Quality control and quality assurance procedures will be carried out on all parameters monitored. The practices to be undertaken in the context of the proposed project are as follows:

1. Monitoring records:
   - Daily readings of all meters will be filled out in paper worksheets and filed consequently. All data collected will also be entered in electronic worksheets and stored in a computer (immediately) and on discs (periodically).
   - Periodic controls of the monitoring records will be carried out to check any deviation from the monitoring plan.

2. Equipment calibration and maintenance:
   - Flow meters, gas analyzers and other sensors will be subject to regular maintenance and testing according to the technical specifications by the manufacturers to ensure accuracy and good performance.
   - Calibration of equipment will be conducted periodically according to technical specifications.

3. Corrective actions:
   - Actions to correct deviations from the monitoring plan will be implemented as these deviations are observed either by the operator or during internal audits.
   - Apart from periodic meetings, additional technical meetings will be held in order to define any necessary corrective actions to be undertaken.

4. Site audits:

- Regular site audits will be conducted to ensure that monitoring and operational procedures are being observed in accordance with the monitoring plan.

5. Training:
    - A training plan will be set out for the operator. This will ensure that monitoring staff involved in the project are properly trained so that they are able to undertake the tasks required by this monitoring plan. Appropriate staff training must be provided before the project starts operating and generating emission reductions.

6. Storage of documents:
    - List of monitoring equipment (e.g. flow meters, gas analyzers, thermometers) including their model numbers, names, manufacturers, specifications, user requirements, etc.
    - Calibration lists and reports, including equipment or parts calibrated, date, method and procedures of calibration, precision following these processes, personnel, devices needed, etc.
    - Maintenance lists and reports, including equipment or parts maintained, date, method and procedure of maintenance, performance following these procedures, personnel, devices needed, etc.
    - Operation manual of the proposed project.
    - Registration of minutes during meetings.
    - Non-conformance reports.
    - Monthly and annual worksheets.
    - Training plan.
    - Internal audit reports, including personnel, time, findings, corrective actions, follow-up inspections, etc.
    - Annual monitoring review

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification
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